



**Verified Carbon
Standard**

WASTEWATER TO BIOGAS GROUPED PROJECT



CONTROLUNION

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VCS Standard Version	4.7
Client	Green Lagoon Technology Sdn Bhd
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Summary:

Control Union Certifications Germany GmbH ("CUCG"), appointed by Green Lagoon Technology Sdn Bhd (GLT), has conducted a validation for greenhouse gas emission reductions related to the "Wastewater to Biogas Grouped Project" with Project ID VCS 4887, adhering to VCS activities' relevant requirements.

The primary objective of this validation was to independently evaluate the project activity against the VCS standard, assess its ex-ante estimation of emission reductions, and establish the monitoring procedures to be followed. Additionally, the validation aimed to independently review ex-post the monitored reductions in greenhouse gas emissions.

The purpose of the validation of the project is to have a thorough and independent assessment of the proposed Project against the following:

- Applicable VCS requirements, in particular the project's baseline, monitoring plan and compliance with relevant VCS rules and host country criteria.
- The application of methodologies AMS-III.H Version 21.0 "Methane Recovery in Wastewater Treatment" and AMS-I.D Version 18.0 "Grid Connected Renewable Electricity Generation"

The validation is to confirm that the project design, as documented, is sound and reasonable and meets the identified criteria and assess the claims and assumptions made without limitation on the information provided by the project proponents.

The scope of the services provided by CUCG is to conduct an independent and objective assessment of the Project Description Document (PDD) to compare the assertions and presumptions in the PDD to the VCS criteria, including but not limited to the VCS Standard, other pertinent guidelines established for Verra, UNFCCC criteria and the applied methodology.

The validation consisted of the following three phases:

- A desk review of the project design and the baseline and monitoring plan;
- Project site visit, onsite validation and follow up interviews with project stakeholders;
- The resolution of outstanding issues and the issuance of the final validation report and opinion.

The "Grouped Project," developed by GLT, aims to achieve biogas recovery, flaring, and/or utilization of recovered biogas for gainful use, such as electricity or heat generation. The proposed first project

activity instance (PAI1) is designed to replace the open lagoon treatment method for POME from Tian Siang Oil Mill (Triang) Sdn Bhd ("Tian Siang POM").

During this validation, 16 Corrective Action Requests (CARs), 02 Clarification Requests (CLs), 01 Forward Action Request (FAR), and 01 Observation (OBS) were identified concerning project details, safeguards and stakeholder engagement, application of the methodology, quantification of estimated GHG emission reductions and removals, and monitoring of the VCS project activity. These CARs and CLs were successfully addressed after necessary corrections/clarifications by the client. The same has been discussed in Appendix IV of this validation report.

In summary, CUCG is of the opinion that the project correctly applies the baseline and monitoring methodology AMS-III.H Version 21.0 "Methane Recovery in Wastewater Treatment" and AMS-I.D Version 18.0 "Grid Connected Renewable Electricity Generation" and meets all the relevant VCS Standard Version 4.7, relevant methodology, tools and guidelines, UNFCCC requirements and the relevant host country criteria.

PAI1 is estimated to achieve an annual average GHG emission reduction of 35,217 tCO₂e/year, totaling 246,518 tCO₂e over the first 7-year crediting period (24-April-2024 to 23-April-2031). Therefore, CUCG requests the registration and issuance of the project as VCS Project.

CONTENTS

1	INTRODUCTION	5
1.1	Objective	5
1.2	Scope and Criteria	5
1.3	Reasonableness of Assumptions	6
1.4	Summary Description of the Project	6
2	VALIDATION PROCESS	7
2.1	Method and Criteria	7
2.2	Document Review	10
2.3	Interviews	10
2.4	Site Visits	12
2.5	Resolution of Findings	13
3	VALIDATION FINDINGS	14
3.1	Project Details	14
3.2	Safeguards and Stakeholder Engagement	36
3.3	Application of Methodology	52
3.4	Non-Permanence Risk Analysis	100
4	VALIDATION OPINION	100
4.1	Validation Summary	100
4.2	Validation Conclusion	102
	APPENDIX I: COMMERCIALLY SENSITIVE INFORMATION	105
	APPENDIX II: REFERENCED DOCUMENTS	106
	APPENDIX III: ABBREVIATIONS	108
	APPENDIX IV: CLARIFICATION (CL), CORRECTIVE ACTIONS (CAR) & FORWARDS ACTION REQUESTS (FAR).	109

1 INTRODUCTION

1.1 Objective

Control Union Certifications Germany GmbH (hereafter referred to as “CUCG”) has been appointed by Green Lagoon Technology Sdn Bhd to conduct the validation assessment of the project – “Green Lagoon Technology Sdn Bhd”, VCS ID 4887 according to the relevant requirements of VCS programme guidelines and standard:

- VCS Standard, v4.7
- ISO 14065
- ISO 14064-3
- VCS Program Definitions, v4.5
- VCS Program Guide, v4.4
- Project Description, v4.4
- Project Methodologies Documents (AMS-III.H & AMS-I.D)

The purpose of the validation is to have an independent third-party assessment of whether the project activity conforms to the qualification criteria presented in the VCS Standard, version v4.7 to attain real, measurable, additional, and permanent emission reductions.

The assessment covered the project activity intended to achieve 246,518 tCO₂e emission reductions in the first 7 years crediting period. The grouped project consists of 2 Sectoral scopes, 4 project activities and 1 Project Activity Instances (using methodologies AMS-III.H, AMS-I.D).

1.2 Scope and Criteria

The validation assessment, "Wastewater to Biogas Group," was conducted at PAI1, Tian Siang Oil Mill (Triang), located in Triang, Pahang, on May 28th and 29th, 2024. The assessment activities included a documentation review, interviews, and a site visit to the project site. Both internal and external stakeholders were interviewed during the assessment.

The validation also includes an assessment of the data and documents provided to Control Union, as well as a review of the contract terms. The scope of the validation is conducted to establish and assess the following:

- The project activity is in accordance with rules, applicable law and legislation of the host country (Malaysia).

- The project activity is in accordance with all relevant VCS rules and requirements, which includes the following: VCS Standard, v4.7, VCS Program Definitions, v4.5, VCS Program Guide, v4.4, Project Description, v4.4, Project Methodologies Documents, VCS Validation and Verification Manual, v3.2, including but not limited to additionality, proof of ownership and compliance with local laws and regulations.
- The project activity is in accordance with the conditions of the latest version of applied VCS Project Methodology: AMS-III.H (Version 19.0) - Methane recovery in wastewater treatment and AMS-I.D (Version 18.0) – Grid-connected renewable electricity generation.

The validation is not intended to provide consulting services to the project participants. However, requests for clarifications and/or corrective actions may provide input for improving the project design.

1.3 Reasonableness of Assumptions

The validation report is based on the VCS Program Guide and VCS Validation and Verification Manual, supporting evidence made available to the validator and information collected through performing interviews and during the on-site audit assessment.

1.4 Summary Description of the Project

Grouped Project Description

Project description and location: ‘The Wastewater to Biogas Grouped Project’, owned by Green Lagoon Technology Sdn Bhd, comprises multiple biogas recovery projects in Malaysia. Each project activity instance (PAI) is implemented by an individual plant operator. The project aims to recover biogas from wastewater treatment processes, preventing greenhouse gas emissions. The recovered biogas will be used for grid upload, bottling and distribution, or flaring excess biogas.

Project Activity Instance 1 (PAI1) Description

The first Project Activity Instance (PAI1) is located at Tian Siang Oil Mill (Triang) Sdn Bhd (“Tian Siang POM”) in the State of Pahang, Malaysia, about 6 KM Off Jalan Triang Ke Kemayang, Triang, Pahang and 180km north of Kuala Lumpur. GLT Maju Sdn Bhd, owned by Green Lagoon Technology Sdn Bhd, is the owner of the biogas plant PAI1. Green Lagoon Technology Sdn Bhd has an agreement/45 with GLT Maju Sdn Bhd for the legal rights to operate PAI1. Additionally, the mill operator, Tian Siang POM, has an agreement with GLT Maju Sdn Bhd for ongoing access to palm oil mill effluent (POME) and permission to construct and operate the PAI1.

Baseline Scenario: Before implementing PAI1, Tian Siang POM treated wastewater in a series of open anaerobic lagoons, a common method used by about 85% of palm oil mills in Malaysia. This process, involving eight treatment ponds (four anaerobic, two aerobic, one settling basin, and one maturation pond), resulted in significant methane generation due to the anaerobic conditions and regional temperature, producing approximately 246,518 tCO₂e/year over seven

years. The treated wastewater was then discharged into the river, and sludge was disposed-off via land application near the project area.

Project Scenario: The project improves wastewater treatment by installing an anaerobic digester with a membrane to capture methane emissions in a new lagoon. The residue undergoes further aerobic treatment before being discharged into the river. The captured biogas is used for power generation and uploaded to the grid, significantly reducing greenhouse gas emissions. The estimated annual average and total GHG emission reductions during the first seven-year crediting period for PA11 are detailed in the project documentation.

Project Title	Wastewater to Biogas Grouped Project
Project Proponent	Green Lagoon Technology Sdn Bhd
Location of the Project	Tian Siang POM, Pahang, Malaysia
Project Start Date	24-Apr-2024
Sectoral Scope	13 - Waste handling and disposal 1 - Energy industries (renewable/non-renewable sources)
Applied Methodology/Version	AMS-III.H (Version 19.0) - Methane recovery in wastewater treatment AMS-I.D (Version 18.0) – Grid-connected renewable electricity generation
Annual Average GHG Emission (tCO2e)	35,217
Total GHG emission reductions for first crediting period (tCO2e)	246,518

2 VALIDATION PROCESS

2.1 Method and Criteria

The validation assessment was conducted by using CUCG's procedures which in line with the requirements specified in the VCS Standard version 4.7 and relevant project methodology

documents and specification with guidance for the verification and validation of greenhouse gas statement ISO 14064-3 Standard.

CUCG completed a strategic analysis and risk assessment of the project's activities and processes in order to gain a full understanding of (if applicable):

- Project Details;
- Application of Methodologies;
- Applicable local and international laws and regulations;
- Estimated GHG Emission Reduction and Removals;
- Monitoring plan;
- Safeguards and Stakeholder Engagement.

CUCG has ascertained that the reported information in the Project Description Document (PDD) is complete and accurate in question. This involved an on-site audit, and a desk review of the Project Design. This Validation Report describes the findings of the assessment. A sample of documents related to administrative and operational activities of the project were requested from GLT during the validation assessment.

Assessment team

According to the sectoral scopes/ technical area and experiences in the sectoral or national business environment, CUCG has composed a project assessment team in accordance with the appointment rules in CUCG. The composition of the assessment team was approved by the CUCG that the required skills are covered by the team. The four qualification levels for team members that are assigned by formal appointment rules as below:

- Leader Validator (Lead Validator)
- Validator (V)
- Technical Reviewer (TR)
- Technical Experts (TE)

Name	Title	Coverage of scope	Technical Area
Supun Sachithra Nigamuni	LV	Project Details	Sectoral Scope & Project Type, Project Eligibility, Project Design, Project Proponent, Other Entities Involved in the Project, Ownership, Project Start Date, Project Crediting Period, Project Scale and Estimated GHG Emission Reduction or Removals, Description of the Project Activity, Project Location, Conditions Prior to Project Initiation, Double Counting and Participation

			under Other GHG Programs, Double Claiming, Other Forms of Credit, and Scope 3 Emissions, Sustainable Development Contributions and Additional Information Relevant to the Project.
Shahizzudin Bin Mohd Said	V	Application of Methodology, Quantification of Estimated GHG Emission Reductions and Removals, and Monitoring	Title and Reference of Methodology, Applicability of Methodology, Project Boundary, Baseline Scenario, Additionality, Methodology Deviations, Baseline Emissions, Project Emissions, Leakage Emissions, Estimated GHG Emission Reductions and Carbon Dioxide Removals, Data and Parameters Available at Validation, Data and Parameters Monitored and Monitoring Plan.
Jasmandy Syahrul	V	Project Details, Safeguard and Stakeholder Engagement Applicable Local and International Laws and Regulations.	Compliance with Laws, Statues and Other Regulatory Frameworks, Stakeholder Engagement and Consultation, Risk to Stakeholders and the Environment, Respect for Human Rights and Equity, and Ecosystem Health,
Tan Zee Boon	TE	Application of Methodology, Quantification of Estimated GHG Emission Reductions and Removals, and Monitoring	Title and Reference of Methodology, Applicability of Methodology, Project Boundary, Baseline Scenario, Additionality, Methodology Deviations, Baseline Emissions, Project Emissions, Leakage Emissions, Estimated GHG Emission Reductions and Carbon Dioxide Removals, Data and Parameters Available at Validation, Data and Parameters Monitored and Monitoring Plan.
Oyeshola Kofoworola	TR	Technical review of Validation report and process	Technical review of Validation report and process

2.2 Document Review

CUCG has conducted a comprehensive desk review that included:

- Evaluating the completeness of the data and information presented.
- Reviewing the VCS Project Description, the estimated GHG emission reduction calculation sheet, the applied methodologies including applicable tools, and the relevant laws and regulations for the host country.
- Examining the supporting documents provided by the Project Proponent.

A complete list of documents/evidence reviewed is included as Appendix III.

2.3 Interviews

The interview session with the project's stakeholders is a part of validation procedure. It was done during the site visit on 28.05.2024 and 29.05.2024. It involved interaction with project representatives and the project team of GLT to assess the information reported in the PDD/**1**. The assessment team has interviewed the representatives of Project Proponent to confirm selected information and to clarify issues identified during the document review. The interviews were conducted through physical and phone conversations.

The details of the interviews conducted during the validation process are as follows:

No	Interviewee		Date	Subject
	Name of Stakeholder	Affiliation		
1	Mr. Chan Sow Keong	Green Lagoon Technology Sdn Bhd (Director)	28.05.2024 - 29.05.2024	Project ownership and management.
2	Ms. Loh Phui Nying	Green Lagoon Technology Sdn Bhd (Sustainability Engineer)	28.05.2024 - 29.05.2024	Project details, stakeholder engagement and project monitoring
3	Ms. Nur Hidayah binti Abdul Samat	Green Lagoon Technology Sdn Bhd (Site Engineer)	28.05.2024 - 29.05.2024	Project Design (project technical drawings)

4	Mr. Faiz	GLT Maju Sdn Bhd (Site Operation Manager)	28.05.2024 - 29.05.2024	Project site management and operation
5	Ms. Chen Saw Ling	Eco Ideal Consulting Sdn Bhd (Chief Technical Officer)	28.05.2024 - 29.05.2024	Title & reference of methodology, applicability of methodology, project boundary, baseline scenario, additionality, methodology deviations, quantification of estimated GHG emission reductions and removals, monitoring plan.
6	Ms. Dahlia binti Dahlan	Eco Ideal Consulting Sdn Bhd (Sr. Environmental Executive)	28.05.2024 - 29.05.2024	Safeguards and stakeholder engagement.
7	Mr. Afizal bin Azemim	GLT Maju Sdn (Workers)	28.05.2024 - 29.05.2024	Contract agreement, wages, training & social issues (child labour, sexual harassment, discrimination and etc.)
8	Mr. Danesh A/L Krishnan	Tian Siang Palm Oil Mill (Mill Manager)	28.05.2024	Land ownership, terms and condition of the land, land agreement between GLT & Tian Siang POM, POME supply & stakeholder consultation and etc.
9	Ms. Aini	Department of Environment (Enforcement officer)	28.05.2024	Applicability of law & regulation related to environment towards biogas plant.
10	Ms. Fazrina	Department of Environment (Enforcement officer)	28.05.2024	Applicability of law & regulation related to environment towards biogas plant.

11	Mr. Suzaimi	Felda Mengkuang (Staff & villages)	29.05.2024	Stakeholder consultation, job opportunity and complaint & grievances.
12	Ms. Hanum	Bera Land Department (Enforcement officer)	28.05.2024	Term and condition of land title, ownership of the land title
13	Mr. Adel	Bera Land Department (Enforcement officer)	28.05.2024	Term and condition of land title, ownership of the land title
14	Ms. Nurul Ain binti Mazlan	Pahang Labour Department (Enforcement officer)	28.05.2024	Applicability of law & regulation related to workers.

2.4 Site Visits

CUCG has performed an on-site validation at Green Lagoon Technology Sdn Bhd at Tian Siang POM located at Triang, Pahang from 28th to 29th May 2024. The site visit was performed to validate the documents established by PP and to inspect the biogas plant as declared in the PDD/1. Details of the activity plan as described below:

Date	Activity	Location	Verifier / TE
28.06.2024	- Opening meeting by Lead Validator - Discussions with Project Proponent, Project Developers & Project Consultant - Overview of the PAI1 project by the company - Site-Visit – Biogas Plant - Stakeholders Interview (internal & external)	GLT Biogas Plant (Office)	- Supun Nigamuni - Shahizzudin Mohd Said - Jasmandy Syahrul - Tan Zee Boon (TE)

	6. Documentation review and validation of the project according to the sub-sections as follows: a. Project documentation b. Project design c. Project ownership d. Project crediting period e. Project scale, location, boundary f. Baseline scenario g. Additionality h. Quantification of GHG Emission Reduction and Carbon Dioxide Removals i. Monitoring j. Sustainability Development Contributions k. Stakeholder Consultation		
29.05.2024	Documentation review and validation of the project according to the sub-sections as follows: a. Safeguards b. Methodology deviations c. Project description deviations d. Methodology grace periods e. Double counting and Participation under Other GHG Programs f. Double Claiming g. Other Forms of Credit h. Scope 3 Emissions Closing meeting by Lead Validator and presentation of finding(s)		

The interviewed personnel and objective of the interview were listed in section 2.3, Interviews.

2.5 Resolution of Findings

The findings from the validation activities were raised according to the following types:

- Forward Action Requests (FARs): The VVB considers that a certain issue regarding project implementation or reporting should be reviewed during the next verification.

- Clarification Requests (CLs): Project reporting lacks transparency and further information is needed to determine if a material discrepancy is present.
- Corrective Action Requests (CARs): The VVB has identified a material discrepancy or non-conformance that the Project proponent must address.
- Observation (OBS): areas being in compliance but very close to becoming a nonconformance or that given additional evidence could transform into a nonconformance.

The objective of this phase of the validation was to resolve the requests for corrective actions and clarifications, as well as any other outstanding issues that needed to be clarified for CUCG to issue a positive conclusion on the project design and validation report.

The CARs and CLs raised by CUCG were resolved during the communications between the Project Proponent and CUCG, which guaranteed the transparency of the validation assessment process. The concerns raised and the responses given are summarized in Appendix IV.

2.5.1 Forward Action Requests

During the validation assessment, one FAR (FAR-01) was raised according to the indicator 3.1.4 of VCS Standard version 4.7 as detailed below:

“The biogas plant is built on the Tian Siang POM’s land Lot PT 7952 and PT 7953, which is for the purpose of planting oil palms. Referring to the Tian Siang POM’s memo to Bera Land Department dated 05.12.2023, the mill has appointed BroUkur as a consultant to change the term and condition of the land. The memo is signed by the mill manager and acknowledged by Bera Land Department on 05.12.2023. The PP need to ensure the land is converted to the industrial land to ensure comply with the National Land Code, Act 828.”

This issue was raised as a FAR as the Tiang Siang POM is in process of converting the current land title of Agricultural to Industrial and also considering the government process takes considerable time to be completed.

3 VALIDATION FINDINGS

3.1 Project Details

Project type, technologies and measures implemented, and eligibility of the project

Grouped Project Description

Green Lagoon Technology Sdn Bhd’s “Wastewater to Biogas Grouped Project” in Malaysia is a series of biogas recovery projects. Each project activity instance (PAI) is managed by a different

plant operator. The goal is to capture biogas from wastewater treatment, reducing greenhouse gas emissions. The collected biogas will be used for grid energy, bottling and flaring (if there is excess).

Project Activity Instance 1 (PAI1) Description

The first Project Activity Instance (PAI1) is at Tian Siang Oil Mill in Pahang, Malaysia, about 180 km north of Kuala Lumpur. The biogas plant is owned by GLT Maju Sdn Bhd, a subsidiary of Green Lagoon Technology Sdn Bhd. Green Lagoon Technology has the legal rights to operate PAI1 through an agreement/4 with GLT Maju. Additionally, Tian Siang Oil Mill has granted GLT Maju access to palm oil mill effluent (POME) and permission to build and run PAI1.

Baseline Scenario

Prior to PAI1 commencement, Tian Siang POM treated its wastewater using a series of open anaerobic lagoons, the method was widely used by most of palm oil mills in Malaysia. The process involved eight (8) treatment ponds consist of four (4) anaerobic, two (2) aerobic, one (1) settling basin, and one (1) maturation pond. This setup generated significant methane, producing around 246,518 tCO₂e/year over seven (7) years due to the anaerobic conditions and regional temperature. The treated wastewater was then discharged into the river, and the sludge was disposed via land application near the project area.

Project Scenario

The project improves wastewater treatment by installing an anaerobic digester with a membrane to capture methane in a new lagoon. The remaining waste is treated aerobically before being released into the river. The captured biogas is used for power generation and fed into the grid, significantly reducing greenhouse gas emissions. The expected annual and total GHG emission reductions during the first seven years for PAI1 are detailed in the project documentation.

No	Project activity instance	Annual Average GHG Emission (tCO ₂ e)	Total GHG emission reductions for first crediting period (tCO ₂ e)
PAI1	Tian Siang POM	35,217	246,518

Item	Evidence gathering activities, evidence checked, and assessment conclusion			
Audit history	Audit type	Date of assessment (on-site)	Program	Validation/verification body name

	<table><tr><td>Validation</td><td>28th – 29th May 2024</td><td>VCS</td><td>Control Certifications GmbH</td><td>Union Germany</td></tr></table>	Validation	28 th – 29 th May 2024	VCS	Control Certifications GmbH	Union Germany			
Validation	28 th – 29 th May 2024	VCS	Control Certifications GmbH	Union Germany					
	The validation team did the cross checking with the previous audit documents and found no previous audit record been conducted to GLT. Hence, the validation team has ascertained that this is the initial validation assessment for GLT.								
Sectoral scope	The PAI1 under consideration is a grouped project activity. The PAI1 as part of the grouped project are defined as follow: <table><tr><td>Sectoral scope</td><td>13 - Waste handling and disposal</td></tr><tr><td>Sectoral scope</td><td>1 - Energy industries (renewable/non-renewable sources)</td></tr></table> Based on the document review of the PDD /1 and site visit conducted at the PAI1 project site, the validation team is assured that the GLT applied the sectoral scope 13 and 1 for its PAI1 project.	Sectoral scope	13 - Waste handling and disposal	Sectoral scope	1 - Energy industries (renewable/non-renewable sources)				
Sectoral scope	13 - Waste handling and disposal								
Sectoral scope	1 - Energy industries (renewable/non-renewable sources)								
AFOLU project category, if applicable	Not applicable								
Project activity type	<u>Grouped Project</u> <table><tr><td>Project activity type</td><td>III - Other project activities AMS-III.H (Version 19.0) - Methane recovery in wastewater treatment</td></tr><tr><td>Project activity type</td><td>I - Renewable energy projects AMS-I.D (Version 18.0) – Grid-connected renewable electricity generation</td></tr></table> <u>Project Activity Instance (PAI1)</u> <table><tr><td>Project activity type</td><td>III - Other project activities AMS-III.H (Version 19.0) - Methane recovery in wastewater treatment</td></tr><tr><td>Project activity type</td><td>I - Renewable energy projects</td></tr></table>	Project activity type	III - Other project activities AMS-III.H (Version 19.0) - Methane recovery in wastewater treatment	Project activity type	I - Renewable energy projects AMS-I.D (Version 18.0) – Grid-connected renewable electricity generation	Project activity type	III - Other project activities AMS-III.H (Version 19.0) - Methane recovery in wastewater treatment	Project activity type	I - Renewable energy projects
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Project activity type	I - Renewable energy projects								

	<table border="1"> <tr> <td data-bbox="584 184 860 294"></td><td data-bbox="860 184 1422 294">AMS-I.D (Version 18.0) – Grid-connected renewable electricity generation</td></tr> </table> Based on desktop and on-site validation assessments conducted, the validation team ascertained that the methodology of AMS-III.H (Version 19.0) and AMS-I.D (Version 18.0) described in the PDD/1 were applied to the PAI1.		AMS-I.D (Version 18.0) – Grid-connected renewable electricity generation
	AMS-I.D (Version 18.0) – Grid-connected renewable electricity generation		
General eligibility of the project to participate in the VCS Program	Grouped Project The new project activity instances are set to apply few methodologies, which are according to requirements of paragraph 3.6.16 and 3.6.17 of the VCS Standards Version 4.7. The project activity instances shall meet applicability conditions for applicable methodologies as defined in section 3.2 of the PDD. There are three (3) potential individual methodologies or combination methodologies for new PAIs, which are listed below: - o AMS-III.H. - o AMS-I.D. - o AMS-III.H. and AMS-I.D. The validation team conducted a cross-check assessment during the document review session and confirmed that all listed methodologies are approved GHG programs under the VCS. Project Activity Instances (PAI1) GLT applied methodologies of AMS-III.H (Version 19.0), and AMS-I.D (Version 18.0) from the CDM Program for Initial Project Instances PAI1. The validation team has conducted cross-checks during the document review and found that the applied methodologies are approved GHG programs under VCS. The validation team once again did confirm that the applied methodologies are in used for PAI1 during the site visit. The applied methodologies are conformed to the general eligibility under section 2.1.1 (3) of the VCS Standard (v4.7) – “Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions)”. As such, the validation team is assured that the general eligibility criteria applied for PAI1 is met.		

AFOLU project eligibility, if applicable	Not applicable as the project does not fall within the AFOLU sector.
Transfer project eligibility, if applicable	Not applicable since the grouped project activity does not involve transfers from other GHG Programs.
Project design	Grouped Project: • Geographical Area: The project activity instance is located in the geographic boundary for the grouped project, i.e., Malaysia (Peninsular Malaysia, Sabah, and Sarawak), and owned by the project owner or the subsidiary of the project owner. The geographical area is defined as per requirements under section 3.6.10 and 3.11.1 of the VCS Standard (version 4.7) and presented in KML file. • Baseline scenario: All Project Activity Instances shall have the following existing conditions at their respective PAI level, following the combination of methodology(ies) outlined above. Future PAIs will only be eligible for inclusion if their existing conditions at the PAI level are consistent with the baseline scenario and applicability conditions that defined in the project description section 3.4, in accordance with Sections 3.6.11 of the VCS Standard. ○ According to AMS-III.H version 19.0, the baseline scenario involves a baseline system with anaerobic wastewater treatment system without biogas recovery, which is the case the baseline scenario is determined to be the continuation of the existing open lagoon-based treatment of wastewater without methane recovery as commonly practiced because it only requires small investment and maintenance cost where the treatment system only consume less energy for pumping and mixing. ○ According to AMS-I.D version 18.0, the baseline scenario for greenfield power plant, electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid. • According to VCS Standard, paragraph 3.6.11, "Determination of baseline scenario and demonstration of

additionality are based upon the initial project activity instances". Hence, for grouped project level, no determination of baseline scenario shall be done.

- **Technology type:** The project activity instances to be included in the grouped project activity is the implementation of biogas recovery, flaring and/or utilization systems in palm oil mills using either an anaerobic digester or covered anaerobic lagoon, and gas engine for power generation.
- **Additionality:** The project activity instances to be added as part of the grouped project shall meet additionality criteria as set out in the methodology at respective PAI level, following the combination of methodology(ies) above.
 - For any combination involves wastewater related methodologies (AMS-III.H.), the respective PAIs will apply the methodological tool 32 Positive lists of technologies - Version 04.0, A positive list of technology and project activity types that are defined as automatically additional if the PAIs apply such technologies and meet specified conditions. The positive list of technology including the methane recovery in wastewater treatment where the specific conditions to be met is listed below:
 - The existing treatment system is an anaerobic lagoon and the wastewater discharged meets the host country regulation. The Project Proponent shall submit the engineering drawings of the open anaerobic lagoon, as well as documentation related to wastewater discharge permits and compliance
 - There is no regulation in the host country that requires the management of biogas from domestic, industrial and agricultural sites. In all three (3) regions (Peninsular Malaysia, Sabah, and Sarawak), the industry is regulated at the federal level by the Department of Environment (DOE), which oversees environmental compliance and standards. In addition, the licensing and operational requirements of palm oil mills are specifically regulated by the Malaysian Palm Oil Board (MPOB) under the Malaysian Palm Oil Board Licensing Regulations. The requirement to install

methane capture facilities under the Malaysian Palm Oil Board Licensing Regulations is determined based on the mill's start date and any subsequent capacity additions. To demonstrate that this project is not subject to a mandatory methane capture requirement, the Project Proponent shall submit documentation confirming the start date of the mill (e.g., trade license/operating permit).

- There is no capacity increase in the wastewater treatment system. The requirement to install methane capture facilities under the Malaysian Palm Oil Board Licensing Regulations is determined based on the mill's start date and any subsequent capacity additions. The Project Proponent shall submit documentation confirming there is no capacity addition to the original approved mill capacity
- No other alternative economic activity is expected to be undertaken on the land of the existing lagoon. The Project Proponent shall submit relevant documents to VVB.
- The biogas is used to generate electricity in one or more power plants, and the total nameplate capacity is below 5 MW. The Project Proponent shall submit the engine specification to the VVB.
- For project activity instances that solely apply AMS-I.D., the additionality criteria will apply the tool "Demonstration of additionality of small-scale project activities" Version 13.1.
- The additionality will be demonstrated for each designated geographic area (i.e., Peninsular Malaysia, Sabah, and Sarawak) for each of the PAIs in accordance with Section 3.6.14 of the VCS Standard. The start date of each project activity instance under the grouped project shall not be prior to the start date of the grouped project. The start date of each project activity instance will be determined through documentary evidence.

According to VCS Standard, paragraph 3.6.11, "Determination of baseline scenario and demonstration of additionality are based upon

the initial project activity instances”. Hence, for grouped project level, no determination of additionality shall be made.

Future project activity instances only can be included if they are located within the defined geographic area, are subject to the established baseline scenario, and meet the additionality demonstration provided at registration, in accordance with the VCS Standard v4.7 (Sections 3.6.10 and 3.6.11).

At this stage, initial PAI of combination of AMS-III.H. and AMS-I.D. have been included.

Project Activity Instances (PAI1):

The project design for PAI1 is assessed as follows:

No.	Eligibility Criteria	Explanation	Evidence Checked
1	Applicability Conditions: The project activity instances shall meet applicability conditions for applicable methodology (AMS-III.H version 19.0 and AMS-I.D version 18.0)	Based on the document review and physical site visit, PAI1 meets respective applicability conditions by applying methodologies of AMS III.H (Version 19.0) and AMS-I.D (Version 18.0).	Project Design in PDD/ 1 , GLT_PAI Project Activity Diagram, Project Emission Calculation Sheet (Tian Siang CER sheet) and eye witnessing during the site visit.
2	Geographical Area: The project activity instance locates in the geographic boundary for the grouped project, i.e., Malaysia, and owned by the	PAI1 is located at Bera, Pahang, Malaysia, which is within geographic boundaries of Malaysia.	Project Location in PDD/ 1 , KML files/ 11 and physical site visit in Bera, Pahang, Malaysia.

		project owner or the subsidiary of the project owner		
	3	Baseline scenario: All Project Activity Instances shall meet the baseline definition as defined in respective methodology	Based on document review and physical site visit, PAI1 meets the baseline of: 1. AMS-III.H – PAI1 baseline scenario involved a baseline system with anaerobic wastewater treatment system without biogas recovery, which is determined to be the continuation of the existing open lagoon-based treatment of wastewater without methane recovery as commonly practiced. 2. AMS-I.D – PAI1 baseline scenario includes the	Baseline Scenario in PDD/1, Baseline Scenario Diagram in the PDD/1, Project Emission Calculation Sheet (Tian Siang CER sheet) and physical witnessing during the site visit.

			electricity delivered to the grid that has been generated during the operation of grid-connected power plants and by the addition of new generation sources (biogas) into the grid.	
	4	Technology type: The project activity instances to be included in the grouped project activity is the implementation of biogas recovery, flaring and/or utilization systems in palm oil mill using either anaerobic digester or covered anaerobic lagoon, and gas engine for power generation.	PAI1 applied the implementation of biogas recovery, flaring and utilization systems in oil palm industry (sludge applied for land application) The recovered biogas is converted for purpose of supplying electricity generated to grid.	Technology type under eligibility criteria in PDD/1, and physical witnessing during the site visit.
	5	Additionality: The project	PAI1 is a small scale project and	1. Engineering diagrams for

		activity instances to be added as part of the grouped project will meet additionality criteria as set out in the methodology meets the additionality of the positive list of technology requirements for “wastewater handling and disposal”. PAI 1 meets the additionality condition as follows: 1. The existing treatment system is an anerobic lagoon and the wastewater discharged meets the regulation of Malaysia. 2. There is no regulation in Malaysia that requires the management of biogas from a site such as the project site. 3. No capacity increase in the wastewater treatment system based on the latest Tian Siang POM’s MPOB license (no	the anaerobic lagoon and documentation relevant to wastewater discharge. 2. Naskah Buku Kriteria Lesen MPOB Malay Version. 3. Lesen MPOB 2024. 4. Lease Agreement, BOOT Agreement, and Addendum of BOOT Agreement/4. 5. Technical Description of Biogas Engine Jenbacher JGS 416 GS-B.L/33 had been reviewed and physical witness the biogas engine during the site visit. The details: a. Brand: Jenbacher Innio
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			increment in the processing / production capacity). 4. No other alternative economic activity is expected to be undertaken on the land of the existing lagoon. 5. The biogas is used to generate electricity for the powerplant capacity is 1.234MW, below 5MW. Hence this condition is fulfilled.	b. Engine type: J416 GS-B25 c. Standard power: 1234KW
	6	Start Date: The start date of each project activity instance under the grouped project should not be prior to the start date of the grouped project. The start date of each project activity instance will be determined	Based on document review and record, the start date of the PAI1 is on 24-Apr-2024	1. DB2304-TST-C-TST-002 - Digester Commissioning letter/19. 2. GLT's POME pumping physical record at project site. 3. Record of SCADA system – POME pumping date at project site.

	<table><tr><td></td><td>through documentary evidence</td><td></td><td></td></tr></table>		through documentary evidence		
	through documentary evidence				
	As such, the validation team has ascertained that the PAI 1 meets the specific eligibility criteria under methodologies of AMS-III.H & AMS-I.D.				
Project ownership	<u>Grouped Project:</u> Based on document review, Green Lagoon Technology Sdn Bhd serves as the sole project owner for this initiative. Any additional PAI integrated into the Grouped Project must provide proof of ownership. <u>Project Activity Instances (PAI1):</u> GLT Maju Sdn Bhd (GLT) is a subsidiary of Green Lagoon Technology Sdn Bhd. Green Lagoon Technology owns the biogas plant. Additionally, Tian Siang POM, the mill operator, has signed few agreements/4 with GLT Maju Sdn Bhd to ensure continuous access to palm oil mill effluent (POME), which supports the construction and operation of PAI1. The validation team has reviewed the agreements/4 (Lease Agreement, BOOT Agreement and Addendum of BOOT Agreement) which had clearly mentioned the lease terms for 21 years and GLT shall have the right to claim for the environmental attributes from the project (PAI1) including the carbon credits, RECs and emissions reductions. As such, the verifier is assured that the project is clearly owned by GLT including the ownership of the carbon credits and emissions reductions generated from PAI1.				
Project start date	The validation team has confirmed that the commissioning of the digester started on 24th April 2024 by pumping in raw POME of PAI1. This was stated in a letter of GLT to Green Lagoon Technology and also validated with the physical record of POME pumping to the project site.				
Project crediting period	Based on the review of the PDD/1 and on-site interviews conducted with GLT’s personnel, it was assessed that the crediting period is set for seven (07) years and twice renewable. The project is started on 24.04.2024. Hence, the validation team is assured that the project crediting period was set for 7 years from 24.04.2024 to 23.04.2031.				

Project scale	Based on the document review on PAI1’s baseline and project emissions calculation sheet the validation team was assured that the GHG emission reduction or removals for PAI1 is estimated at 246,518 tCO2e/year (project).	
Likelihood of achieving estimated GHG emission reduction or removals	Calendar year of crediting period	Estimated GHG emission reductions or removals(tCO ₂ e)
	24.04.2024 – 31.12.2024	23,335
	01.01.2025 – 31.12.2025	35,372
	01.06.2026 – 31.12.2026	35,372
	01.06.2027 – 31.12.2027	35,372
	01.06.2028 – 31.12.2028	35,372
	01.06.2029 – 31.12.2029	35,372
	01.06.2030 – 31.12.2030	35,372
	01.06.2031 – 31.12.2031	10,951
	Total estimated ERRs during the first or fixed crediting period	246,518
	Total number of years	7
	Average annual ERRs	35,217
	Based on the document review on PAI1’s PDD/ 1 , methodologies and GHG’s baseline and project emission calculation sheet, the validation team is assured that the information of above GHG reduction or removals of PAI is valid.	

Technologies and measures implemented by the project activity or activities

During the site visit to PAI1, the validation team has confirmed that biogas digester is installed in a new lagoon, and a membrane is installed over the lagoon to capture the methane emitted.

By capturing methane emissions and converting it into biogas, the PAI helps reduce greenhouse gas (GHG) emissions. The biogas is then processed and used for electricity generation through combustion.

The validation team is also assured that the following technologies are incorporated for PAI1:

1. **Anaerobic Digester:** The In-Ground Bio Reactor (IGBR) System, with dimensions of approximately 104m × 44m × 8m depth, offers an effective volume of around 30,000m³. The reactor features an in-ground pond covered by a High-Density Polyethylene Geomembrane (HDPE) membrane, which also serves as a biogas storage facility with a capacity of up to 21,000m³. It operates with a hydraulic retention time of 35 days and is equipped with hydraulic sludge mixers and a sludge removal system. Based on the technology supplier, the digester is highly resistant to chemical, UV, and mechanical degradation, with a technical lifespan of up to 50 years.
2. **Biogas Treatment System:** This system includes a scrubber, dehumidifier, and chiller. The biological scrubbers, with a capacity of 650m³/hr, oxidize hydrogen sulphide (H₂S) to sulphate, reducing H₂S content to less than 200 ppm. The biogas is then passed through a dehumidifier and chiller system, lowering its water content from 75-90% to 40% and cooling it to make it suitable for the biogas engine.
3. **Biogas Engine:** A Jenbacher biogas engine (model JGS 416 GS-B.L) is installed, delivering a gross electrical output of 1,202 kW. Based on the supplier, the engine has a technical lifespan of over 20 years, with major maintenance typically required every 60,000 to 80,000 operating hours. An open flare system is also installed but is not expected to be used due to the biogas membrane's ample storage capacity.
4. **Control and Monitoring System:** All necessary operational and management parameters of the biogas plant are integrated into

Moreover, an illustration of the process flow is available as figure below:



Implementation
schedule of the project
activity or activities

The assessment of the implementation status of the project activity had been physically conducted at PAI1 by the validation team on 28th & 29th April 2024. The assessment flow is as follows:

Date	Time	Activity	Assessor
28/4/2024	1000 - 1100	Site visit – Biogas Plant	All
	1100 - 1300	Documentation Review, Interviews, - 1. Project Details	SS
	1100 - 1300	Stakeholders Interview (internal) Documentation Review, Interviews,	JS

		- 1.15. Compliance with laws, statutes, and other regulatory frameworks	
	1100 - 1300	Documentation Review, Interviews, - 3. Application of Methodology	SS & TZ
	1400 - 1530	Stakeholders Interview (external)	SN & JS
	1400 - 1800	Continue documentation review, and interviews	All
	29/04/2024 0900 - 1300	Documentation review & interviews - 4. Quantification of estimated GHG emission reductions and removals	SS & TZ
	0900 - 1300	Documentation review & interviews - 2. Safeguards, Stakeholder Engagement	SN & JS
	1400 - 1500	Documentation review & interviews - 5. Monitoring	SS & TZ
Project location	<u>PAI1:</u> During the site visit, the validation team has confirmed that PAI1 is located at Tian Siang POM in the State of Pahang, Malaysia, about 6 KM Off Jalan Triang Ke Kemayang, Triang, Pahang and 160km north of Kuala Lumpur. The geographical coordinates for the site are N 3° 10'58.6" and E 102° 27'27.3".		
Conditions prior to project initiation	Based on the physical site visit to PAI1, the validation team has confirmed that the conditions prior to project initiation as follows: wastewater was stored in uncovered open lagoons, leading to		

	methane emissions directly into the atmosphere without any methane recovery or destruction facility. These conditions align with the baseline scenario detailed in Section 3.4.
Project compliance with applicable laws, statutes and other regulatory frameworks	<u>Grouped Project:</u> All industry in Malaysia is regulated at the federal level by the Department of Environment (DOE), which oversees environmental compliance and standards across the 3 regions (Peninsular, Sabah & Sarawak). In addition, the licensing and operational requirements of the Palm Oil Industry including the palm oil mills are specifically regulated by the Malaysian Palm Oil Board (MPOB) under the Malaysian Palm Oil Board Licensing Regulations. While general frameworks such as the National Guidance on Voluntary Carbon Market Mechanisms (applicable nationwide), the Sabah Climate Change and Carbon Governance Enactment 2025 (sub-national) and the Environment (Reduction of Greenhouse Gases Emission) Ordinance 2023 exist, however these instruments do not directly regulate and apply to the project activity, <u>Project Activity Instances (PAI1)</u> The validation team has conducted a thorough examination on the relevant applicable legal obligations related to the implementation of PAI1. The following are among applicable laws and regulations identified by PAI1: • Environmental Quality Regulations (Prescribed Premises) (Crude Palm Oil) 1977 – P.U.(A) 342/77 Regulation 6 - Making Changes that change the quality of the effluent • Written Declaration on the Design and Construction of Air Pollution Control Systems [Regulation 7(5)] DOE ASPUB • Construction Site Registration - DOSH JKJ 103 • Environmental Quality (Clean Air) Regulations 2014 • Environmental Quality (Scheduled Wastes) Regulations 2005 • Act A1648 OSH Act 2022 • Minimum Wage Order 2022 • Employment Act 1955 • Employees Provident Fund Act 1991 (Act 452)

- ACT 177, Industrial Relations Act 1967
- Act 840 Anti-Sexual Harassment Act 2022
- Act 800 Employment Insurance System Act 2017
- Act 4 Employees' Social Security Act 1969 (Amended 2003)

While the following are among licenses, approvals, guidelines of PAI1 reviewed and validated during the assessment date:

- Company registration with Companies Commission of Malaysia (SSM)/**23**
- Licensing from Energy Commission (ST)/**3**
- Registration of construction site with CIDB/**26**
- Feed-In Approval (FiAH)/**3**
- Feed-in Tariff (FiT)/**43**
- Approved plan for POME discharge under Section 18(1), Act 127, Regulation 14, P.U (A) 342/1977 (JP/KKS/2018/2019/003238) dated 12.06.2018 signed by DOE, Pahang/**46**
- Grid connection feasibility study and power purchase agreement from TNB/**3**

At the validation assessment date, there is no regulation mandating the management of biogas from PAI1. Despite that, Malaysian Palm Oil Board (MPOB) requires new palm oil mills constructed after 01.01.2014 to install the wastewater treatment systems with methane capture prior to licensing application. This requirement also applies to existing mills that applied for capacity extensions after the same date, except for the mills applying for extensions below 270,000 MT FFB per year, with exemptions granted starting May 22, 2021, though this threshold has undergone revisions over time. As Tian Siang POM is established in 2000 and there is no capacity expansions since the date, hence, the new criteria are not applicable. In addition, there is no obligation to recover methane or utilize biogas at this facility.

Based on the site visit, documentation review and interview with the relevant stakeholders such as government agencies found, PAI1 is comply to the applicable law and regulations in Malaysia.

Double counting and participation under other GHG programs	Not applicable						
No double claiming with emissions trading programs or binding emission limits	Not applicable						
No double claiming with other forms of environmental credit	Tian Siang Oil Mill (Triang) Sdn. Bhd., where PAI1 is located, is ISCC EU certified under scope 'Point of Origin' (PO) for material Palm Oil Mill Effluent (POME) Oil. The certificate number is EU-ISCC-Cert-MY231-2023154 valid from 15.11.2023 to 14.11.2024. The Tiang Siang POM certified with ISCC. However, the scope of the ISCC EU certification is applicable for POME Oil only, thus there is no risk of double claiming with other forms of environmental credits for PAI1.						
Supply chain (Scope 3) emissions double claiming	Not applicable						
Sustainable development contributions	The project captures methane emissions to produce biogas for electricity generation, thereby achieving substantial reductions in GHG emissions that would otherwise be vented directly into the atmosphere. In addition to its environmental benefits, the project creates long-term employment opportunities for local residents and contributes to economic prosperity. The PP has explained the contribution of the project towards the sustainable development in Section 1.18 of the PDD/1: <table border="1"> <thead> <tr> <th>SDG</th><th>Project Activity Contribution</th></tr> </thead> <tbody> <tr> <td>SDG 7 "Ensure access to affordable, reliable, sustainable and modern energy for all"</td><td>The biogas generated from the project activity will be used for electricity generation for the project activity and upload to the grid. The biogas plant will capture the methane to convert into renewable energy and turn in producing valuable climate and clean air energy solution</td></tr> <tr> <td>SDG 8 "Promote sustained,</td><td>During the construction and operation of the project will provide new job opportunities</td></tr> </tbody> </table>	SDG	Project Activity Contribution	SDG 7 "Ensure access to affordable, reliable, sustainable and modern energy for all"	The biogas generated from the project activity will be used for electricity generation for the project activity and upload to the grid. The biogas plant will capture the methane to convert into renewable energy and turn in producing valuable climate and clean air energy solution	SDG 8 "Promote sustained,	During the construction and operation of the project will provide new job opportunities
SDG	Project Activity Contribution						
SDG 7 "Ensure access to affordable, reliable, sustainable and modern energy for all"	The biogas generated from the project activity will be used for electricity generation for the project activity and upload to the grid. The biogas plant will capture the methane to convert into renewable energy and turn in producing valuable climate and clean air energy solution						
SDG 8 "Promote sustained,	During the construction and operation of the project will provide new job opportunities						

inclusive and sustainable economic growth, full and productive employment and decent work for all”	which will increase the income of local residences and accelerate economy development in the areas.
SDG 13 “Take urgent action to combat climate change and its impacts	The project will achieve a GHG emission reduction during the crediting period. The project will recover and destroy methane gas that would otherwise be released directly into the atmosphere, The expected total emission reductions are 246,518 tCO ₂ e during the first 7-year crediting period, with an average annual emission reduction 35,217 tCO ₂ e

During the monitoring period, the Project will contribute to sustainable development in the following ways.

SDG 7 – Affordable and Clean Energy

SDG Target: 7.2

SDG Indicator: Amount of electricity generation

Net Impact on SDG Indicator	Current Project Activity Contribution	Contributions over project lifetime
Implemented activities to increase	An average of 7,884 MWh per year of power is uploaded to the grid.	A total of 55,188 MWh of power will be uploaded to the grid during the first crediting period

SDG 8 – Decent Work and Economic Growth

SDG Target: 8.5

SDG Indicator: Number of job opportunities provided

	Net Impact on SDG Indicator	Current Project Activity Contribution	Contributions over project lifetime
	Implemented activities to increase	A total number of 8 job opportunity will be provided to the local community	At least total number of 8 job opportunity will be provided to the local community

SDG 13 – Climate Action

SDG Target: 13.0

SDG Indicator: Tons of greenhouse gas emissions avoided

Net Impact on SDG Indicator	Current Project Activity Contribution	Contributions over project lifetime
Implemented activities to increase	An annual average emission reduction of 35,217 tCO ₂ e was achieved from the project activity	A total of 246,518 tCO ₂ e of emission reduction will be achieved during the first crediting period

As such, based on the site visit, documentation review and interview conducted with relevant stakeholders, the validation team found the element of SDG identified by GLT is realistic and credible.

Additional information relevant to the project

The validation team has confirmed that there is no additional information for PA11 based on the document review, interviews and the site visit. The validation team decision is as follows:

- No leakage emissions for the project
 - No leakage emissions are considered for the project under AMS-III.H, Ver19 since the technology does not utilize equipment transferred from other activities.
 - Since the PA11 do not show any competing use of biomass, leakage emissions are not required to be taken into account under AMS-I.D, Ver18.
- No commercially sensitive information

	This is not applicable to PAI1, as there is no commercially sensitive information that needs to be withheld from the public project description. 3. <u>No Further Information</u> Not applicable. There is no further relevant legislative, technical, economic, sectoral, social, environmental, geographic, site-specific, or temporal information that would impact the project's eligibility, GHG emission reductions, carbon dioxide removals, or the quantification of its reductions or removals.
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3.2 Safeguards and Stakeholder Engagement

3.2.1 Stakeholder Engagement and Consultation

3.2.1.1 Stakeholder Identification

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder identification	GLT has established a procedure of 'Stakeholder Consultation Procedure' that outlined the process of stakeholder consultation as below. Noted that, this procedure will be used as a guideline for future PAIs as well. • Stakeholders' identification. The identification can be through Google Maps, EIA and SIA reports. The stakeholders may come from government agencies, settlements or other institutions. Moreover, the identification is extended up to 10km radius if necessary. • Planning. The consultation process will be conducted either through physical or online consultation with a specific timeline and relevant resources have been determined. • Realization of consultation. Announcement between 7 to 30 days will be made through various channels; website, newsletter and social media. Also, the invitation is carried out by using a phone call, email fax or postal service. • Conduct the consultation. During the consultation process, clear and accessible material will be disseminated to the stakeholders. Any feedback will be recorded accordingly. GLT has identified their stakeholders within 10-km radius from the project site (Tian Siang Oil Mill (Triang) Sdn. Bhd.) assuming that this is the area likely to be impacted by the project. Sighted the stakeholders identified include local residents, schools, government

	agencies, housing estates and neighboring plantation estates. Among stakeholders identified are as below: • Department Of Environment (DOE) Pahang • Labour Department (Kuantan), • Department of Occupational Safety and Health (DOSH) • Pejabat Daerah Dan Tanah Bera • Communities residing within MRSM Bera • Communities residing within SK LKTP Triang II • Communities residing within SJK (C) Mengkuang • Nearby local residents – Felda Triang II • Felda Triang 01 and Felda Mengkuang millworkers and community residing within the mills • Tian Siang Oil Mill (Triang) Sdn Bhd workers and community residing within the mill • Bera town and housing estates residents The stakeholders identified above are considered adequate by the validation team to fulfil the criteria to represent communities affected by the project activity within 10-km radius of the project site.
Legal or customary tenure/access rights	The land for the project activity is owned by Tian Siang Oil Mill (Triang) Sdn Bhd. The surrounding area includes plantation land owned by Felda plantation and its settlement. A small town, Bera, is located at the northeastern part of the 10-km perimeter border of the area. The stakeholders within the area are from different ethnicities, reflecting the diverse cultural mix typical of Malaysia. Despite the varied cultural, economic, and social backgrounds, there is no known conflict among the community members. Therefore, interactions between the stakeholder groups are harmonious. The nearest indigenous settlements are located near Tasek Bera, which is 40 km away from the site. Thus, there is no legal or customary tenure/ access rights involved or expected.

Stakeholder diversity and changes over time	Review the Pahang’s Profile Report, a survey report of each district in Pahang, for the year 2018 found, there are 11 districts registered in Pahang state which are Kuantan, Temerloh, Bentong, Bera, Cameron Highlands, Jerantut, Lipis, Maran, Raub, Rompin and Pekan. Total area for all districts is 3,596,585 hectares.																					
	Triang, where the PAI1 is located, is under Bera district. Research on external references was performed by the validation team to analyze the population development in Bera. It was found that, as of 2023, estimated population is 100,100 with male to female ratio at 46.6% and 53.4% respectively. Below is the estimated population by ethnic in Bera for 2020:																					
	<table><tr><th>Ethnic Group</th><th>Total</th><th>Percentage (%)</th></tr><tr><td>Bumiputera</td><td>63,400</td><td>63</td></tr><tr><td>Chinese</td><td>25,400</td><td>25</td></tr><tr><td>Indian</td><td>3,900</td><td>4</td></tr><tr><td>Others</td><td>500</td><td>1</td></tr><tr><td>Foreigner</td><td>6900</td><td>7</td></tr><tr><td>Total</td><td>100,100</td><td>100</td></tr></table>	Ethnic Group	Total	Percentage (%)	Bumiputera	63,400	63	Chinese	25,400	25	Indian	3,900	4	Others	500	1	Foreigner	6900	7	Total	100,100	100
	Ethnic Group	Total	Percentage (%)																			
	Bumiputera	63,400	63																			
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	Others	500	1																			
	Foreigner	6900	7																			
	Total	100,100	100																			
While the following is the estimated population by ethnic in 2005:																						
<table><tr><th>Ethnic Group</th><th>Total</th><th>Percentage (%)</th></tr><tr><td>Bumiputera</td><td>48,184</td><td>54.2</td></tr><tr><td>Chinese</td><td>24,981</td><td>28.1</td></tr><tr><td>Indian</td><td>4978</td><td>5.6</td></tr><tr><td>Others</td><td>10,757</td><td>12.1</td></tr><tr><td>Total</td><td>88,900</td><td>100</td></tr></table>	Ethnic Group	Total	Percentage (%)	Bumiputera	48,184	54.2	Chinese	24,981	28.1	Indian	4978	5.6	Others	10,757	12.1	Total	88,900	100				
Ethnic Group	Total	Percentage (%)																				
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Total	88,900	100																				
Based on the above, the population has increased by 13% over the 15 years, 2005 to 2020. There has been a noticeable decrease in the 'others' group and additional group for ‘foreigner’. Often, these group are brought into Malaysia to work in plantation or construction sector.																						
Expected changes in well-being	There have been no significant changes in the well-being and characteristics of stakeholders in the Bera district, where the project																					

		is situated. This district, located in the central part of Peninsular Malaysia, is primarily agricultural. Due to its rural nature, it is anticipated that it will experience less development compared to more centrally located cities in Peninsular Malaysia over the next decade. While some minimal development may occur, the project area is expected to remain less developed than urban areas. Consequently, the project will positively impact stakeholders by introducing investment and creating increased employment opportunities within the local community.
Location of stakeholders	of	GLT has identified their stakeholders within 10-km radius from the project site. During the site visit, observed that the PAI is surrounded by nearby oil palm plantations, housing area, and other education facilities.
Location of resources		GLT has identified their stakeholders within 10-km radius from the project site. During the site visit, observed that the PAI is surrounded by nearby oil palm plantations, housing area, and other education facilities. The validation team was informed by GLT that Google Maps is one of the applications used to identify their stakeholders within the said radius. Therefore, the same application was used by the validation team to verify the stakeholders identified by GLT. Moreover, the validation team used another application, Calcmaps, to ensure relevant stakeholders were appropriately identified within 10-km radius. The validation team also has reviewed the Pahang's Profile Report for the year 2018 which consists of information on villages under Triang, location of GLT. Further confirmation on the existence of nearby stakeholders was obtained during site interviews with internal and external stakeholders. Based on the above approaches, the validation team has confirmed that stakeholders identified by GLT are reliable.

3.2.1.2 Stakeholder Consultation and Ongoing Communication

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder engagement process	According to the PDD/1, GLT has organised the stakeholders meeting on 31.10.2023 at Pahang Rest House @ Bera. Further informed by GLT that a series of online publication was made through GLT's website, social media such as Facebook and LinkedIn on 21.10.2023.

	Additionally, claimed by GLT that an invitation letter through email was submitted to relevant stakeholders to join the meeting. Based on the above information, the validation team has reviewed the 'Stakeholder Consultation Report'/14 developed by GLT in October 2023. Public announcement through GLT's website and social media such as Facebook and LinkedIn as well as photo of the consultation are available in the report. Also, the attendance form with hand signature by stakeholders is also available as evidence of the attendance. Representative from Bursa Malaysia Berhad, Felda Mengkuang, Tian Siang POM, SK LKTP TRG 2 and Felda Triang 01 are among stakeholders attended the stakeholders meeting. The stakeholder's consultation encompassed a project presentation followed by question and answer (Q&A) session. Inquiries raised during the consultation were appropriately addressed by GLT as sighted in the 'Summary of Comment Received' under section 3 of the meeting report. As for the online publication, the validation team has verified the GLT's website and social media as mentioned in the PDD/1 and stakeholders meeting report and confirmed the public announcement on stakeholders meeting was performed and information on stakeholders meeting is available on the GLT's website and social media such as Facebook and LinkedIn. Additionally, a sample of invitation letter through email to government agencies such as Labour Department and Safety and Health Department were verified during the validation assessment. Site and phone call interview were also conducted during the validation assessment. Confirmed by the interviewed stakeholders that, they have attended the meeting organised by GLT. Referring to the above, the validation team has ascertained that the stakeholders meeting was conducted on 31.10.2023 at Pahang Rest House @ Bera. Prior to the stakeholders meeting, GLT has made a public announcement on their website including social media account and sending the invitation letter through email.
Consultation outcome	The validation team has reviewed the 'Stakeholder Consultation Report'/14 developed by GLT in October 2023. The following information obtained from the consultation report, among others: • Date: 31.10.2023

	• Time: 10.00 am – 12.00pm • Location: Pahang Rest House @ Bera, Bera Pahang • Participant: 23 participants - Green Lagoon Technology Sdn Bhd - Eco Ideal Consulting Sdn Bhd - Tian Siang POM - Felda Triang 01 - Felda Mengkuang - SK LKTP TRG 2 - Bera Land Department - Bursa Malaysia Berhad • Agenda - Overview of biogas plant (project design, implementation, regulations and benefits etc.) - Introduction to climate change and GHGs - Introduction to VCS project activities, validation & verification process - Questions and Answers session Observed that, there are four (04) inquiries were raised during the consultation and appropriately addressed by GLT as sighted in the 'Summary of Comment Received' under section 3 of the meeting report. Confirmed during site interview with both internal external stakeholders, the development of the project resulting in positive impact. Besides creating a job opportunity to nearby residents, the project also creates a positive impact to the environmental especially within the project area. Hence, the validation team has ascertained that the stakeholders have not expressed by adverse remarks towards the implementation of the project.
Ongoing communication	During the document review it was found that GLT had adopted the GLT Sdn Bhd's Customer Complaint Handling Procedure/15 (Reference Number: GLTSB-CM-04) dated 01.01.2021 as a communication channel to their stakeholders. According to the procedure, any feedback, whether verbal or written, can be submitted to any GLT personnel and will be forwarded to the Management Representatives (MR) for further action. During the site interviews with GLT's personnel it was found that they are aware of the procedure and able to demonstrate their knowledge of the complaint procedure.

	Besides that, all relevant stakeholders may submit their feedback through GLT's website, Contact Us - Green Lagoon (glt.my). A phone number (+60327332960) is also available on the company's website. The validation team was informed by GLT during the validation assessment that the stakeholders may submit their feedback through phone call. Confirmed during site interview with the stakeholders, complaint channel is explained during the stakeholder's consultation. Additionally, sighted during site visit, a complaint form and complaint box were available for stakeholders submit their feedback.
Stakeholder input	The review and PDD/1 and stakeholders' consultation report showed that there are four (04) inquiries raised by the stakeholders during the meeting conducted on 31.10.2023 at Pahang Rest House @ Bera. It was observed that the inquiries were appropriately addressed by GLT. The following comments were received and feedback by GLT during the stakeholder's consultation. <u>Comment 1</u> Q. Will biogas generated be connected to Tenaga National Berhad (TNB) grid or micro-grid? A. It will be connected to the grid after the voltage goes through step-down. This is a normal process to reduce voltage before connecting to a grid. <u>Comment 2</u> Q. What is the type of gas released during flaring? A. Burning of methane by flaring in high temperature >500 degree centigrade will release carbon dioxide <u>Comment 3</u> Q. Where is the location of the project and when will it be expected to start operation? Will the project be open to the public/ school children to visit? A. The biogas plant is in Triang and is expected to start operation in April 2024. As for the possibility of visiting the plant, they (the stakeholders or any interested parties) are more than welcome to visit.

	However, prior approval is required and therefore they need to contact the plant manager. There is no safety issue that the public need to worry because methane capture projects are common these days, and this is not the first project in Malaysia. No untoward incidence has been recorded so far in Malaysia. <u>Comment 4</u> Q. Is the project status under Verra Registry? A. Not yet. The PDD/1 is still currently under development, pending the information related to stakeholder consultation before being uploaded to VERRA website. The validation team has performed the on-site and phone call interviews during the validation assessment with the sampled stakeholders. It was evident that the stakeholders were sufficiently briefed on the project implementation and the associated environmental and social advantages that would be accomplished.
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3.2.1.3 Free, Prior, and Informed Consent

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Obtaining consent	Based on the survey map, the nearest IPs and LCs settlements are located at Tasik Bera which is 40-km away from the project site. As there is no impact of the PAI1 to any IPs, LCs, and customary rights holders, thus there is no FPIC process take place.
Outcome of FPIC discussion	PAI1 is constructed on the Tian Siang POM's land who has the legal ownership of the land. In addition, the nearest IPs and LCs settlements are located at Tasik Bera which is 40-km away from the project site. As there is no impact of the PAI1 to any IPs, LCs, and customary rights holders, thus there is no FPIC process take place.

3.2.1.4 Grievance Redress Procedure

Item	Evidence gathering activities, evidence checked, and assessment conclusion
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Development process	During the document review it was found that GLT had adopted the GLT Sdn Bhd's Customer Complaint Handling Procedure/15 (Reference Number: GLTSB-CM-04) dated 01.01.2021 as a communication channel to their stakeholders. According to the procedure, any feedback, whether verbal or written, can be submitted to any GLT personnel and will be forwarded to the Management Representatives (MR) for further action. During the site interviews with GLT's personnel it was found that they are aware of the procedure and able to demonstrate their knowledge on the complaint procedure. Besides that, all relevant stakeholders may submit their feedback through GLT's website, <u>Contact Us - Green Lagoon (glt.my)</u>. A phone number (+60327332960) is also available on the company's website. The validation team was informed by GLT during the validation assessment that the stakeholders may submit their feedback through phone call. It was confirmed during the site interviews with the stakeholders that the complaint channel was explained during the stakeholder's consultation. Additionally, as sighted during the site visit, a complaint form and complaint box were available for stakeholders to submit their feedback. Based on the above, the development process of Grievance Redress Procedure is considered sufficiently explained.
Grievance redress procedure	During the document review it was found that GLT had adopted the GLT Sdn Bhd's Customer Complaint Handling Procedure/15 (Reference Number: GLTSB-CM-04) dated 01.01.2021 as a communication channel to their stakeholders. According to the procedure, any feedback, whether verbal or written, can be submitted to any GLT personnel and will be forwarded to the Management Representatives (MR) for further action. The process flow of the grievance procedure is explained under section 7.0 of the Customer Complaint Handling Procedure/15. The following is the summary of the complaint procedure: 1. Receives and Receipt of comment/ feedback 2. Record Complaints and Communicate 3. Review Feedback and Request Information 4. Initiative Corrective

	5. Follow- up & Verify actions & effectiveness 6. Update QMS & Risks/ Opportunities 7. Feedback to respondent 8. Compile Data 9. Annual Review & Suggests Improvements In addition, during site interviews with the internal and external stakeholders it was found that they can demonstrate their knowledge of how to lodge a complaint or grievances. Additionally, as sighted during site visit, a complaint form and complaint box were available for stakeholders to submit their feedback. Based on the above assessment, the validation team concluded that the procedure established and implemented is in line with standard requirements.
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3.2.1.5 Public Comments

GLT received one comment on 15.02.2024 during the public comment period, which lasted from 22.01.2024 to 21.02.2024, as shown in the table below:

Comments received	Actions taken by the project proponent	Evidence gathering activities, evidence checked, and assessment conclusion
Public comment requested more details about the baseline scenario description	The details description for the baseline scenario is included under section 1.1 of the VCS PDD/1	The validation team reviewed the PDD/1 and proof of GLT reply to the public comment/39, and found that the baseline scenario was explained and defined under section 1.1 Summary Description of the Project. Referring to the above, the validation team has ascertained that GLT has appropriately addressed the public comment received.

3.2.2 Risks to Local Stakeholders and the Environment

3.2.2.1 Management Experience

Green Lagoon Technology Sdn Bhd is a Malaysian business entity which was established in May 2010. The Company had initially participated in Clean Development Mechanism ("CDM") projects in the palm oil mill industry by providing design, construction, operations, and management services. The company experienced in converting biogas from palm oil milling into various uses, such as domestic consumption and electricity sales through Malaysia's Feed-in-Tariff (FiT) program with involvement in more than 60 projects, including investment and ownership in these power assets, the company has become one of Malaysia's top biogas developers. On top of that GLT has engaged Eco-Ideal Consulting Sdn Bhd as a consultant for the project which the company consists of over 25 highly qualified professionals including environmental managers, engineers, executives and supporting staff with expertise in the renewable energy including biogas plants. Based on above justification, the GLT have expertise in, and prior experience implementing similar carbon project and demonstrated that they partnered with experienced local organization, Eco-Ideal Consulting Sdn Bhd.

3.2.2.2 Risk Assessment

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Natural and human-induced risks to stakeholders' wellbeing	GLT adopted GLT Sdn Bhd's 'Environmental Policy' / 10 dated 10.06.2020 signed by CEO which include a commitment on minimizing the environmental aspect that resulted from their activity as example below: 1. Integrate the consideration of environmental concerns and impacts into our decision making and activities 2. Train, educate and inform the employees about environmental issues that may affect their work 3. Promote environmental awareness among their employees and encourage them to work in an environmentally responsible manner Based on the PAI1 activity, PAI1 is not expected to cause any natural and human-induced risks to the stakeholder's wellbeing such as droughts, desertification, floods, landslides, wildfires, earthquakes and tsunamis.
Risks to stakeholder participation	Based on the invitation letter, information posted on the GLTSB's website found and Stakeholder Consultation

	Report/14 found, the consultation is conducted on 31.10.2023 at Pahang Rest House @ Bera. During the consultation, project activity is informed to the stakeholders including safety and environmental aspect.
Working conditions	GLT adopted GLTSB's policy/9 on safety working conditions/7. In addition, the Emergency Plan and First Aid Management are available in the PAI1 area as sighted during the site visit.
Safety of women and girls	GLT has identified the <i>Anti-Sexual Harassment Act 2022</i> as one of the applicable laws to their operation. In addition, a 'Sexual Harassment' poster/38 is available in the PAI1 area. During the validation assessment it was sighted that there are no women employed by GLT. Further confirmation obtained by review the list of employees/6 of GLT.
Safety of minority and marginalized groups, including children	The project necessitates skilled personnel to operate, maintain, and manage PAI1. Consequently, it neither employs nor is complicit in any form of child labor. Based on the GLT list of employees/6, the project does not employ child labor. Additionally, safety procedures for visitors/40, including children, are in place, ensuring that safety issues are not a concern for anyone visiting the biogas plant.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials, and chemical pesticides and fertilizers)	During both the construction and operational phases, the disposal of scheduled waste will adhere to the Environmental Quality (Scheduled Wastes) Regulations 2005. In addition, it is anticipated that PAI's activities will not emit any harmful pollutants that could harm the environment such as air, noise, chemical, fertilizer and release of hazardous materials.

3.2.3 Respect for Human Rights and Equity

3.2.3.1 Labor and Work

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Discrimination	GLT adopted GLTSB's 'Employee Policy'/9 which include a commitment on no discrimination. Section 8.0 'Equal Employment Opportunity' of the 'Employee Policy' has specified the following:

	4. To provide equal employment and advancement opportunities to all individuals. 5. The company does not discriminate in employment opportunities or practices based on race, colour, religion, sex, sexual orientation, national origin, age, or any characteristic protected by law. The validation team was informed by the workers during the site interviews that there is no discrimination practiced by PAI1. Referring to the above, element of discrimination is absent within the PAI1 activity.
Sexual harassment	GLT has identified Anti-Sexual Harassment Act 2022 as one of the applicable laws to their operation. In addition, a 'Sexual Harassment' poster is available in the PAI1 area. Based on the list of employees/6 reviewed and observed during the validation assessment, there is no woman working in the PAI1. The validation team was informed by the workers during the site interviews that there is no discrimination practiced by PAI1.
Equal pay for equal work	GLT adopted GLTSB's 'Employee Policy'/9 which specify that equal employment and advancement opportunities is provided to all employees based on merit, qualifications and abilities. In addition, GLT adheres to the Minimum Wage Order 2022 for the worker's salary payment. The review of the list of employees/6 showed that there are 5 employees working as an operator and were paid equally.
Gender equity in labor and work	GLT adopted GLTSB's 'Employee Policy'/9 which include a commitment on no discrimination. Equal job opportunities and pay were provided to both genders. At GLT Headquarters, the staff ratio is 60% male to 40% female. However, at the Group level, the gender ratio is 95% male to 5% female, with men predominantly operating the biogas plants. Men are primarily involved in upstream and technical activities, such as loading feedstock into the digester, cleaning clogged channels, and maintaining biogas pipe distribution. Additionally, since most operational plants are in rural areas where safety is a significant concern, all trained personnel at these plants are men. Based on the list of employees/6 reviewed and observed during the validation assessment, there is no woman working in the PAI1.
Forced labor	GLT adheres to the Employment Act 1955 to safeguard workers' legitimate rights and interests and to prohibit forced labor. Site interview with sampled workers found, there is no element of forced labor practiced in PAI1.

Child labor	The project necessitates skilled personnel to operate, maintain, and manage PAI1. Consequently, it neither employs nor is complicit in any form of child labor. Based on the GLT list of employees/6, the project does not employ child labor.
Human trafficking	GLT has identified and adheres to the Employment Act 1955. Based on the onsite interview with the sampled workers and observed during the validation assessment, there is no human trafficking within the PAI1 area.

3.2.3.2 Human Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
Low risk identified	PAI1 is constructed on the Tian Siang POM who have legal ownership of the land. In addition, the nearest IPs and LCs settlements are located at Tasik Bera which is 40-km away from the project site. Looking to the location of the Ips and LCs with PAI1, there is no impact of the PAI1 to any IPs, LCs, and customary rights holders. Furthermore, GLT has identified relevant laws and regulations to safeguards their workers right such as <i>Employment Act 1955</i>, <i>Minimum Wages Order 2022</i> and <i>Anti-Sexual Harassment Act 2022</i>. Moreover, GLT adopted GLTSB's Employee Policy/9 that prohibit any form of discrimination.

3.2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risk identified	PAI1 is constructed on the Tian Siang POM who have legal ownership of the land. In addition, the nearest IPs and LCs settlements are located at Tasik Bera which is 40-km away from the project site. Looking to the location of the Ips and LCs with PAI1, there is no impact of the PAI1 to any IPs, LCs, and customary rights holders.

3.2.3.4 Property Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risk identified	PAI1 is constructed on the Tian Siang POM who have legal ownership of the land. In addition, relevant contract agreements/4 were established and signed between Tian Siang POM and GLTSB as well as GLT Maju Sdn Bhd prior to construction of the biogas plant. Based on the contract agreement reviewed there is no issue on property rights at PAI1.

3.2.3.5 Benefit Sharing

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Process used to design the benefit sharing plan	PAI1 does not implement any benefit sharing to their relevant stakeholders. Thus, planning design for benefit sharing is not established.
Summary of the benefit sharing plan	PAI1 does not implement any benefit sharing to their relevant stakeholders. Thus, planning design for benefit sharing is not established.
Approval and dissemination of benefit sharing plan	PAI1 does not implement any benefit sharing to their relevant stakeholders. Thus, planning design for benefit sharing is not established.

3.2.4 Ecosystem Health

Item	Evidence gathering activities, evidence checked, and assessment conclusion
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Impacts on biodiversity and ecosystems	PAI1 is constructed on the Tian Siang POM that has been developed since 2000. Based on the on-site observation and interview with the sampled workers found, there is no sighting of any endangered species and no expansion of the new area prior to the construction of PAI1. Referring to the above, there are no or low impacts on biodiversity and ecosystem due to the development of PAI1.
Soil degradation and soil erosion	PAI1 is constructed on the Tian Siang POM that has been developed since 2000. Reviewed the '40. Hydroseeding Layout' established by GLT Design Build Sdn Bhd on 07.04.2023 found, a potential of soil degradation and erosion area has been identified to install the hydroseeding within the PAI1 area. Thus, low potential for soil degradation and erosion within PAI1 was identified.
Water consumption and stress	Water consumption is mainly for daily usage only as observed during the validation assessment. Thus, water consumption and water stress are graded as low impact.

3.2.4.1 Rare, Threatened, and Endangered species

Risk identified	Evidence gathering activities, evidence checked, and assessment conclusion
Species and habitat	PAI1 is constructed on the Tian Siang POM that has been developed since 2000. Based on the on-site observation and interview with the sampled workers found, there is no sighting of any RTE species within the PAI1 area.
Areas needed for habitat connectivity	Based on the on-site observation and interview with the sampled workers found, there is no sighting of any RTE species within the PAI1 area. Thus, there are no areas needed for habitat connectivity.

3.2.4.2 Introduction of Species

Species introduced	Evidence gathering activities, evidence checked, and assessment conclusion
No	There is no introduction of new species resulting from the PAI1 activities. Thus, it is not applicable.
Existing invasive species	Evidence gathering activities, evidence checked, and assessment conclusion

No	There is no introduction of new species resulting from the PAI1 activities. Thus, it is not applicable.
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Evidence gathering activities, evidence checked, and assessment conclusion	
Invasive species	There is no introduction of new species resulting from the PAI1 activities. Thus, it is not applicable.

3.2.4.3 Ecosystem conversion

Risks Identified	Evidence gathering activities and evidence checked
Ecosystem conversion	No risk identified as the Grouped Project and PAI1 does not involve ARR, ALM, WRC or ACoGS projects.

3.3 Application of Methodology

3.3.1 Title and Reference

Grouped Project

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS-III.H	AMS-III.H.: Methane recovery in wastewater treatment	19.0
Tool	TOOL03	Tool to calculate project or leakage CO ₂ emissions from fossil fuel combustion	03.0
Tool	TOOL04	Emissions from solid waste disposal sites	08.1
Tool	TOOL05	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	03.0
Tool	TOOL06	Project emissions from flaring	04.0

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Tool	TOOL32	Positive lists of technologies	04.0
Methodology	AMS-I.D	AMS-I.D: Grid connected renewable electricity generation	18.0
Tool	TOOL03	Tool to calculate project or leakage CO ₂ emissions from fossil fuel combustion	03.0
Tool	TOOL07	Tool to calculate the emission factor for an electricity system	07.0
Tool	TOOL10	Tool to determine the remaining lifetime of an equipment	01
Tool	TOOL16	Project and leakage emissions from biomass	05.0
Tool	TOOL11	Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period	03.0.1

During the validation activity, the validation team was briefed by the GLT that PAIs within the Grouped Project may not require the use of all the above approved methodologies and methodological tools. Relevance and/or applicability should be assessed on a project-by-project basis during its commencement. At this stage, initial PAI of combination of AMS-III.H. and AMS-I.D. have been included.

Project Instance 1 (PAI1)

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS-III.H	AMS-III.H.: Methane recovery in wastewater treatment	19.0
Tool	TOOL05	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	03.0
Tool	TOOL06	Project emissions from flaring	04.0

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Tool	TOOL32	Positive lists of technologies	04.0
Methodology	AMS-I.D	AMS-I.D: Grid connected renewable electricity generation	18.0
Tool	TOOL07	Tool to calculate the emission factor for an electricity system	07.0
Tool	TOOL10	Tool to determine the remaining lifetime of an equipment	01

The validation team had confirmed during on-site validation process through project site visit, document reviewed and interviewed with the focal persons that the methodologies listed above were applied and GLT had applied the latest version of the methodologies and tools for PAI1.

3.3.2 Applicability

Grouped Project

At this stage, initial PAI of combination of AMS-III.H. and AMS-I.D. have been included.

Project Instance 1 (PAI1)

The validation team has conducted the validation and assessment of the applicability of the listed methodologies below for PAI1. The validation was conducted through the document review and cross checking with the applicability conditions of each methodology, confirmation through a site visit and interviews with focal persons from GLT and Eco-Ideal Consulting Sdn Bhd. The assessment and conclusion are as follows:

Methodology ID	Applicability condition	VVB assessment and conclusion
AMS-III.H	As per paragraph 2 of AMS-III.H. The methodology comprises measures that recover biogas from biogenic organic matter in wastewater by means of one, or a combination, of the following options: (a) Substitution of aerobic wastewater or sludge treatment systems with anaerobic systems with biogas recovery and combustion;	The validation team has reviewed the AMS-III.H Version 19.0 under Scope of applicability and entry force and the team has conducted the site visit of the biogas plant including verify the scope applicability through the biogas project drawing. The validation team has verified that the methodology applied by PAI1 is conformed with option (f) by

Methodology ID	Applicability condition	VVB assessment and conclusion
	(b) Introduction of anaerobic sludge treatment system with biogas recovery and combustion to a wastewater treatment plant without sludge treatment; (c) Introduction of biogas recovery and combustion to a sludge treatment system; (d) Introduction of biogas recovery and combustion to an anaerobic wastewater treatment system such as anaerobic reactor, lagoon, septic tank or an on-site industrial plant; (e) Introduction of anaerobic wastewater treatment with biogas recovery and combustion, with or without anaerobic sludge treatment to an untreated wastewater stream; (f) Introduction of a sequential stage of wastewater treatment with biogas recovery and combustion, with or without sludge treatment, to an anaerobic wastewater treatment system without biogas recovery (e.g. introduction of treatment in an anaerobic reactor with biogas recovery as a sequential treatment step for the wastewater that is presently being treated in an anaerobic lagoon without methane recovery).	introduction of treatment in an anaerobic reactor with biogas recovery as a sequential treatment step for the wastewater that is presently being treated in an anaerobic lagoon without methane recovery. The conformation was supported by DOA approval letter for Biogas Plant /41 (including anaerobic digester construction) and supported by interview with focal persons from GLT. Based on that, we conclude that the GLT Biogas Project conformed with the methodology applicability condition option (f).
AMS-III.H	As per paragraph 3 of AMS-III.H. In cases where baseline system is anaerobic lagoon, the methodology is applicable if: (a) The lagoons are ponds with a depth greater than two meters, without aeration. The value for depth is obtained from engineering design documents,	The validation team has verified that the anaerobic lagoon of GLT is conformed with the condition in the AMS-III.H through: (a) lagoon drawing plan approved by Department of Environment (DOE)/46

Methodology ID	Applicability condition	VVB assessment and conclusion
	or through direct measurement, or by dividing the surface area by the total volume. If the lagoon filling level varies seasonally, the average of the highest and lowest levels may be taken; (b) Ambient temperature above 15°C, at least during part of the year on a monthly average basis; (c) The minimum interval between two consecutive sludge removal events shall be 30 days.	(b) verify the source of information of “average ambient temperature in Pahang, Malaysia which reported at 30 °C” via https://www.accuweather.com/en/my/temerloh/229429/june-weather/229429 (c) through the lagoon desludging report from Tia Siang POM/42.
AMS-III.H	As per paragraph 4 of AMS-III.H. The recovered biogas from the above measures may also be utilized for the following applications instead of combustion/flaring: (a) Thermal or mechanical, electrical energy generation directly; (b) Thermal or mechanical, electrical energy generation after bottling of upgraded biogas, in this case, additional guidance provided in Annex 1 shall be followed; or (c) Thermal or mechanical, electrical energy generation after upgrading and distribution, in this case, additional guidance provided in Annex 1 shall be followed: (i) Upgrading and injection of biogas into a natural gas distribution grid with no significant transmission constraints; (ii) Upgrading and transportation of biogas via a dedicated piped network to a group of end users; or	The validation team has verified the recovered biogas from PAI1 project will be utilized for electrical energy generation where the PAI1 will generate the electricity by using the Jencbacher biogas engine. The availability of the biogas engine was physically witnessed during the site visit. In the same time it was confirmed through document review of the application of AMS-I.D methodology for PAI1, which specifically for Grid connected renewable electricity generation in the PDD/1. Hence, the validation team is assured that the PAI1 is conformed with the (a) “Thermal or mechanical, electrical energy generation directly”.

Methodology ID	Applicability condition	VVB assessment and conclusion
	(iii) Upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users. (d) Hydrogen production; (e) Use as fuel in transportation applications after upgrading.	
AMS-III.H	As per paragraph 5 of AMS-III.H. If the recovered biogas is used for project activities covered under paragraph 4 (a), that component of the project activity can use a corresponding methodology under Type I.	Based on the document review of the project details in the PDD/1 on applied methodologies, and physical confirmation during the site visit, the validation team verified that the PA11 applied the corresponding methodology Type 1 (AMS-I.D methodology, Version 18.0, Grid connected renewable electricity generation). Hence the validation team is assured that the PA11 conformed with the conditions under paragraph 5 of AMS-III.H.
AMS-III.H	As per paragraph 6 of AMS-III.H. For project activities covered under paragraph 4 (b), if bottles with upgraded biogas are sold outside the project boundary, the end use of the biogas shall be ensured via a contract between the bottled biogas vendor and the end-user. No emission reductions may be claimed from the displacement of fuels from the end use of bottled biogas in such situations. If, however, the end use of the bottled biogas is included in the project boundary and is monitored during the crediting period, CO₂ emissions avoided by the displacement of fossil fuel can be claimed under the corresponding Type I methodology, e.g. AMS-I.C Thermal energy production with or without electricity.	<u>Not applicable</u> PA11 does not involve bottling of upgraded biogas

Methodology ID	Applicability condition	VVB assessment and conclusion
AMS-III.H	As per paragraph 7 of AMS-III.H. For project activities covered under paragraph 4 (c) (i), emission reductions from the displacement of the use of natural gas are eligible under this methodology, provided the geographical extent of the natural gas distribution grid is within the host country boundaries.	<u>Not applicable</u> PAI1 does not involve upgrading and injection of biogas into a natural gas distribution grid.
AMS-III.H	As per paragraph 8 of AMS-III.H. For project activities covered under paragraph 4 (c) (ii), emission reductions for the displacement of the use of fuels can be claimed following the provision in the corresponding Type I methodology, e.g. AMS-I.C.	<u>Not applicable</u> PAI1 does not involve the transportation of natural gas distribution via a dedicated piped network to a group of end-user (s).
AMS-III.H	As per paragraph 9 of AMS-III.H. In particular, for the case of 4 (b) and (c) (iii), the physical leakage during storage and transportation of upgraded biogas as well as the emissions from fossil fuel consumed by vehicles for transporting biogas shall be considered. Relevant procedures in paragraph 18 of the appendix of AMS-III.H Methane recovery in wastewater treatment shall be followed in this regard.	<u>Not Applicable</u> PAI1 does not involve thermal or mechanical electrical energy generation after bottling of upgraded biogas or upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users
AMS-III.H	As per paragraph 10 of AMS-III.H. For project activities covered under paragraph 4 (b) and (c), this methodology is applicable if the upgraded methane content of the biogas is in accordance with relevant national regulations (where these exist) or in the absence of national regulations, a minimum of 96% (by volume).	<u>Not Applicable</u> PAI1 does not involve bottling, upgrading or transportation of biogas.

Methodology ID	Applicability condition	VVB assessment and conclusion
AMS-III.H	As per paragraph 11 of AMS-III.H. If the recovered biogas is utilized for the production of hydrogen (project activities covered under paragraph 4 (d)), that component of the project activity shall use the corresponding methodology AMS-III.O Hydrogen production using methane extracted from biogas.	<u>Not applicable</u> PAI1 does not involve hydrogen production
AMS-III.H	As per paragraph 12 of AMS-III.H. If the recovered biogas is used for project activities covered under paragraph 4 (e), that component of the project activity shall use the corresponding methodology AMS-III.AQ Introduction of Bio-CNG in road transportation.	<u>Not applicable</u> PAI1 does not involve use of recovered biogas as fuel in transportation applications after upgrading.
AMS-III.H	As per paragraph 13 of AMS-III.H. New facilities (Greenfield projects) and project activities involving a change of equipment resulting in a capacity addition of the wastewater or sludge treatment system compared to the designed capacity of the baseline treatment system are only eligible to apply this methodology if they comply with the relevant requirements in the General guidelines to SSC CDM methodologies. In addition, the requirements for demonstrating the remaining lifetime of the equipment replaced as described in the general guidelines shall be followed.	<u>Not Applicable</u> PAI1 is not a Greenfield project. The project activity is implemented in an existing wastewater treatment plant. There is also no capacity addition of wastewater treatment system compared to the designed capacity of the baseline treatment system.
	As per paragraph 2.2.14 of AMS-III.H. The location of the wastewater treatment plant as well as the source generating the wastewater shall be	Based on the site visit, the validation team confirmed that the PAI1 is located at Tian Siang Palm Oil Mill (Triang) Sdn Bhd and the geographical coordinates same as provided in the PDD/1. Hence it is

Methodology ID	Applicability condition	VVB assessment and conclusion
	uniquely defined and described in the Project Design Document (PDD).	considered that the methodological condition is met. The geographical coordinates for the site are N 3°10'58.6" and E 102°27'27.3".
AMS-III.H	As per paragraph 2.2.15 of AMS-III.H Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 ktCO ₂ equivalent annually from all Type III components of the project activity.	The validation team has validated that GLT demonstrated the total emission reductions achieved are estimated as 37.071 tCO ₂ e (type III) per annum, which is less than the identified threshold i.e., 60ktCO ₂ e per annum in the PDD/1 and Baseline Emission Calculation Sheet.
AMS-I.D	As per paragraph 2 of AMS-I.D This methodology comprises renewable energy generation units such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass: (a) Supplying electricity to a national or a regional grid; or (b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling.	The validation team has validated that the PAI1 utilizes the captured biogas from Palm Oil Mill Effluent (POME) for electricity generation own use and upload to national which is conformed with condition (a) of para 2 AMS-I.D, through site visit, document review of PDD/1, project engineering drawing and interviews with focal persons from GLT and Eco Ideal Consulting Sdn Bhd.
AMS-I.D	As per paragraph 4 of AMS-I.D This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/ unit(s); or (e) Involve a replacement of (an) existing plant(s).	Through the site visit and interviews the focal persons from GLT and Eco Ideal Consulting Sdn Bhd, the PAI 1 PAI1 is a new power generation plant that qualifies as greenfield plant which conform with condition (a) of para 4 AMS-I.D.

Methodology ID	Applicability condition	VVB assessment and conclusion
AMS-I.D	As per paragraph 5 of AMS-I.D Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4W/m²; or (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4W/m².	<u>Not Applicable</u> PAI1 is not a hydro power plant project.
AMS-I.D	As per paragraph 6 of AMS-I.D If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15MW.	Based on the validation activity through site visit, project document review and interview with focal persons from GLT. The validation is assured that the capacity of the renewable component for PAI1 is 1.2MW which is less than 15MW which conforms to the condition in para 6 of AMS-I.D.
AMS-I.D	As per paragraph 7 of AMS-I.D Combined heat and power (co-generation) systems are not eligible under this category.	<u>Not Applicable</u> PAI1 is not co-generation project
AMS-I.D	As per paragraph 8 of AMS-I.D In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power	<u>Not Applicable</u> PAI1 is not capacity addition of renewable energy generation units.

Methodology ID	Applicability condition	VVB assessment and conclusion
	generation facility, the added capacity of the units added by the project should be lower than 15MW and should be physically distinct from the existing units.	
AMS-I.D	As per paragraph 9 of AMS-I.D In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/ unit shall not exceed the limit of 15MW.	<u>Not Applicable</u> PAI1 does not involve retrofit, rehabilitation or replacement of existing facilities.
AMS-I.D	As per paragraph 10 of AMS-I.D In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to the grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as 'AMS-I.C: Thermal energy production with or without electricity' shall be explored.	Based on the document review and site visit, PAI1 is a wastewater treatment methane recovery project, and it is eligible under Type III category. Hence, the validation team is assured that the PAI1 is conformed to the condition under para 10 of AMS-I.D.
AMS-I.D	As per paragraph 11 of AMS-I.D In case biomass is sourced from dedicated plantations, the applicability criteria in the tool 'Project emissions from cultivation of biomass' shall apply.	<u>Not Applicable</u> PAI1 does not use biomass sourced from dedicated plantations.

3.3.3 Project

Grouped Project

At this stage, initial PAI of combination of AMS-III.H. and AMS-I.D. have been included.

PAI 1 Project Boundary

In Section 3.3 of the Project Design Document (PD) elaborates the paragraph 22 of the applied methodology AMS-III.H, providing a clear distinction between treatment systems in the baseline scenario and project activity. This distinction has been verified through a site visit and documentation review, and is visually represented as follows:

PAI1 Project and Baseline Scenario

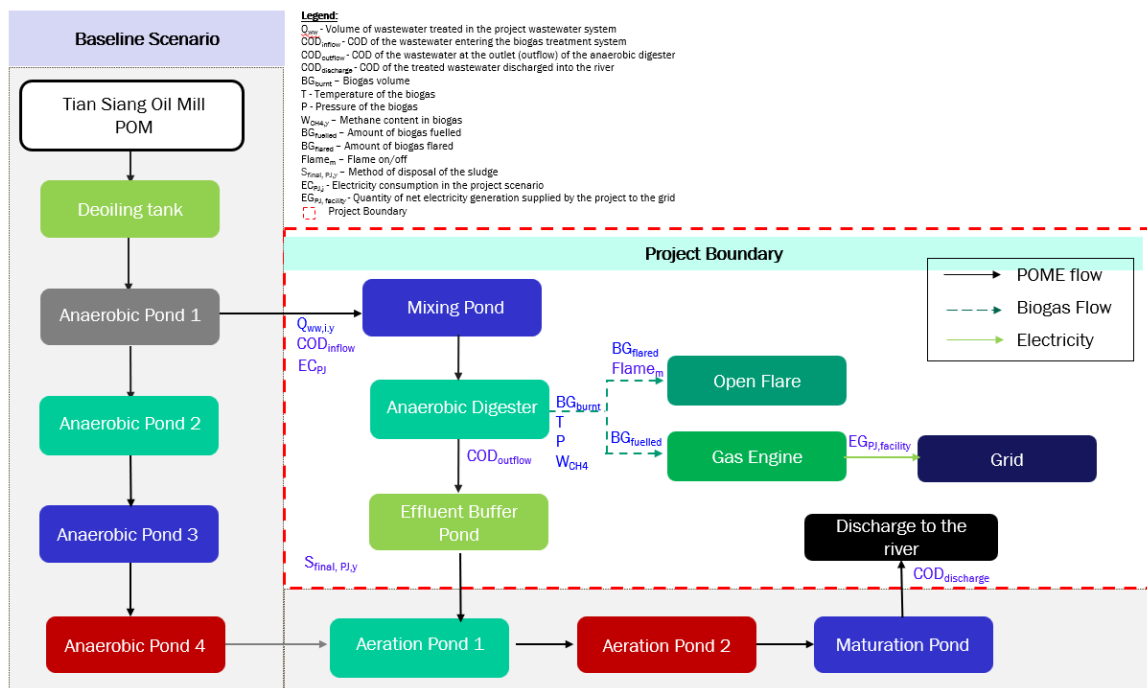


Figure: PAI1 Baseline Scenario and Project Boundary

Above diagram also illustrates the PAI1 baseline scenario where the typical industry practice for handling sludge before discharge into natural waterways:

- **Tian Siang Oil Mill POM:** The origin point where sludge is generated during palm oil processing.

- **Deoiling Tank:** Removes residual oil from the sludge to reduce oil content before biological treatment.
- **Anaerobic Treatment Ponds:**
 - **Anaerobic Pond 1 to 4:** Sequential anaerobic digestion ponds where organic matter is broken down in the absence of oxygen, reducing BOD and COD levels.
- **Aerobic Treatment Ponds:**
 - **Aeration Pond 1 & 2:** Introduce oxygen to further degrade organic pollutants through aerobic microbial activity.
- **Maturation Pond:** Final polishing stage where treated effluent undergoes natural processes like sedimentation and sunlight exposure.
- **Discharge to River:** The treated effluent is released into a nearby river, expected to meet environmental discharge standards.

While the project boundary illustrates in the same above diagram is realized through the diversion of sludge from Anaerobic Pond 1 into a Mixing Pond, which marks the entry point into the enhanced treatment system. From there, the sludge is directed into an Anaerobic Digester, where methane gas is captured under controlled anaerobic conditions.

The captured methane is then utilized in a Gas Engine, where it undergoes a methane combustion process to generate electricity. This electricity is subsequently supplied to the national grid, contributing to renewable energy generation.

Any excess methane that is not used for power generation is safely combusted via an Open Flare, ensuring that unutilized biogas does not escape into the atmosphere.

The residual sludge from the digester flows into the Effluent Pond, *continuing through the conventional treatment stages:*

- Aeration Pond 1
- Aeration Pond 2
- Maturation Pond

These stages complete the biological treatment process. Finally, the treated effluent is discharged into a river or natural waterway, in compliance with environmental discharge standards.

Verification activities have confirmed the accuracy and consistency of key operational parameters across both the Baseline Scenario and the Project Boundary. The parameters verified include:

- Chemical Oxygen Demand (COD) inflow
- Palm Oil Mill Effluent (POME) flow rates
- Biogas volume
- Temperature data

The consistency of these readings across both scenarios supports the validity of the methodological approach and reinforces the credibility of the emission reduction calculations. These verified data points and their corresponding visualizations substantiate compliance with the requirements outlined in AMS-III.H, Paragraph 22.

The validation team has confirmed that the treatment system under the PAI1 project activity remains unaffected. This determination is based on the fact that the system continues to operate with the same feed inflow quality, volume (retention time), and temperature (heating conditions) as in the baseline scenario.

As such, the system is deemed to conform with the conditions outlined in Paragraph 22 of the AMS-III.H methodology,

Emission Sources for PAI1

Based on the document review of the methodologies (AMS-III.H, Ver19 and AMS-I.D, Ver18) and project GHG emissions calculation sheets and physical site visit concurrently conducted with the interview session with the GLT's focal person, the validation team is assured that the project boundary includes all geographical areas where wastewater and sludge treatment take place as demonstrated in the diagram above.

The validation team had also checked and verified the PAI1's greenhouse gas (GHG) sources and emissions for both the baseline and project scenarios for applied methodologies AMS-III.H, Ver19 and AMS-I.D, Ver18. During the assessment, the validation team conducted the document review on the following documents:

- PDD/1 Document (project boundary section)
- GHG's project baseline and emissions calculation sheet
- Interview with the focal persons

As a result, the validation team is assured that the sources and emissions included are only Methane (CH₄) and Carbon Dioxide (CO₂). The included emission sources for PAI1 are as follows:

Emission sources included for AMS-III.H for PAI1

Source		Gas	Included?	VVB assessment and conclusion
Baseline		CO ₂	No	Excluded for simplification

Source		Gas	Included?	VVB assessment and conclusion
	Methane emissions from baseline wastewater treatment systems	CH4	Yes	Main emission source
		N2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions on account of inefficiencies in the baseline wastewater treatment systems and presence of degradable organic carbon in the treated wastewater discharged into river/lake/sea	CO2	No	Excluded for simplification
		CH4	Yes	Main emission source
		N2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	CO2 emissions from electricity and fuel used by the project facilities	CO2	Yes	Main emission source
		CH4	No	Excluded for simplification
		N2O	No	Excluded for simplification
		Other	No	Excluded for simplification
Project	Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery in the project scenario	CO2	No	Excluded for simplification
		CH4	Yes	Main emission source
		N2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions on account of inefficiency of the project activity wastewater treatment systems and presence of degradable organic carbon in treated wastewater	CO2	No	Excluded for simplification
		CH4	Yes	Main emission source
		N2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane fugitive emissions due to	CO2	No	Excluded for simplification
		CH4	Yes	Main emission source

Source		Gas	Included?	VVB assessment and conclusion
	inefficiencies in capture systems	N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane fugitive emissions due to inefficiencies in capture systems	CO ₂	No	Excluded for simplification
		CH ₄	Yes	Main emission source
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions due to incomplete flaring	CO ₂	No	Excluded for simplification
		CH ₄	Yes	Main emission source
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification

Emission sources included for AMS- I.D for PAI1

Source		Gas	Included?	Assessment/Explanation
Baseline	CO ₂ emissions from electricity generation in power plants that are displaced due to the project activity	CO ₂	Yes	Main emission source
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
Project	CO ₂ emissions from on-site consumption of fossil fuel due to project activity	CO ₂	Yes	Main emission source
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification

3.3.4 Baseline Scenario

Grouped Project

According to VCS Standard, paragraph 3.6.11, “Determination of baseline scenario and demonstration of additionality are based upon the initial project activity instances”. Hence, for grouped project level, no determination of baseline scenario shall be done.

At this stage, initial PAI of combination of AMS-III.H. and AMS-I.D. have been included.

Project Instance 1 (PAI1)

According to the AMS-III.H guidelines, the baseline scenario involves using an anaerobic wastewater treatment system without biogas recovery. During the onsite validation visit, a comprehensive evaluation of site conditions and legal compliance was conducted. It was confirmed that, in the absence of the project, the mill would continue operating its wastewater treatment facility without implementing technological advancements. The current treatment method aligns with Malaysia's regulatory framework, which does not mandate specific wastewater technologies but requires effective management of mill effluent discharge. The baseline scenario describes the current situation where methane is emitted into the atmosphere without any restrictions or containment measures.

At PAI1 the applied project methodologies are AMH-III.H and AMS-I.D. During site visit, the baseline condition of the PAI1 Baseline Scenario has been verified. It has been confirmed that the wastewater is treated in a series of lagoons. Lagoons 1-4 of this system are anaerobic lagoons, with each of the depth deeper than two (2) meters and the average atmospheric temperature in the project location is 31° Celsius. The verification of this condition was conducted by cross referring with the pond engineering drawing from Tian Siang Palm Oil Mill. These conditions create an anaerobic environment in lagoons 1-4, resulting in methane generation. Once the wastewater has completed processing through the baseline treatment system, this wastewater is within the acceptable legal limits applicable to Malaysia and is then discharged into a river.

The sludge removal from lagoons was conducted on an “as-needed” basis. The sludge removed is discarded on the nearby land of the palm oil mill and, as a result, does not produce any emissions of greenhouse gases. The verification of sludge removal was verified by checking the Tian Siang Palm Oil Mill desludging records where the latest removal recorded from the anaerobic ponds 1 – 4 was conducted on 30/09/2010, more than 30 days.

In the absence of PAI1, the baseline scenario will be the continuation of the present wastewater treatment, including ongoing methane emissions to the atmosphere from the anaerobic lagoons.

Hence the validation team affirms that the PAI1 is met the additionality of VCS Standard.

The baseline scenario for PAI1 is presented below:

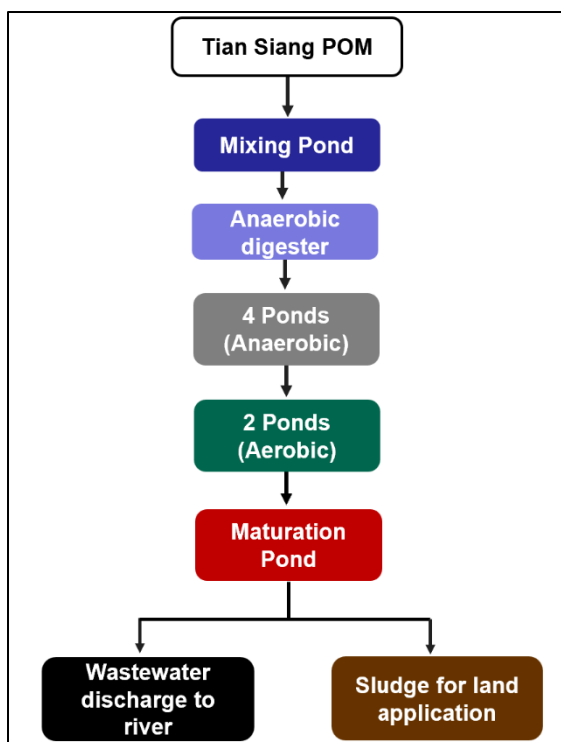


Figure: PAI1 Baseline Scenario

The baseline scenario has been assessed by the validation team through physical site visit to the 4 anaerobic ponds of Tian Siang POM and the biogas plant ordained by the PP and the supporting documents in order to confirm the description provided in the PDD/1.

Hence the validation team confirms that the baseline scenario is in accordance with the applied methodologies AMS-III.H, v.19.0 and AMS-I.D, version 18.0.

3.3.5 Additionality

Grouped Project

According to VCS Standard, paragraph 3.6.11, “Determination of baseline scenario and demonstration of additionality are based upon the initial project activity instances”. Hence, for grouped project level, no determination of baseline scenario shall be done.

At this stage, initial PAI of combination of AMS-III.H. and AMS-I.D. have been included.

Project Instance 1 (PAI1)

GLT has demonstrated the additionality for PAI1 as per tools 32 Positive lists of technologies - Version 04.0. A positive list of technology and project activity types defined as automatically additional if the PAIs apply such technologies and meet specified conditions. The validation team had conducted an assessment of each additionality requirement based on the document review and physical site visit including interview session with the GLT’s focal persons.

The positive list of technology requirements for 'waste handling and disposal' for PAI1 and VVB assessment is tabulated below:

Requirements for Additionality	PAI Scenario	VVB Assessment
Tool 32 As per paragraph 12(a): The existing treatment system is an anaerobic lagoon and the wastewater discharged meets the host country regulation	The existing treatment system is an anaerobic lagoon and the wastewater discharged meets the regulations of Malaysia.	Based on the Validation site visit it is confirmed that the existing treatment system involved series of anaerobic lagoons meets the regulation of Malaysia. The validation team has verified the pond engineering drawing/46 approved by DOE Malaysia on 12 th June 2018, where all the anaerobic ponds constructed according to the host country regulation with a depth of 5 meter (exceeding the requirement of 2 meters).
Tool 32 As per paragraph 12(b): No regulation in the host country requires the management of biogas from domestic, industrial and agricultural sites	Yes There is no regulation in Malaysia that requires the management of biogas from a site such as the project site. Condition in the criteria and guidelines for an application for licensing by the Malaysian Palm Oil Board (MPOB) require new palm-oil mills constructed after 01-Jan-2014 to install wastewater treatment systems with methane capture. The requirement was also placed on existing mills who applied for capacity extensions after 01-Jan-2014, although an exemption effective 22-May-2021 was given to those existing mills who applied	Based on the MPOB Guidance for licensing titled "Naskah Buku Kriteria Lesen MPOB" stated that all existing oil palm fruit mills (MF) that are applying for increased processing capacity oil palm fruit must install a device to trap or avoid the release of methane gas. However effective on 2 May 2021 the exception of the installation of trapping devices or the avoidance of methane gas emissions given to factories that process oil palm fruits 270,000 tons per year and below. Any existing plant applying for power addition annual processing of MORE than 270,000 tons of palm oil per year is required to install biogas capture facilities or methane gas emission avoidance technology or both with a POME usage rate of 80% (±5%) of the POME capacity produced. A confirmation assessment was conducted on the Tian Siang Palm Oil Mill's 3 years FFB process volume record found; the highest volume processed

Requirements for Additionality	PAI Scenario	VVB Assessment
	for capacity extensions below 270,000 MT FFB per year, noting this threshold has been revised several times. Tian Siang POM was established in 2002 and was not a mill constructed after 01-Jan-2014 and has not undergone a capacity expansion since 01-Jan-2014. Therefore, the new criteria do not apply and there is no obligation to recover the methane and/or use the biogas	was 251,900 tons / year in 2022. Furthermore, the Tian Siang Palm Oil Mill has been in operation since 2002. Hence, the Tian Siang Palm Oil Mill are not restricted to any regulation requires management of the biogas plant or methane capture facility. Based on that, validation team confirms that the PAI1 is conformed with the condition of paragraph 12.
Tool 32 As per paragraph 12(c): There is no capacity increase in the wastewater treatment system	Yes There has been no capacity increase in the wastewater treatment system.	It is confirmed that there is no capacity increase for the wastewater treatment system proposed in PAI1 based on the previous 3 years FFB process volume by Tian Siang Palm Oil Mill demonstrated that the processed FFB is within permissible limit by the MPOB. Hence validation team is of the opinion that the condition of paragraph 12(c) is met.
Tool 32 As per paragraph 12(d): No other alternative economic activity is expected to be undertaken on the land of the existing lagoon	Yes No other alternative economic activity is expected to be undertaken on the land of the existing lagoon.	The PAI1 located on the abandon land of Tian Siang Palm Oil Mill. During the site visit, based on the confirmation obtains through site interview with the mill's manager, the validation team confirmed that there are no economic activities carried out on the PAI1.
Tool 32	Yes	Yes, the validation team confirmed that the biogas plant is used to generate

Requirements for Additionality	PAI Scenario	VVB Assessment
As per paragraph 12(e): The biogas is used to generate electricity in one or more power plants, and the total nameplate	The biogas is used to generate electricity in a power plant and the total nameplate capacity is below 5MW.	electricity based on the confirmation obtained by witnessing the maximum capacity output from the gas engine is 1.234MW during the site visit.

As PAI1 meets all five (5) conditions under Tool 32, the validation team is of the opinion that PAI1 is in conformance with the additionality requirements of the VCS Standard.

3.3.6 Quantification of GHG Emission Reductions and Carbon Dioxide Removals

The project participant has utilized algorithms and formulas in accordance with the applied baseline and monitoring methodologies, specifically AMS-III.H, Version 19.0, and AMS-I.D, Version 18.0, as well as the "Tool to calculate baseline, project, and/or leakage emissions from electricity consumption" and the "Tool to calculate project or leakage CO2 emissions from fossil fuel combustion." Detailed explanations of the algorithms and formulas used in emission calculations are provided in sections 4.1, 4.2, 4.3, and 4.4 of the revised PDD/1. The validation team confirms that these formulas have been correctly applied in line with the applied baseline and monitoring methodologies.

Baseline Emissions for the PAI1

AMS-III.H, Version 19.0

The validation team has conducted a validation assessment through a documentation review of the PAI1's GHG baseline emission calculation sheet provided by the GLT and physical site visit . The table below summarises the baseline emissions applied for PAI1:

Source	Included/Excluded	VVB Justification
BE _{power,y}	Excluded	No electricity or fossil fuel used for operation of PAI1. PAI1 will be used electricity generated from the biogas engine resulted of its biogas production from its from captured methane of POME.
BE _{ww,treatment,y}	Included	Main emissions source for the PAI1 is from open lagoons.
BE _{s,treatment,y}	Excluded	No sludge treatment activity involved in PAI1.

Source	Included/Excluded	VVB Justification
BE _{ww,discharge,y}	Included	The effluent from the wastewater will be discharged into river.
BE _{s,final,y}	Excluded	No sludge treatment activity involved in PAI1.

Hence, the validation team is assured that only BE_{ww,treatment,y} and BE_{ww,discharge,y} will be included under AMS-III.H calculation for PAI1's baseline emissions.

The validation team has conducted a validation assessment on the baseline emissions calculation for AMS-III.H based on PAI1's GHG baseline emissions calculation sheet provided by the GLT. The validation team has also asked the GLT's focal person to demonstrate the calculation during the site visit and the reference of the equation was referring to the CDM AMS-III.H, Version 19.0 document.

The validation team assessed the BE_{ww,treatment,y} of PAI1 calculated by GLT.

BE_{ww,treatment,y} - Baseline emission of the wastewater treatment systems affected by the project activity:

Parameter	24-Apr-2024 – 31-Dec-2024	01-Jan-2025 – 31-Dec-2025	01-Jan-2026 – 31-Dec-2026	01-Jan-2027 – 31-Dec-2027	01-Jan-2028 – 31-Dec-2028	01-Jan-2029 – 31-Dec-2029	01-Jan-2030 – 31-Dec-2030	01-Jan-2031 – 23-Apr-2031
$\Sigma Q_{ww,i,y}$	92,002	133,257	133,257	133,257	133,257	133,257	133,257	41,255
COD inflow _{i,y}	0.0657	0.0657	0.0657	0.0657	0.0657	0.0657	0.0657	0.0657
h _{cod, BL, i}	85%	85%	85%	85%	85%	85%	85%	85%
MCF _{ww, treatment, BL,i}	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
B _{o,ww}	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF _{BL}	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89
GWP _{CH4}	28	28	28	28	28	28	28	28
BE _{ww, treatment, y}	25,594	37,071	37,071	37,071	37,071	37,071	37,071	11,477

The validation team had also validated the parameters' description and its reference values for above calculation of BE_{ww,treatment,y} as follows:

1. $\Sigma Q_{ww,i,y}$ – volume of wastewater treated in baseline wastewater treatment system i in year y (m^3).

During the validation it was confirmed that the volume of wastewater treated corresponds to a 3-year average of the POME generated by the Tian Siang Palm Oil Mill. The annual average is 133,257 m^3 .

2. $COD_{inflow,i,y}$ - Chemical oxygen demand of the wastewater inflow to the baseline treatment system i in year y (tonnes/ m^3).

The validation team assessed that the number of COD is derived from average COD values of a 10-days campaign took at the inlet point to the anaerobic lagoon pond (Point A – Raw POME), with the average value of 0.0657 tCOD/ m^3 . Additional sample cross-checks have been conducted to ensure the COD values of the 10-days campaign are aligned with the COD lab result reports/**18** provided by the GLT. As this is the initial project and there is no historical data available, PAI1 is using a 10-day ex-ante measurement campaign to determine the parameters.

3. $h_{COD, BL, i}$ – COD removal efficiency of the baseline system i.

The validation team assessed that the $h_{COD, BL, i}$ value is derived from the value of COD removed and divided with the average COD inlet to anaerobic lagoon times by model correction factor to account for model uncertainties (0.89) as per guided in the AMS-III.H, Version 19 document. Therefore, the value of $h_{COD, BL, i}$ is confirmed to be 85.0%.

4. $MCF_{ww,treatment,BL,i}$ - Methane correction factor for baseline wastewater treatment systems i (MCF value as per table III.H. I).

The validation team assessed that the default value of $MCF_{ww,treatment,BL,i}$ used for PAI1's baseline calculation is 0.8, referring to the IPCC default values for Methane Correction Factor, for type of wastewater treatment of "Anaerobic deep lagoon (depth more than 2 meters)" in the AMS-III.H, Version 19.0 document.

5. $B_{0,ww}$ - Methane producing capacity of the wastewater.

The validation team confirmed that the default IPCC value of 0.25 kg CH_4 /kg COD is applied for $B_{0,ww}$ for the calculation of PAI1's baseline emissions.

6. UF_{BL} – Model correction factor to account for model uncertainties.

The validation team confirmed that the default value applied (0.89) for model correction factor to account for model uncertainties is as per guided in the AMS-III.H, Version 19 document.

7. GWP_{CH_4} - Global Warming Potential for methane.

The validation team confirmed the latest GWP applied is 28 based on the latest IPCC AR5 Report for PAI1's baseline emissions calculations.

Hence, the validation team is assured that the $BE_{ww,treatment,y}$ calculation for PAI1 is in conformance with the AMS-III.H, Version 19.0 guideline document.

Concurrently, the validation team has also assessed $BE_{ww,discharge,y}$ for PAI1.

BE_{ww,discharge,y} - Baseline methane emissions from degradable organic carbon in treated wastewater discharged into river:

Parameter	24-Apr-2024 – 31-Dec-2024	01-Jan-2025 – 31-Dec-2025	01-Jan-2026 – 31-Dec-2026	01-Jan-2027 – 31-Dec-2027	01-Jan-2028 – 31-Dec-2028	01-Jan-2029 – 31-Dec-2029	01-Jan-2030 – 31-Dec-2030	01-Jan-2031 – 23-Apr-2031
Q _{ww,y}	92,002	133,257	133,257	133,257	133,257	133,257	133,257	41,255
GWP _{CH4}	28	28	28	28	28	28	28	28
B _{0,ww}	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF _{BL}	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89
COD _{ww,discharge,BL,y}	0.00122	0.00122	0.00122	0.00122	0.00122	0.00122	0.00122	0.00122
MCF _{ww,BL,discharge}	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
BE _{ww,discharge,y}	70	101	101	101	101	101	101	31

The validation team has validated the parameters' description and its reference values for above calculation of BE_{ww,discharge,y} as follows:

1. $\Sigma Q_{ww,y}$ – Volume of treated wastewater discharged in year y (m³)

The validation team confirmed that the forecasted of wastewater treated by PAI1 is derived from a 3-year average of POME generated by the Tian Siang Palm Oil Mill. The annual average is 133,257 m³. The $\Sigma Q_{ww,y}$ value for 2024 is calculated based on 252 days.

2. GWP_{CH4} - Global Warming Potential for methane.

The validation team had validated the latest GWP applied is 28 based on the latest IPCC AR5 Report for the PAI1's project emissions calculations.

3. B_{0,ww} - Methane producing capacity of the wastewater.

The validation team confirmed that the default IPCC value of 0.25 kg CH₄/kg COD is applied for B_{0,ww} for the calculation of PAI1's baseline emissions.

4. UF_{BL} – Model correction factor to account for model uncertainties.

The validation team confirmed that the default value applied (0.89) for model correction factor to account for model uncertainties is as per guided in the AMS-III.H, Version 19 document.

5. COD_{ww,discharge,BL,y} - Chemical oxygen demand of the treated wastewater discharged into river in the baseline situation in the year y (tonnes / m³)

The validation team assessed that the number of chemical oxygen demand is derived from the average COD value of 10 samples taken at the discharge point (Point D – Final Discharge), with the average value of 0.0012 tCOD/m³. Additional sample cross-checks have been conducted to ensure the COD values of the 10-days campaign are aligned with the COD

lab result reports/**18** provided by the GLT. As this is the initial project and there is no historical data available, PAI1 is using a 10-day ex-ante measurement campaign to determine the parameters.

6. $MCF_{ww,treatment,BL,i}$ – Methane correction factor for baseline wastewater treatment systems i (MCF value as per table III.H. I).

The validation team had validated that the default value of $MCF_{ww,treatment,BL,i}$ used for PAI1’s baseline calculation is 0.8, referred to the IPCC default values for Methane Correction Factor, for type of wastewater treatment of “Anaerobic deep lagoon (depth more than 2 meters)” in the AMS-III.H, Version 19.0 document.

Hence, the validation team is assured that the $BE_{ww,discharge,y}$ calculation for PAI1 is conforming to the AMS-III.H, Version 19.0 guideline document.

AMS-I.D, Version 18.0

For AMS-I.D, the validation team had assessed the baseline emissions for PAI1 through documentation review, interview session and physical witnessing during the site visit.

Source	Included/Excluded	VVB Justification
BE_y	Included	The power generated from PAI1 will be supplied to the grid.

Therefore, the validation team is assured that BE_y will be included under AMS-I.D calculation for PAI1’s baseline emissions.

AMS-I.D

BE_y - Baseline emissions from electricity generation in power plants that are displaced due to the project activity

Parameter	19-July-2024 – 31-Dec-2024	01-Jan-2025 – 31-Dec-2025	01-Jan-2026 – 31-Dec-2026	01-Jan-2027 – 31-Dec-2027	01-Jan-2028 – 31-Dec-2028	01-Jan-2029 – 31-Dec-2029	01-Jan-2030 – 31-Dec-2030	01-Jan-2031 – 23-Apr-2031
EG_{Bly}	3,586	7884	7884	7884	7884	7884	7884	2,441
EF_{CO2}	0.585	0.585	0.585	0.585	0.585	0.585	0.585	0.585
BE_y	2,098	4,612	4,612	4,612	4,612	4,612	4,612	1,428

The validation team had also validated the parameters’ description and its reference values for above calculation of BE_y as follows:

7. EG_{Bly} – Quantity of net electricity generation that is produced and fed into the grid (MWh)

The validation team assessed that the forecasted annual value of power uploaded to grid from PAI1 is 7,884 MWh. The EG_{Bly} value for 2024 is calculated based on 166 days, expected date for commissioning of biogas engine is 19/07/2024.

8. EF_{CO2} - CO2 emission factor for grid connected power generation (tCO₂/MWh).

The validation team assessed that the value of EF_{CO2} applied is 0.585 tCO₂/MWh, a Combined Margin (CM) GEF which refers to the latest 2017 Grid Emission Factor published by Malaysian Green Technology and Climate Change Corporation (MGTC), a Malaysia government agency under Ministry of Energy, Science, Technology, Environment and Climate Change. Despite the GEF is published in 2017, it is the most recent figure available in Malaysia for a CM's type of GEF. On 25 Nov 2024, Energy Commission of Malaysia has updated the GEF where the value is 0.774 [data up to 2022 only] for Peninsular Malaysia. However, it is exclusively for Operating Margin (OM) GEF which is unsuitable for the CDM and PP's project approach.

Therefore, the validation team is assured that the BE_y calculation for PAI1 is conforming to the AMS-III.H, Version 19.0 guideline document.

Referring to the validation assessment conducted for AMS-III.H and AMS-I.D of PAI1 baseline emissions by the validation team, the PAI1's total baseline emissions value were validated to be as follows:

Year	Total BE _y	AMS IIIH		AMS ID
		BE _{ww,treatment,y}	BE _{ww,discharge,y}	BE _y
24-Apr-2024 – 31-Dec-2024 (252 days)	27,762	25,594	70	2,098
01-Jan-2025 – 31-Dec-2025	41,784	37,071	101	4,612
01-Jan-2026 – 31-Dec-2026	41,784	37,071	101	4,612
01-Jan-2027 – 31-Dec-2027	41,784	37,071	101	4,612
01-Jan-2028 – 31-Dec-2028	41,784	37,071	101	4,612
01-Jan-2029 – 31-Dec-2029	41,784	37,071	101	4,612
01-Jan-2030 – 31-Dec-2030	41,784	37,071	101	4,612
01-Jan-2031 - 23-Apr-2031 (113 days)	12,936	11,477	31	1,428
Total	291,404	259,498	708	31,198

The validation team is assured that the estimated PAI1's baseline emission conforms to the VCS Standard.

Project Emissions Project Instance 1 (PA1)

Project emissions accordance to AMS-III.H, Version 19.0

The validation team has conducted an assessment through a documentation review of the PAI1's GHG project emission calculation sheet provided by the GLT and physical site visit. The table below summarises the sources of project emissions considered for PAI1:

Source	Included/Excluded	Justification
PE _{power,y}	Excluded	No electricity or fossil fuel used in the project scenario
PE _{ww,treatment,y}	Excluded	The wastewater from the anaerobic digester is discharged to aerobic pond
PE _{s,treatment,y}	Excluded	No sludge treatment activity involved in the project activity
PE _{ww,discharge,y}	Included	The effluent is discharged into river
PE _{s,final,y}	Excluded	No sludge treatment activity involved in the project activity
PE _{fugitive,y}	Included	Project emissions due to potential inefficiencies of biogas capture system
PE _{flaring,y}	Included	Project emission due to incomplete flaring
PE _{biomass,y}	Excluded	No storage of additional biomass under anaerobic conditions in project activity instances

The validation team is assured that only PE_{ww,discharge,y}, PE_{fugitive,y} and PE_{flaring,y} sources of emissions are applicable for PAI1's project emissions calculation under AMS-III.H, Version 19.0.

Project emissions accordance to AMS-III.H, Version 19.0

The validation team has conducted an assessment of PAI1's project emissions in accordance with AMS-III.H, Version 19.0 through a document review, cross-checking the PAI1's GHG project emissions calculation sheet and interview with the GLT's focal persons.

The detailed calculation for project emissions is tabulated below:

PE_{ww,discharge,y} - Methane emissions from degradable organic carbon in treated wastewater

Para meter	24-Apr- 2024 – 31-Dec- 2024	01-Jan- 2025 – 31-Dec- 2025	01-Jan- 2026 – 31-Dec- 2026	01-Jan- 2027 – 31-Dec- 2027	01-Jan- 2028 – 31-Dec- 2028	01-Jan- 2029 – 31-Dec- 2029	01-Jan- 2030 – 31-Dec- 2030	01-Jan- 2031 – 23-Apr- 2031
ΣQ _{ww,i,y}	92,002	133,257	133,257	133,257	133,257	133,257	133,257	41,255
COD ww,discharge, PJ,y	0.00403 5	0.00403 5	0.00403 5	0.00403 5	0.00403 5	0.00403 5	0.00403 5	0.00403 5
MCF _{ww,dis charge,PJ,k}	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1

$B_{0,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{PJ}	1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.12
GWP_{CH4}	28	28	28	28	28	28	28	28
$PE_{ww,discharge,y}$	291	422	422	422	422	422	422	131

The validation team has also validated the parameters' description and its reference values for above calculation of $PE_{ww,discharge,y}$ as follows:

1. $\Sigma Q_{ww,y}$ – Forecasted volume of wastewater treated in project wastewater treatment system i in year y (m^3)

The validation team has confirmed that the forecasted volume of wastewater treated by PAI1 is derived from a 3-year average of POME generated by the Tian Siang Palm Oil Mill. The annual average is 133,257 m^3 . The $\Sigma Q_{ww,y}$ value for 2024 is calculated based on 252 days.

2. $COD_{ww,discharge,PJ,y}$ - Chemical oxygen demand of the wastewater outflow to the river in year y (tonnes/ m^3)

The validation team assessed that the number of chemical oxygen demand is derived from the value of COD leaving the anaerobic digester for the project, times the COD removal efficiency of the project wastewater treatment system (System-Anaerobic) and the value is validated at 0.0040 tCOD/ m^3 .

3. $MCF_{ww,discharge,PJ,k}$ - Methane correction factor based on the discharge pathway of the wastewater in the project scenario (Discharge into river).

The validation team assessed that the default value of $MCF_{ww,treatment,PJ,k}$ Used for PAI1's project emissions calculation is 0.1 and referring to the IPCC default values for Methane Correction Factor, for type of wastewater treatment of "Discharge of wastewater to river" in the AMS-III.H, Version 19.0 document.

4. $B_{0,ww}$ - Methane producing capacity of the wastewater.

The validation team confirmed that the default IPCC value of 0.25 kg CH_4 /kg COD is applied for $B_{0,ww}$ for calculation of PAI1's project emissions

5. UF_{PJ} – Model correction factor to account for model uncertainties.

The validation team confirmed that the default value applied (1.12) for model correction factor to account for model uncertainties is as per guided in the AMS-III.H, Version 19 document.

6. GWP_{CH4} - Global Warming Potential for methane.

The validation team confirmed the GWP applied is 28 based on the latest IPCC AR5 Report for PAI1's baseline emissions calculations.

Therefore, the validation team is assured that the $PE_{ww,discharge,y}$ calculation for PAI1 is in conformance with the AMS-III.H, Version 19.0 guideline document.

$PE_{fugitive,y}$ - Emissions on account of inefficiencies in capture systems:

Parameter	24-Apr-2024 – 31-Dec-2024	01-Jan-2025 – 31-Dec-2025	01-Jan-2026 – 31-Dec-2026	01-Jan-2027 – 31-Dec-2027	01-Jan-2028 – 31-Dec-2028	01-Jan-2029 – 31-Dec-2029	01-Jan-2030 – 31-Dec-2030	01-Jan-2031 – 23-Apr-2031
CFE _{ww}	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
GWP _{CH4}	28	28	28	28	28	28	28	28
Q _{ww,y}	92,002	133,257	133,257	133,257	133,257	133,257	133,257	41,255
B _{o,ww}	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF _{PJ}	1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.12
COD _{removed,PJ,k,y}	0.0558	0.0558	0.0558	0.0558	0.0558	0.0558	0.0558	0.0558
MCF _{ww,treatment,PJ,k}	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
MEP _{ww,treatment,y}	1,256	1,667	1,667	1,667	1,667	1,667	1,667	411
PE _{fugitive,y}	3,222	4,667	4,667	4,667	4,667	4,667	4,667	1,445

The validation team has validated the parameters' description and its reference values for above calculation of PE_{fugitive,y} as follows:

1. CFE_{ww} – Capture efficiency of the biogas recovery equipment in the wastewater treatment systems
The validation team assessed that PAI1 applied the default value of 0.9 for CFE_{ww} as per guided in the AMS-III.H Version 19.0 document for PAI1's GHG project emissions.
2. GWP_{CH4} - Global Warming Potential for methane.
The validation team confirmed the GWP applied is 28 based on the latest IPCC AR5 Report for PAI1's baseline emissions calculations.
3. Q_{ww,y} - Volume of treated wastewater discharged in year y (m³)
The validation team confirmed that the forecasted volume of wastewater treated by PAI1 is derived from a 3-year average of POME generated by the Tian Siang Palm Oil Mill. The annual average is 133,257 m³. The ΣQ_{ww,y} value for 2024 is calculated based on 252 days
4. B_{o,ww} - Methane producing capacity of the wastewater.
The validation team confirmed that the default IPCC value of 0.25 kg CH₄/kg COD is applied for B_{o,ww} for the calculation of PAI1's project emissions.
5. UF_{PJ} – Model correction factor to account for model uncertainties.
The validation team confirmed that the default value applied (1.12) for model correction factor to account for model uncertainties is as per guided in the AMS-III.H, Version 19 document.
6. COD_{removed,PJ,k,y} - The chemical oxygen demand removed by the treatment system k of the project activity equipped with biogas recovery in the year y (tonnes/m³)

The validation team assessed that the number of chemical oxygen demand is derived from the average value of COD inlet to the anaerobic lagoon for baseline, times the value of COD removal efficiency of biogas digester in baseline case. The validated value is 0.0558 tCOD/m³.

7. $MCF_{ww,discharge,PJ,k}$ - Methane correction factor for the project wastewater treatment and discharge pathway or system “type anaerobic digester for sludge without methane recovery (MCF values as per table AMS-III.H.1)

The validation team assessed that the default value of $MCF_{ww,treatment,PJ,k}$ used for PAI1’s project emissions calculation is 0.8 referring to the IPCC default value for Methane Correction Factor, for type of wastewater treatment of “Anaerobic deep lagoon (depth more than 2 metres)” in the AMS-III.H, Version 19.0 document.

8. $MEP_{ww,treatment,y}$ - Methane emission potential of wastewater treatment systems equipped with biogas recovery system

The validation team confirmed that the value of $MEP_{ww,treatment,y}$ is 1,667 tCH₄/year (for 365 day).

Therefore, the validation team is assured that the $PE_{fugitive,y}$ calculation for PAI1 ‘s project emissions is in conformance with the AMS-III.H, Version 19.0 guideline document.

$PE_{flaring,y}$ - Methane emissions due to incomplete flaring

Parameter	24-Apr-2024 – 31-Dec-2024	01-Jan-2025 – 31-Dec-2025	01-Jan-2026 – 31-Dec-2026	01-Jan-2027 – 31-Dec-2027	01-Jan-2028 – 31-Dec-2028	01-Jan-2029 – 31-Dec-2029	01-Jan-2030 – 31-Dec-2030	01-Jan-2031 – 23-Apr-2031
$\Sigma Q_{ww,i,y}$	92,002	133,257	133,257	133,257	133,257	133,257	133,257	41,255
$COD_{inflow,i,y}$	0.0657	0.0657	0.0657	0.0657	0.0657	0.0657	0.0657	0.0657
$h_{COD, BL, i}$	85.0%	85.0%	85.0%	85.0%	85.0%	85.0%	85.0%	85.0%
$MCF_{ww,treatment,BL,i}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{BL}	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89
$PE_{flaring,y}$	914	1,324	1,324	1,324	1,324	1,324	1,324	410

The validation team also validated the parameters’ description and its reference values for above calculation of $PE_{flaring,y}$ as follows:

1. $\Sigma Q_{ww,i,y}$ – Forecasted volume of wastewater treated in project wastewater treatment system i in year y (m³)

The validation team confirmed that the forecasted volume of wastewater treated by PAI1 is derived from a 3-year average of POME generated by the Tian Siang Palm Oil Mill. The annual average is 133,257 m³. The $\Sigma Q_{ww,y}$ value for 2024 is calculated based on 252 days.

2. $COD_{inflow,i,y}$ - Chemical oxygen demand of the wastewater inflow to the baseline treatment system i in year y (tonnes/m³)

The validation team assessed that the number of chemical oxygen demand derived from an average COD value of a 10-days campaign which took place at the inlet point to the anaerobic lagoon pond (Point A – Raw POME), with an average value of 0.0657 tCOD/m³. Additional sample cross-checks have been conducted to ensure the COD values of the 10-days campaign are aligned with the COD lab result reports/**18** provided by GLT. As this is the initial project and there is no historical data available, PAI1 is using a 10-day ex-ante measurement campaign to determine the parameters.

3. $h_{COD, BL, i}$ - COD removal efficiency of the baseline system i

The validation team assessed that the $h_{COD, BL, i}$ value is derived from the value of COD removed and divided with the average COD inlet to anaerobic lagoon times by model correction factor to account for model uncertainties (0.89) as per guided in the AMS-III.H, Version 19 document. Therefore, the value of $h_{COD, BL, i}$ is confirmed to be 85.0%.

4. $MCF_{ww, treatment, BL, i}$ - Methane correction factor for baseline wastewater treatment systems i (MCF value as per table AMS-III.H.).

The validation team confirmed that the default value of $MCF_{ww, treatment, BL, i}$ used for PAI1's baseline calculation is 0.8 referring to the IPCC default values for Methane Correction Factor, for type of wastewater treatment of "Discharge of wastewater to river" in the AMS-III.H, Version 19.0 document.

5. $B_{0, ww}$ - Methane producing capacity of the wastewater.

The validation team confirmed that the default IPCC value of 0.25 kg CH₄/kg COD is applied for $B_{0, ww}$ for the calculation of PAI1's project emissions.

6. UF_{BL} – Model correction factor to account for model uncertainties.

The validation team confirmed that the default value applied (0.89) for model correction factor to account for model uncertainties is as per guided in the AMS-III.H, Version 19 document.

Therefore, the validation team is assured that the $PE_{flaring, y}$ estimated calculation for PAI1 is in conformance to the AMS-III.H, Version 19.0 guideline document.

Referring to the validation assessment had been conducted for AMS-III.H and AMS-I.D of PAI1 by the validation team, the PAI1's total project emissions value is validated as follows:

Year	PE _y	PE _{ww, discharge, y}	PE _{fugitive, y}	PE _{flaring, y}
24-Apr-2024 – 31-Dec-2024 (252 days)	4,427	291	3,222	914
01-Jan-2025 – 31-Dec-2025	6,412	422	4,667	1,324
01-Jan-2026 – 31-Dec-2026	6,412	422	4,667	1,324
01-Jan-2027 – 31-Dec-2027	6,412	422	4,667	1,324
01-Jan-2028 – 31-Dec-2028	6,412	422	4,667	1,324
01-Jan-2029 – 31-Dec-2029	6,412	422	4,667	1,324

Year	PE _y	PE ww,discharge,y	PE fugitive,y	PE flaring,y
01-Jan-2030 – 31-Dec-2030	6,412	422	4,667	1,324
01-Jan-2031 – 23-Apr-2031 (113 days)	1,985	131	1,445	410
Total	44,886	2,951	32,667	9,268

Project emissions accordance to AMS-I.D, Version 18.0

According to AMS-I.D paragraph 39, the project emissions are 0, and CO₂ emissions from on-site consumption of fossil fuel due to project activity shall be calculated using the latest version of “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion”.

Hence, the validation team is assured that the total estimated PAI1’s project emission with application of AMS-III.H, Version 19.0 and AMS-I.D, Version 18.0 is in conformance with the VCS Standard.

Leakage Emissions Project Instance 1 (PAI1)

No leakage emissions are considered for the project under AMS-III.H, Ver19 since the technology does not utilize equipment transferred from other activities.

Emissions Reduction Project Instance 1 (PAI1)

The estimated emissions reduction for all the PAIs are calculated as per AMS-III.H, Version 19.0, which is tabulated as below:

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
24-Apr-2024 – 31-Dec-2024	27,762	4,427	0	0	23,335
01-Jan-2025 – 31-Dec-2025	41,784	6,412	0	0	35,372
01-Jan-2026 – 31-Dec-2026	41,784	6,412	0	0	35,372
01-Jan-2027 – 31-Dec-2027	41,784	6,412	0	0	35,372

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
01-Jan-2028 – 31-Dec-2028	41,784	6,412	0	0	35,372
01-Jan-2029 – 31-Dec-2029	41,784	6,412	0	0	35,372
01-Jan-2030 – 31-Dec-2030	41,784	6,412	0	0	35,372
01-Jan-2031 – 23-Apr-2031	12,936	1,985	0	0	10,951
Total	291,404	44,886	0	0	246,518

The verification team confirms the following:

- The project description includes all relevant assumptions and data, along with their references and sources.
- The data and parameter values used in the project description are deemed reasonable within the project's context.
- The baseline emissions estimates can be replicated using the provided data and parameter values in the project description.
- The validation team affirms that the methodology and referenced tools have been correctly applied to calculate baseline emissions, project emissions, leakage, and net GHG emission reductions.

3.3.7 Methodology Deviations

Project Instance 1 (PA1)

There is no methodology deviations observed during this validation.

3.3.8 Monitoring Plan

Monitoring Plan Grouped Project

a) Organizational structure and responsibilities

The project owner organises a specific team in assigning responsibility for data collection, supervision, and witness the whole process of data measuring and recording. The organizational structure, responsibilities and competencies of the personnel that will carry out the monitoring and management of the project activities are as follows:

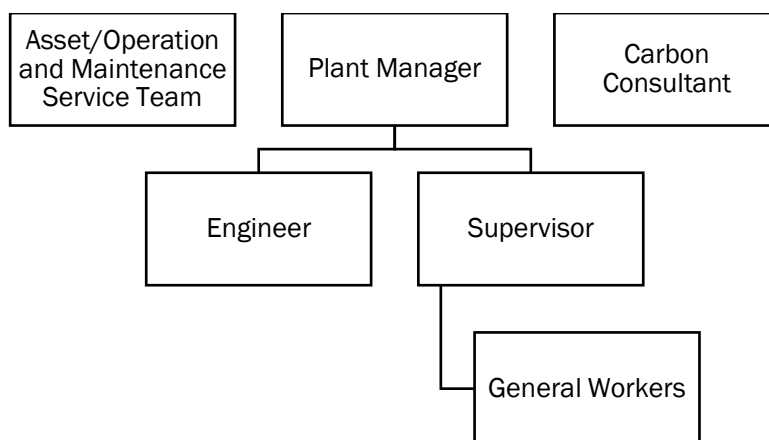


Figure: Grouped PAI VCS Organization Chart

The following table describes the position and overall responsibility of the operation and monitoring team:

Table: Description of roles for PAI VCS team

Personnel	Description of Roles in Monitoring Plan	VVB assessment / conclusion
Asset/Operation and Maintenance Service Team	Overall managing the carbon project	Interviewed with GLT's focal persons and confirmed the roles of the personnel for the monitoring plan.
Plant Manager	- Report and seek approval from the asset/operation and maintenance service team on VCS- related matters - Take full responsibility in the overall monitoring of the project - Conduct internal meetings on VCS matters - Oversees and coordinate the VCS monitoring plan data gathering - Monitor & ensure that data QA/QC is carried out by appointing (internal) independent verifiers to do cross-checking on irregular basis - Ensure all documentations were prepared accordingly based on VCS-PDD/1 requirement. - Coordinate with VCS consultant for matters (data/documents/records etc) that require rectification from plant - Point of contact for the mill, to respond to grievance from external parties.	Interviewed with GLT's focal persons and confirmed the roles of the personnel for the monitoring plan as per stated in the PDD/1.

Personnel	Description of Roles in Monitoring Plan	VVB assessment / conclusion
Engineer	- Assists the VCS Manager in performing the VCS monitoring works - Assisting in the daily and monthly monitoring records for all the VCS related equipment. - Prepare monthly summary record of the operations of plants. - Ensure maintenance and calibration is carried out for proper running of machinery and equipment	Interviewed with GLT's focal persons and confirmed the roles of the personnel for the monitoring plan as per stated in the PDD/1.
Supervisors	- Report to the Engineer on any VCS monitoring issues - Conduct daily checking of data that have been identified for collection. - Ensuring on the functionality of the data recording devices and calibrated as planned (including performing QA/QC) - Responsible with the security of locked Programmable Logic Controller (PLC) control room.	Interviewed with GLT's focal persons and confirmed the roles of the personnel for the monitoring plan as per stated in the PDD/1.
General Workers	- Perform regular operational and maintenance tasks. - Conduct the daily recordings in daily monitoring log sheets and request verification from the supervisors on the log sheets. - Report any fault to supervisor-in-charge when necessary	Interviewed with GLT's focal persons and confirmed the roles of the personnel for the monitoring plan as per stated in the PDD.
VCS Consultant	- Provides advice on all VCS - related matters - Prepare monitoring reports for verification audit. - Liaises with verifier on verification processes - Conducts data audits on VCS monitoring	Interviewed with GLT's focal persons and confirmed the roles of the personnel for the monitoring plan as per stated in the PDD/1.

b) Data and parameters to be monitored

GLT has identified all the monitoring parameters required for the grouped projects, which must be monitored according to the approved methodology AMS-III.H, version 19.0, and AMS-I.D,

version 18.0. This has been clearly described in section 5.3 of the VCS PDD/**1**. These parameters are essential for monitoring baseline emissions, project emissions, and emission reductions during the crediting period of the grouped project and any subsequent project activity instances (PAIs) included in the grouped project during the specified crediting period. The monitoring plan or procedure followed to measure the emission reduction is applied accurately and with a conservative approach.

Data and parameters in accordance to AMS-III.H, version 19.0

The parameters to be monitored in the monitoring plan:

Data to be monitored	Description	VVB Assessment
$Q_{ww,i,y}$	Volume of wastewater treated in the project wastewater system in year y (m^3 /month)	Checked and verified with AMS-III.H, Version 19.0 document.
$COD_{ww,untreated,PJ,y}$ or $COD_{inflow,PJ,y}$	Chemical oxygen demand of the wastewater entering the biogas treatment system in year y (tonnes/ m^3)	Checked and verified with AMS-III.H, Version 19.0 document.
$COD_{ww,treated,y}$ or $COD_{ww,outflow,y}$	Chemical oxygen demand of the wastewater after the treatment system affected by the project activity (tonnes/ m^3)	Checked and verified with AMS-III.H, Version 19.0 document.
$COD_{ww,discharge,y}$	Chemical oxygen demand of the treated wastewater discharged into sea, river or lake (tonnes/ m^3)	Checked and verified with AMS-III.H, Version 19.0 document.
$S_{final,PJ,y}$	Method of disposal of the sludge (t)	Checked and verified with AMS-III.H, Version 19.0 document.
$BG_{burnt,y}$	Biogas volume in year y (m^3)	Checked and verified with AMS-III.H, Version 19.0 document.
$BG_{fuelled,y}$	Amount of biogas fuelled in year y (m^3)	Checked and verified with AMS-III.H, Version 19.0 document.
$BG_{flared,y}$	Amount of biogas flared in year y (m^3)	Checked and verified with AMS-III.H, Version 19.0 document.
$W_{CH4,y}$	Methane content in biogas in the year y (%)	Checked and verified with AMS-III.H, Version 19.0 document.
T	Temperature of the biogas at the flow measurement site ($^{\circ}C$)	Checked and verified with AMS-III.H, Version 19.0 document.

Data to be monitored	Description	VVB Assessment
P	Pressure of the biogas at the flow measurement site (Pa)	Checked and verified with AMS-III.H, Version 19.0 document.
Flame _m	Flame detection of flare in the minute m (flame on and off)	Checked and verified with AMS-III.H, Version 19.0 document.
T _{EG,m}	Temperature in the exhaust gas of the enclosed flare in minute m (°C)	Checked and verified with AMS-III.H, Version 19.0 document.
EC _{PJ,j,y}	Electricity consumption in the project scenario in the year y (MWh/year)	Checked and verified with AMS-III.H, Version 19.0 document.

Data and parameters in accordance to AMS-I.D, version 18.0

The parameters to be monitored in the monitoring plan:

Data to be monitored	Description	VVB Assessment
EC _{PJ, facility,j,y}	Quantity of net electricity generation supplied by the project to the grid in year y (MWh)	Checked and verified with AMS-I.D, Version 18.0 document.
EG _{PJ,add,y}	Quantity of net electricity generation supplied to the grid in year y by the project plant/unit that has been added under the project activity	Checked and verified with AMS-I.D, Version 18.0 document.

Data and parameters in accordance to TOOL06: Project emissions from flaring

The parameters to be monitored in the monitoring plan:

Data to be monitored	Description	VVB Assessment
F _{CH4,EG,t}	Mass flow of methane in the exhaust gas of the flare on a dry basis at reference conditions in the time period t (kg)	Checked and verified with TOOL06 document.
T _{EG,m}	Temperature in the exhaust gas of the enclosed flare in the minute m (°C)	Checked and verified with TOOL06 document.

Data to be monitored	Description	VVB Assessment
$V_{i,RG,m}$	Volumetric fraction of component i in the residual gas on a dry basis in the minute m where $i = CH_4, CO, CO_2, O_2, H_2, H_2S, NH_4, N_2$	Checked and verified with TOOL06 document.
$V_{RG,m}$	Volumetric flow of the residual gas on a dry basis at reference conditions in the minute m (m^3)	Checked and verified with TOOL06 document.
$M_{RG,m}$	Mass flow of the residual gas on a dry basis at reference conditions in the minute m (kg)	Checked and verified with TOOL06 document.
$V_{O_2,EG,m}$	Volumetric fraction of O_2 in the exhaust gas on a dry basis at reference conditions in the minute m	Checked and verified with TOOL06 document.
$f_{CH_4,EG,m}$	Concentration of methane in the exhaust gas of the flare on a dry basis at reference conditions in the minute m (mg/m^3)	Checked and verified with TOOL06 document.
$Flame_m$	Flame detection of flare in the minute m (on or off)	Checked and verified with TOOL06 document.
$Maintenance_y$	Maintenance events completed in year y (calendar dates)	Checked and verified with TOOL06 document.

Data and parameters in accordance to TOOL05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation

The parameters to be monitored in the monitoring plan includes:

Data to be monitored	Description	VVB Assessment
$EF_{grid,CM,y}$	Grid Emission Factor for Peninsula Malaysia, Sabah, and Sarawak (tCO_2/MWh)	Checked and verified with TOOL05 document.
$TDL_{j,y}$ and $TDL_{k,y}$ and $TDL_{l,y}$	Average technical transmission and distribution losses for providing electricity to source j , k or l in year y	Checked and verified with TOOL05 document.
$FC_{n,i,t}$	Quantity of fossil fuel type i fired in the captive power plant n in the time period t (m^3 , ton or l)	Checked and verified with TOOL05 document.

Data to be monitored	Description	VVB Assessment
$EG_{n,t}$	Quantity of electricity generated in captive power plant n in the time period t (MWh)	Checked and verified with TOOL05 document.
$NCV_{i,t}$	Average net calorific value of fossil fuel type i used in the period t (GJ / mass or volume unit)	Checked and verified with TOOL05 document.
$EF_{CO_2,i,t}$	CO2 emission factor of fossil fuel type i used in the period t (t CO ₂ / GJ)	Checked and verified with TOOL05 document.
$EC_{PJ,j,y}$	Quantity of electricity consumed by the project electricity consumption source j in year y	Checked and verified with TOOL05 document.
$EC_{BL,k,y}$	Quantity of electricity that would be consumed by the baseline electricity consumption source k in year y (MWh/yr)	Checked and verified with TOOL05 document.
$EG_{PJ,grid,y}$ or $EG_{PJ,facility,l,y}$	Quantity of electricity generated and supplied by the project power plant to the grid in year y. Quantity of electricity generated and supplied by the project power plant to the consumers/electricity consuming facility i in year y (MWh/yr).	Checked and verified with TOOL05 document.

Data and parameters in accordance to Tool 03: Tool to calculate project or leakage CO2 emissions from fossil fuel combustion:

Data to be monitored	Description	VVB Assessment
$FC_{i,j,y}$	Quantity of fuel type i combusted in process j during the year y (ton/yr or m ³ /yr)	Checked and verified with TOOL03 document.
$WC_{i,y}$	Weighted average mass fraction of carbon in fuel type i in year y (tC/mass unit of the fuel)	Checked and verified with TOOL03 document.
$\rho_{i,y}$	Weighted average density of fuel type i in year y (Mass unit/volume unit)	Checked and verified with TOOL03 document.

Data to be monitored	Description	VVB Assessment
$NCV_{i,y}$	Weighted average net calorific value of fuel type i in year y (GJ/m ³ , GJ/ton)	Checked and verified with TOOL03 document.
$EF_{CO_2,i,y}$	CO ₂ emission factor of fossil fuel type i used in the period t (t CO ₂ / GJ)	Checked and verified with TOOL03 document.

c) Training of monitoring personnel

Description	VVB Assessment
All plant staff will be trained by Green Lagoon Technology Sdn Bhd, prior to the commissioning of the plants. The involved personnel shall receive adequate and regular training on the project performances and the monitoring plan involving methane capture up to utilization of the biogas. Training of the running of day-to-day operations is under the responsibility of the individual PAIs. This include collection of data and ensuring data integrity is not tolerated. A VCS monitoring Manual will be drafted and brief training will be provided by the appointed VCS consultant on the monitoring parameters.	The training of monitoring personnel plan has been confirmed by interviews with the focal persons of GLT during the site visit. During the monitoring phase of the grouped project, the implementation of the training plan/5 will be verified by the verification team per project activity instance basis.

d) Procedures for handling non-conformances with the validated monitoring plan

In case of a temporary situations such as the installed flow meter(s)/gas analyser malfunctioned or gave unrepresentative results, the data shall be measured according to the method listed below:

Description	VVB Assessment
1. Manually with a back-up flow meter/gas analyser. The number of samples needed for the calculation will be calculated by using raosoft calculator, assuming the	The procedures for handling non-conformances plan have been confirmed by interviews with the focal persons of GLT during the site visit. During the monitoring

95% confidence interval, 10% margin of error. The lower bound of the 95% confidence interval will be applied in the calculation sheet as a conservative approach.	phase of the grouped project, the implementation of the procedure will be verified by the verification team per project basis
2. Alternatively, if a back-up meter is not available, the historical records and future data (at least one (1) month before and after the malfunction period) will be used to estimate the quantity. The 95% confidence interval will be calculated from the data obtained. The lower bound of the 95% confidence interval will be applied in the calculation sheet as a conservative approach.	
3. In relation to malfunctions of measurement of COD, the following will be observed. In case of temporary situation such as the COD sampling device/meter malfunctioned, the amount of COD shall be based on the most recent testing result.	

e) Sampling approaches and precision level

Description	VVB Assessment
For AMS-III.H related PAIs, all data that are collected in accordance with the VCS-PDD monitoring frequency and data collection, are included and considered in the emission reduction calculation.	Checked and verified with the AMS-III.H document reference and the monitoring plan reported under section 5.1, 5.2 and 5.3 of the PDD/1.
In relation to the measurement of COD: According to Raosoft the total number of months in a year is 12 months and the total number of samples needed to be collected is	Checked and verified Raosoft Calculator the sample size calculator. 90% of confident level to be applied by GLT for the COD measurement also aligned with EB 86 document "Sampling and surveys for CDM

11 samples to meet the 90% confidence interval.	project activities and programme of activities'
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f) Quality Assurance and Quality Control & maintenance procedures

Description	VVB Assessment
Maintenance of the equipment shall be conducted regularly to ensure a good and proper functioning of the equipment. Calibration of monitoring equipment and instruments will be done periodically according to the standard and procedures to maintain the accuracy of the measuring instrument Emergency/Contingency preparedness for monitoring system is presented as below: 1. Data collection If the person-in-charge for daily raw data logging is unavailable, there will be another back-up person to take over the job. 2. Data storage, transfer and backup The monitored parameters that are collected and will be stored in the following ways: (a) Hard copy - Log sheets/ logbooks will be kept in proper manner, and available for audit verification purpose, (b) Soft copy - Raw data from log sheets/ logbooks will be stored as soft copy; together with monthly CER calculation, the entire data of the project activity will be back-up to an external hard drive or another computer every quarterly.	The quality and maintenance procedures has been confirmed by interviews with the focal persons of GLT during the site visit. During the monitoring phase of the grouped project, the implementation of the procedure will be verified by the verification team per project basis. Despite that, with the capacity and resources owned by GLT, the validation team has ascertained that GLT is capable to implement the quality assurance and quality control as specified in the PDD/1.

Immediate action shall be taken immediately if any matters/ uncertainties affecting the data quality. In case of leakage occurrences in any parts of the biogas plant system, the team shall take immediate action to put the plant operation back on track. The plant operation manual shall include the procedure on safety and always made available on site to ensure the operators are fully aware of emergency procedures.	
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Monitoring Plan Project Instance 1 (PAI1)

The monitoring plan for this PAI1 includes all monitoring requirements following the procedures set by AMS-III.H (version 19.0) and AMS-I.D (version 18.0). The overall monitoring plan can be illustrated in the figure below:

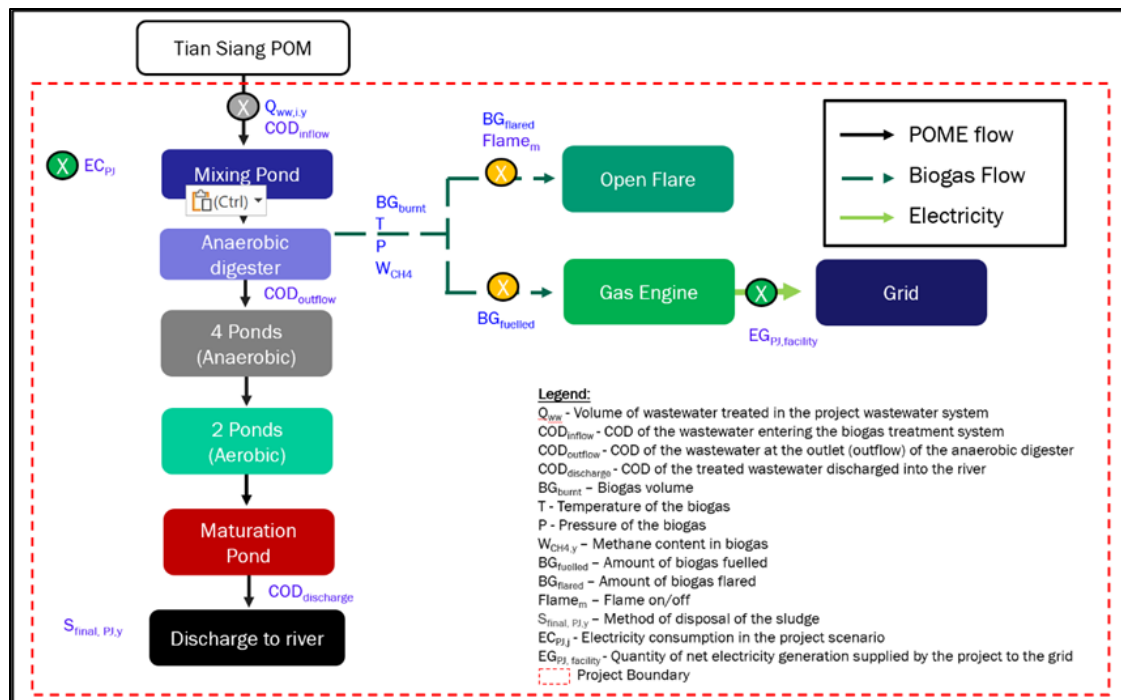


Figure: Monitoring Plan for PAI1

Implementation of the Monitoring Plan

The biogas plant operator will adopt the instructions presented in the monitoring plan and ensure all activities related to the biogas plant operation and monitoring are implemented. The main responsibilities of monitoring include:

Description	VVB Assessment
Data reading and handling: maintaining an adequate system for collecting, recording and storing data in accordance with the protocols determined by the monitoring plan as well as checking data quality, collection and record-keeping procedures on a regular basis. This also includes making sure that all relevant electronic data is backed-up electronically for at least 2 years after the end of the crediting period in case of an unanticipated loss of data.	The monitoring plan for PAI1 has been reviewed under section 5.3 of VCS PDD/1 and confirmed during the interviews with the focal person of GLT during the site visit. However, during the monitoring phase of the PAI1, the implementation of the plan need to be verified during verification assessment. Despite that, with the capacity and resources owned by GLT, the validation team has ascertained that GLT is capable to implement the monitoring plan as specified in the PDD/1.
In the case of erroneous data recorded for the monitored parameter occurred during short period, data for the affected period shall be derived based on the approach based on 95% confidence interval principles (source: “IPCC Good Practice Guidance and Uncertainty Management in National Greenhouse Gas Inventories”, page 6.6). A total of 10 readings before and after the constant period will be used to calculate the lower bound of 95% confidence interval with reference to the above-mentioned guideline. The lower bound of the interval boundaries calculated will be applied to the period for the constant data as a conservative approach. Alternately, if the 10 readings before and after the erroneous period is unavailable, the record from the daily manual log-sheet will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the period for the constant data as a conservative approach.	
Reporting: preparing periodic summaries and reports that include emission reductions	

generated and observations regarding monitoring plan procedures.	
Training: assuring assigned personnel received adequate and regular training on project performances and the monitoring plan. Any new staff taking over the role of monitoring and reporting will be fully trained as well; and	
Quality control and quality assurance: complying with quality control and quality assurance procedures to facilitate periodical audits and verification. Any observation of matters affecting data quality will be reported to the management immediately.	

Responsibility of Monitoring Plan

A dedicated VCS monitoring team will be assigned to be responsible for undertaking the monitoring as well as the quality control and assurance tasks. The following are the position and responsibility of the VCS monitoring team:

Table: Description of roles for PAI1 VCS team

Personnel	Description of Roles in Monitoring Plan	VVB Assessment
Plant Manager	- Report and seek approval from the higher management on VCS- related matters - Take full responsibility in the overall monitoring of the project - Conduct internal meetings on VCS matters - Oversees and coordinate the VCS monitoring plan data gathering - Monitor & ensure that data QA/QC is carried out by appointing (internal) independent verifiers to do cross-checking on irregular basis - Ensure all documentations were prepared accordingly	Interviewed with GLT's focal persons and confirmed the roles of the personnels for the monitoring plan of PAI1.

Personnel	Description of Roles in Monitoring Plan	VVB Assessment
	based on VCS-PDD requirement. Coordinate with VCS consultant for matters (data/documents/records etc) that require rectification from plant	
Engineer	- Assists the VCS Manager in performing the VCS monitoring works - Assisting in the daily and monthly monitoring records for all the VCS related equipment. - Prepare monthly summary record of the operations of plants. - Ensure maintenance and calibration is carried out for proper running of machinery and equipment	
Supervisors	- Report to the Engineer on any VCS monitoring issues - Conduct daily checking of data that have been identified for collection. - Ensuring on the functionality of the data recording devices and calibrated as planned (including performing QA/QC) - Responsible with the security of locked Programmable Logic Controller (PLC) control room.	
General Workers	- Perform regular operational and maintenance tasks. - Conduct the daily recordings in daily monitoring log sheets and request verification from the supervisors on the log sheets. - Report any fault to supervisor-in-charge when necessary	

Personnel	Description of Roles in Monitoring Plan	VVB Assessment
VCS Consultant	- Provides advice on all VCS - related matters - Prepare monitoring reports for verification audit. - Liaises with verifier on verification processes - Conducts data audits on VCS monitoring	

Quality Control, and Quality Assurance Procedures

Quality control and quality assurance procedures will be carried out on all parameters monitored. The practices to be undertaken in the context of the proposed project are as follows:

Description	VVB Assessment
1. Monitoring records: - Daily readings of all meters will be filled out in paper worksheets and filed consequently. All data collected will also be entered in electronic worksheets and stored in a computer (immediately) and on discs (periodically). - Periodic controls of the monitoring records will be carried out to check any deviation from the monitoring plan.	GLT has established a 'Monitoring Plan' procedure, version 1.0 on April 2024. Observed the monitoring plan has explained the following: 3. Organization Structure for Monitoring 4. Training Plan/5 5. Description of Monitoring Plan 6. Data To Be Monitored 7. Data Recording and Documentation 8. Quality Assurance and Quality Control 9. Data Processing and Reporting
2. Equipment calibration and maintenance: Flow meters, gas analyzers and other sensors will be subject to regular maintenance and testing according to the technical specifications by the manufacturers to ensure accuracy and good performance.	Also, a sample of the monitoring plan obtained from VCS template (AMS-I.D, Version 18.0) and "Daily Checksheet" /17 are available in the procedure as a reference for monitoring process.

Calibration of equipment will be conducted periodically according to technical specifications.	Further review the above procedure under section 8, Quality Assurance and Quality Control has specified the following: - Equipment/Instrument calibration and maintenance - Internal Audit and Corrective Actions - Management Review - Data Security Site interview was conducted with GLT's focal person and found, they are aware and able to demonstrate their knowledge on the monitoring plan. Interviewed with GLT's focal persons and confirmed the Quality procedures of the monitoring plan of PAI1 by checking the procedural instructions, templates and calibration list and related documents of PAI1's monitoring plan.
3. Corrective actions: - Actions to correct deviations from the monitoring plan will be implemented as these deviations are observed either by the operator or during internal audits. - Apart from periodic meetings, additional technical meetings will be held in order to define any necessary corrective actions to be undertaken.	
4. Site audits: Regular site audits will be conducted to ensure that monitoring and operational procedures are being observed in accordance with the monitoring plan.	
5. Training: A training plan will be set out for the operator. This will ensure that monitoring staff involved in the project are properly trained so that they are able to undertake the tasks required by this monitoring plan. Appropriate staff training must be provided before the project starts operating and generating emission reductions.	
6. Storage of documents: List of monitoring equipment (e.g. flow meters, gas analyzers, thermometers) including their model numbers, names, manufacturers, specifications, user requirements, etc.	

• Calibration lists and reports, including equipment or parts calibrated, date, method and procedures of calibration, precision following these processes, personnel, devices needed, etc. • Maintenance lists and reports, including equipment or parts maintained, date, method and procedure of maintenance, performance following these procedures, personnel, devices needed, etc. • Operation manual of the proposed project. • Registration of minutes during meetings. • Non-conformance reports. • Monthly and annual worksheets. • Training plan. • Internal audit reports, including personnel, time, findings, corrective actions, follow-up inspections, etc. • Annual monitoring review	
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3.4 Non-Permanence Risk Analysis

This is not applicable to the Project as the Project is not an AFOLU project.

4 VALIDATION OPINION

4.1 Validation Summary

Control Union Certifications Germany GmbH (hereafter referred to as “CUCG”) has been contracted by Green Lagoon Technology Sdn Bhd to conduct the validation of the project – “Green Lagoon Technology Sdn Bhd”, VCS ID 4887 regarding the relevant requirements of VCS programme guidelines and standard:

- VCS Standard, v4.7
- ISO 14065
- ISO 14064-3
- VCS Program Definitions, v4.5
- VCS Program Guide, v4.4
- Project Description, v4.4
- Project Methodologies Documents (AMS-III.H & AMS-I.D)

The purpose of this validation is to have an independent third-party assessment of whether the project activity conforms to the qualification criteria set out in the VCS Standard, version v4.7 to attain real, measurable, additional, and permanent emission reductions.

The management of the project proponent/owner is responsible for the preparation of the GHG emissions data and the reported/estimated GHG emissions reductions on the basis set out within the project’s Monitoring Plan in the VCS Project Description and the approved VCS Project Methodology: AMS-III.H (Version 19.0) - Methane recovery in wastewater treatment and AMS-I.D (Version 18.0) – Grid-connected renewable electricity generation.

The validation assessment, "Wastewater to Biogas Group," was conducted at PAI 1, Tian Siang Oil Mill (Triang), located in Triang, Pahang, on May 28th and 29th, 2024. The assessment activities included a documentation review, interviews, and a site visit to the project site. A sample of internal and external stakeholders was interviewed during the assessment.

The validation also included an assessment of the data and documents provided to CUCG. The scope of the validation was conducted to establish and confirm the following:

- The project activity is in accordance with rules, applicable law and legislation of the host country (Malaysia).
- The project activity is in accordance with all relevant VCS rules and requirements, which includes the following: VCS Standard, v4.7, VCS Program Definitions, v4.5, VCS Program Guide, v4.4, Project Description, v4.4, Project Methodologies Documents, VCS Validation and Verification Manual, v3.2, including but not limited to additionality, proof of ownership and compliance with local laws and regulations.
- The project activity is in accordance with the conditions of the latest version of applied VCS Project Methodology: AMS-III.H (Version 19.0) - Methane recovery in wastewater treatment and AMS-I.D (Version 18.0) – Grid-connected renewable electricity generation.

The validation is not intended to provide consulting services to the project participants. However, requests for clarifications and/or corrective actions may provide input for improving the project design.

The validation report is based on the VCS Program Guide and VCS Validation and Verification Manual, supporting evidence made available to the validator and information collected through performing interviews and during the on-site audit assessment.

A risk-based approach has been followed to perform this validation. During the validation 16 Corrective Action Requests (CARs), 2 Clarification requests (CLs), 1 Forward Action Requests (FARs) and 1 Observation (OBS) were raised and successfully closed. No limitations or doubts were identified related to the validation.

Project Activity Instance 1 (PAI1) Description

The first Project Activity Instance (PAI1) is located at Tian Siang Oil Mill (Triang) Sdn Bhd (“Tian Siang POM”) in the State of Pahang, Malaysia, about 6 KM Off Jalan Triang Ke Kemayang, Triang, Pahang and 180km north of Kuala Lumpur. GLT Maju Sdn Bhd, owned by Green Lagoon Technology Sdn Bhd, is the owner of the biogas plant PAI1. Green Lagoon Technology Sdn Bhd has an agreement/4 with GLT Maju Sdn Bhd for the legal rights to operate PAI1. Additionally, the mill operator, Tian Siang POM, has an agreement with GLT Maju Sdn Bhd for ongoing access to palm oil mill effluent (POME) and permission to construct and operate PAI1.

4.2 Validation Conclusion

Control Union Certification Germany GmbH (CUCG) has performed a validation of the “Wastewater to Biogas Grouped Project” by Green Lagoon Technology Sdn Bhd for a project activity in the host country Malaysia. The validation was performed based on the VCS Standard, CDM methodologies documents, latest version of ISO 14064-3 and host country criteria and also on the criteria given to provide for consistent project operations, monitoring and reporting.

The validation consisted of the following three phases:

- i. a desk review of the project design and the baseline and monitoring plan;
- ii. project site visit, on-site validation and follow-up interviews with project stakeholders;
- iii. the resolution of outstanding issues and the issuance of the final validation report and opinion.

In the course of the validation, 16 Corrective Action Requests (CARs), 2 Clarification Requests (CLs), 1 Forward Action Request (FAR), and 1 Observation (OBS) were identified concerning project details, safeguards and stakeholder engagement, application of the methodology, quantification of estimated GHG emission reductions and removals, and monitoring of the VCS project activity. All CARs and CLs were successfully addressed after necessary corrections/clarifications by the client. More details are discussed in Appendix IV of this validation report.

CUCG's analysis has shown that the project is not a likely baseline scenario. The Project participant used the latest tool for the demonstration of the additionality. In line with this tool, the PDD/1 provides clear information on how the project meets the positive list of technologies identified in this tool to demonstrate additionality for the grouped project as well as for PAI1. Emission reductions attributable to the project are hence additional to any that would occur in the absence of the project activity. The monitoring plan has been assessed to be transparent and adequate. It is CUCG's opinion that the Project Proponent is able to implement the monitoring plan.

In summary, it is CUCG's opinion that the project correctly applies the baseline and monitoring methodology AMS-III.H Version 21.0 "Methane Recovery in Wastewater Treatment" and AMS-I.D Version 18.0 "Grid Connected Renewable Electricity Generation" and meets all the relevant VCS Standard Version 4.7, relevant methodology, tools and guidelines, UNFCCC requirements and the relevant host country criteria. PAI1 is estimated to achieve an annual average GHG emission reduction of 35,217 tCO₂e/year, totalling 246,518 tCO₂e over the first 7-year crediting period (24-April-2024 to 23-April-2031). Given that the project is implemented and maintained as designed, the project is likely to achieve the estimated total amount of emission reductions. The project results in reductions of CO₂ emissions that are real, measurable and give long-term benefits to the mitigation of climate change.

Therefore, CUCG requests the registration and issuance of the project as VCS Project.

Crediting period: From 24-April-2024 to 23-April-2031

Validated estimated GHG emission reductions and carbon dioxide removals for the project crediting period:

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
24-Apr-2024 – 31-Dec-2024	27,762	4,427	0	23,335	0	23,335
01-Jan-2025 – 31-Dec-2025	41,784	6,412	0	35,372	0	35,372
01-Jan-2026 – 31-Dec-2026	41,784	6,412	0	35,372	0	35,372
01-Jan-2027 – 31-Dec-2027	41,784	6,412	0	35,372	0	35,372
01-Jan-2028 – 31-Dec-2028	41,784	6,412	0	35,372	0	35,372
01-Jan-2029 – 31-Dec-2029	41,784	6,412	0	35,372	0	35,372

01-Jan-2030 - 31-Dec-2030	41,784	6,412	0	35,372	0	35,372
01-Jan-2031 - 23-Apr-2031	12,936	1,985	0	10,951	0	10,951
Total	291,404	44,886	0	246,518	0	246,518

APPENDIX I: COMMERCIALLY SENSITIVE INFORMATION

Use the table below to describe the commercially sensitive information included in the project description to be excluded in the public version.

Section	Information	Justification	Assessment conclusion	Method	and
Not Applicable					

APPENDIX II: REFERENCED DOCUMENTS

Reference numbers	Referenced Documents
/1	(a) VCS Green Lagoon PD Version 1.0_201223 (b) VCS Green Lagoon PD Version 2.0_200624 (c) VCS Green Lagoon PD Version 3.0_120724
/2	Project ID 4887 - WASTEWATER TO BIOGAS GROUPED PROJECT - VCS - Pipeline listing requested (under validation)
/3	Documentation demonstrating the existence and legality of the project activity: (a) GLTM-FIAH Certificate (b) GLTM – Boot Agreement 4 Jun 2021 (c) GLTM Lease Agreement 4 Jun 2021 (d) GLTM – Addendum BOOT Agreement
/4	Quality Management Manual, Organization Charts, quality management procedures and related documents: GLTM – GL_ADMIN- Organization Chart
/5	Training Plan, Training Material and record of training - GLTM PAI1 monitoring training
/6	- GLTM – List of employees
/7	(a) Workers Health and Safety Procedures (b) Emergency Response Plan Procedure – GLT Biogas Plant (c) Records of training procedures – Safety Induction - GLT Maju Attendance
/8	Incident Investigation and Reporting Form
/9	GLT Employee Policy – Equal Employment Opportunity
/10	GLT Environmental Policy
/11	KML file with geodetic coordinates of all project activity locations Wastewater to Biogas Grouped Project - Malaysia Boundary
/12	Standard Operating Procedure (SOP) document(s) for the project activity operations GLTM OM - 01 Routine Operations 240422 R1
/13	(a) Photos of facility & equipment - Photos of Monitoring Instrument
/14	Stakeholder engagement procedure and records of consultation/engagement of (local) stakeholders - Stakeholder Consultation Report - Stakeholder Consultation Report
/15	GLTSB Customer Complaint Procedure

/16	GLT_VCS Monitoring Manual 1.0_050424
/17	a) GLTTP17F01-F04 Daily Checksheet_Maju b) SCADA System – Tian Siang
/18	Daily 10 days campaign - COD Lab Result from 05/07/2023 to 25/07/20223 (a) COD_050723 (b) COD_060723 (c) COD_070723 (d) COD_110723 (e) COD_120723 (f) COD_150723 (g) COD_180723 (h) COD_210723 (i) COD_220723 (j) COD_250723
/19	DB2304-TST-C-TST-002 - Digester Commissioning
/20	Tian Siang POME 2020 – 2023 record
/21	DOE license indicating mill capacity & method of effluent discharge on-site 1) KB Kerja Membina 1 Unit Loji Biogas Tian Siang Triang 2) Engineering Design for Rectification of the existing palm oil mill effluent treatment system
/22	COD sampling location
/23	Green Lagoon's Business Licence a) GLT - Form 9 b) GLTM - Section 17
/24	GLTDB - CIDB Certificate valid 14042025
/25	SIA Report TSOMT 2023
/26	FFB 2018 – 2023 records
/27	Workplan EIA TSOMT
/28	Consultant application Letter for land title change
/29	DOE confirmation letter for installation of flaring equipment
/30	DOE confirmation letter for installation of gas engine
/31	Tian Siang POM Business Registration with SSM 2. 37(a) 504643-A_9_02022000_CTC - PENDAFTARAN SYARIKAT – PMSB 3. 37(b) LICENSE (MAJLIS DAERAH)
/32	Tian Siang POM sludge processing flow
/33	Genset Technical Description - Appendix 2 - TD_JGS416_B25_415V
/34	Copy of GLTM employee offer letter

/35	Hydroseeding Layout
/36	Stakeholder meeting invitation letter 1. 41(a) - Stakeholder Consultation Invitation Letter to DOE 2. 41(b) - External Stakeholder Meeting Invitation to DOE
/37	Anti-Sexual Harassment Act 2022 (Act 840)
/38	Anti-Sexual Harassment Poster 1. 43(a)- Sexual Harassment poster (GLT) 2. 43(b)- Anti- Sexual Harassment poster on site
/39	GLT reply to public comment ID4887
/40	Mesyuarat Koordinasi Hentitugas IOM and IOD Projek FIT Biogas-GLT Maju Sdn Bhd 1MW
/41	GLT Visitors SOP 46.GLT-SHE-11-GP-L2-003_Visitor_Procedure
/42	a) Tian Siang POM desludging records b) Tian Siang Ponds Installation Date
/43	Average power uploaded to the grid (Seda requirement)
/44	Profil Daerah Pahang 2018
/45	Ownership Transfer Agreement - GLT2304-MCH_GLTM-Ownership Transfer Agreement_R0
/46	Effluent Plan approved by DOE

APPENDIX III: ABBREVIATIONS

Abbreviations	Full Text
BE	Baseline Emission
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CO ₂	Carbon Dioxide
CL	Clarification Request
CP	Crediting Period
CUCG	Control Union Certification Germany GmbH
DR	Desk Review
EB	Executive Board
EF	Emission Factor
FAR	Forward Action Request
GHG	Green House Gas
GLT	Green Lagoon Technonolog Sdn Bhd
MWh	Mega Watt per Hour
PDD	Project Description Document
PP	Project Proponent
tCO _{2e}	Tonnes of Carbon Dioxide equivalent
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Verra Carbon Standard
VCU	Verra Carbon Units
VV	Validation and Verification
VVB	Validation and Verification Body

APPENDIX IV: CLARIFICATION (CL), CORRECTIVE ACTIONS (CAR) & FORWARDS ACTION REQUESTS (FAR).

Table 1: Clarification Requests (CL)

CL ID	CL-01	Reference to Section:	VCS PDD Template V4.4	Date: 29.05.2024
Description of CL				
The following are stated in the section 2.4 of PDD:				
	Risk Identified	Mitigation or preventive measure taken		
Impacts on biodiversity and ecosystems	No risk identified	No formal impact on biodiversity due to the sitting of the PA1 which is located on the existing mill. No new areas were explored.		
Soil degradation and soil erosion	No risk identified	No formal impact on the soil applicable to the project.		
From the PDD, observed that multiple instances are specified with ‘No formal impact’ and ‘no risk identified’. However, GLT needs to elucidate more why there is ‘no formal impact’ affect the element of Ecosystem Health as required by the ‘VCS-Project-Description-Template-v4.4-FINAL2’.				
Project Proponent response				Date: 05.06.2024
The risk identified has been changed to’ Low Risk Identified’ based on the interviews done by GLT. The mitigation or preventive measures taken have been amended in the revised VCS PDD, version 2.0 section 2.4.				
Impacts on biodiversity and ecosystems: There will be low risk on the impacts on biodiversity and ecosystems since the sitting of the PA1 is located on the existing mill, i.e. has been developed many years ago. Furthermore, there were no new areas explored for PAI1. Based on interviews with the mill workers, only snakes are commonly seen. No appearances of rare, threatened, and endangered (RTE) species were observed.				
Soil degradation and soil erosion: There will be low risk on soil degradation and soil erosion since the sitting of the PA1 is located on the existing mill, i.e. has been developed many years ago. Furthermore, there were no new areas explored for PAI1. Hydroseeding has been implemented at erosion-prone areas within the PAI1 boundary. The hydroseeding layout is included in the Engineering Drawings which are attached in 40. Hydroseeding Layout Engineering Drawing				
Documents submitted by Project Proponent				
40. Hydroseeding Layout Engineering Drawing				
VVB assessment				Date: 25.06.2024
Based on the revised PDD, version 2.0 under section 2.4 Ecosystem Health, GLT has revised the type of risk identified for both ‘Impacts on biodiversity and ecosystems’ and ‘Soil degradation and soil erosion’ elements from ‘No risk identified’ to ‘Low risk identified’.				

In addition, GLT has explained the rationale of the type of risk identified as below:

	Risk Identified	Mitigation or preventive measure taken
Impacts on biodiversity and ecosystems	Low	PAI1 is located on the existing mill and there is no new area explored by GLT. On site observation and interview with mill workers found that there is no RTE species identified within the project area and only snakes are commonly seen. GLT will conduct the regular monitoring to identify the existence of RTE species within the PAI1 area.
Soil degradation and soil erosion	Low	PAI1 is located on the existing mill and there is no new area explored by GLT. Hydroseeding/ turfing will be installed at erosion-prone area within the PAI1 area.

Based on the file '40. Hydroseeding Layout' established by GLT Design Build Sdn Bhd on 07.04.2023 found, a potential of soil degradation and erosion area has been identified to install the hydroseeding within the PAI1 area.

Referring to the above, corrective action established is considered appropriate and accepted. Therefore, CL-01 identified during the validation activities is successfully closed.

CL ID	CL-02	Reference to Section:	Indicator 3.16.2	Date:29.05.2024
Description of CL				
Section 5.3(a) 'Organizational structure and responsibilities' specifies that the overall monitoring and management of the project is on the Plant Manager's responsibility. While Table 9 'Description of roles for PAI VCS team' indicates that the role of the Plant Manager is to report and seek approval from the higher management on VCS-related matters. However, there is no clear indication who is the higher management and relation with the project as well as the accountability for monitoring of the project.				
Project Proponent response				Date: 05.06.2024
The higher management is the asset/operation and maintenance service team. The roles and responsibilities of the asset/operation/maintenance service team are included in the revised VCS PDD, version 2.0 section 5.3, i.e. Figure 5 and Table 9.				
Documents submitted by Project Proponent				

Not applicable	
VVB assessment	Date: 25.06.2024
The Asset/ Operation and Maintenance Service Team is identified as the higher management as verified in the revised PDD, version 2.0 under section 5.3(a) 'Organizational structure and responsibilities'. The role of the Asset/ Operation and Maintenance Service Team is having overall responsibility in managing the carbon project as stated in the Table 9 'Description of roles for PAI VCS team'. Plant Manager will report and seek an approval from the higher management on VCS-related matters. Based on the corrective action established, it is considered appropriate and accepted. Therefore, CL-02 identified during the validation activities is successfully closed.	

Table 2: Corrective Action Requests (CAR)

CAR ID	CAR-01	Reference to Section:	VCS PDD Template V4.4	Date: 29.05.2024
Description of CAR				
1). Point 1 under subsection 1.4.1 General eligibility of the PDD stated the following: <i>The proposed grouped project is designed to treat the wastewater by the installation of new biogas digester and avoid methane emission from open anaerobic lagoons during the baseline scenario.</i> As there is substitution of grid emission applied in the project, thus <i>AMS-I.D – Grid connected renewable electricity generation</i> should be specified in the respective section. 2). Indicator 2.3.1, 2.3.2 and 2.3.3 of the PDD have stated the following: 2.3.1 – <i>There are no discrimination or sexual harassment occurred or expected to occur in the project site</i> 2.3.1 – <i>Not applicable as the project activity does not involve any Human trafficking, force labour and child labour in the operation.</i> 2.3.2 – <i>Not applicable as the project activity does not involve any rights of Ips, LCs, and customary rights holders.</i> 2.3.3 – <i>Not Applicable as the project activity does not involve any cultural heritage</i> According to the <i>VCS-Project-Description-Template-v4.4-FINAL2</i> stated that <i>where a section is not applicable, explain why the section is not applicable.</i> Hence, more explanation is required to ensure comply with the PDD template requirement justifying the company's management approach on the particular matters using i.e; policies, procedures, work instructions, training capacity buildings and related evidence including the history of complaints.				

Project Proponent response	Date: 05.06.2024
1) The description for the utilization of the biogas recovered (Upload to the grid, use in transportation applications, bottled and distributed to end users) and the respective greenhouse gas was included under point 1, subsection 1.4.1 of the revised VCS PDD, version 2.0. 2) All the indicators, i.e. 2.3.1, 2.3.2 and 2.3.3 of the VCS PDD are not applicable. The description is revised accordingly to include the justification in the revised VCS PDD, version 2.0 section 2.3.1, 2.3.2 and 2.3.3 which is described as below: <i>2.3.1 – There are no discrimination or sexual harassment occurred or expected to occur in the project site</i> - In the GLT Employee Policy, it has been outlined that there will be no discrimination in employment opportunities. There is also low risk of sexual harassment to occur on site as GLT strictly abide to the Anti-Sexual Harassment Act 2022. General safety reminder (e.g. posters) on sexual harassment can be found on-site. <i>2.3.1 – Not applicable as the project activity does not involve any Human trafficking, force labour and child labour in the operation.</i> - Employment regulations are described in the Malaysia Employment Act 1955. PAI1 will not employ and complicit with victims of human trafficking, forced labour and child labour in their operation. Based on GLT's list of employees, it can be confirmed that the project does not employ any child labour. <i>2.3.2 – Not applicable as the project activity does not involve any rights of Ips, LCs, and customary rights holders.</i> - PAI1 does not involve any Ips, LCs, and customary rights holders. The land of PAI1 is located within the Tian Siang Palm Oil Mill boundary and there is no customary land involved. <i>2.3.3 – Not Applicable as the project activity does not involve any cultural heritage</i> - There is no risk identified as the project activity does not involve any indigenous peoples and also cultural heritage in or within the project area. PAI1 is sited within the Tian Siang Palm Oil Mill boundary.	
Documents submitted by Project Proponent	
1. 42. Anti-Sexual Harassment Act 2022 2. 43. Anti-Sexual Harassment Poster 3. 12. GLT Employee Policy 4. 8. GLT List of Employees with IC	
VVB assessment	Date: 25.06.2024

Section 1.4.1 'General eligibility'

Referring to the revised PDD, version 2.0, section 1.4.1 indicates that AMS-I.D is one of the methodology applied in the project activity and approved by VCS. Sighted under section 1.4.1, GLT has explained that the recovered biogas will be utilized for electricity generation to be uploaded to the grid for transportation applications, bottled and distributed to end users. Hence, the concern related to section 1.4.1 is successfully closed.

Section 2.3.1, 2.3.2 and 2.3.3

Referring to the revised PDD, version 2.0, section 2.3.1, 2.3.2 and 2.3.3, GLT has amended and elucidate each element identified as required by the 'VCS-Project-Description-Template-v4.4-FINAL2':

Section	Risk identified	Mitigation or preventive measure taken
2.3.1	Low	Discrimination There are no discrimination or sexual harassment occurred or expected to occur in the project site. In the GLT Employee Policy, it has been outlined that there will be no discrimination in employment opportunities. Sexual Harassment There are no discrimination or sexual harassment occurred or expected to occur in the project activity. As to date, there is no woman working in the biogas plant. GLT has strictly abide to the Anti-Sexual Harassment Act 2022. General safety reminder (e.g. posters) on sexual harassment can be found on-site.
2.3.1	No risk	Forced Labour GLT comply with the Employment Act 1955 to protect the legitimate rights and interests of workers and prohibit forced labor. Child Labour Employment regulations are described in the Malaysia Employment Act 1955. The Project requires skilled employees to operate, maintain and manage the biogas plant. Therefore, the project does not employ and is not complicit in any form of child labor. Based on GLT's list of employees, it can be confirmed that the project does not employ any child labor. Human Trafficking

		The project activity will not employ victims of human trafficking in their operation. GLT abide with the Employment Act 1955.
2.3.2	Low	This project does not discriminate with regards to participation and inclusion. The project abides by the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. GLT Employee Policy has an outline on equality and discrimination.
2.3.3	No risk	There is no risk identified as the project activity does not involve any indigenous peoples and also cultural heritage in or within the project area. PAI1 is sited within the Tian Siang Palm Oil Mill boundary.

In addition, the following document established by GLT were reviewed and verified to ensure align with the GLT's mitigation plan;

1. 42. Anti-Sexual Harassment Act 2022
2. 43. Anti-Sexual Harassment Poster
 - GLT has established a no discrimination poster with a motto 'See It, Feel It, Report It'
3. 11. GLT Employee Policy-Equal Employment Opportunity (12. GLT ENVIRONMENT POLICY – Wrongly cited)
 - Section 8.0 Equal Employment Opportunity has specified the following:
 - o To provide equal employment and advancement opportunities to all individuals.
 - o The company does not discriminate in employment opportunities or practices based on race, colour, religion, sex, sexual orientation, national origin, age, or any characteristic protected by law.
4. 8. GLT List of Employees with IC
 - Verify the list of employees found, there are 7 male employees employed permanently by GLT. The youngest worker is 20 years old and working as an operator.

In addition, GLT has identified the Employment Act 1955 as one of the laws that applicable to their operation as sighted under section 1.15 of the revised PDD and will comply to the requirement of the law. Informed by the sampled workers during the site interview, there is no element of discrimination, sexual harassment, forced labour, child labour and human trafficking practiced by GLT.

Stated in the section 2.3.1, 'Sexual Harassment', *as to date there is no woman working in the biogas plant. GLT has strictly abide to the Anti-Sexual Harassment Act 2022.* Noted that, section 2,

'Safeguards and Stakeholder Engagement' of the PDD is applicable not only to the PAI1, but for the group level as well. However, during the validation assessment, there are female employees working as a site engineer and sustainability engineer under GLT Sdn Bhd. Moreover, under Section 1.15, Act 840 Anti-Sexual Harassment Act 2022 is not identified as one of the applicable laws to the GLT.

Based on the above, CAR-01 remains open.

Project Proponent response II

Date: 17.07.2024

No woman is working in the biogas plant during the validation assessment. The female employees working as a site engineer and sustainability engineers are under a different company which is not GLT Maju Sdn Bhd (the biogas plant). The Act 840 Anti-Sexual Harassment Act 2022 is included under Section 1.15 in the revised PDD, version 3.0

Documents submitted by Project Proponent II

VCS PDD, version 3.0

VVB Response II

Date: 22.07.2024

Sighted in the revised PDD, version 3.0 under section 1.15 found, the Act 840 Anti-Sexual Harassment Act 2022 is identified as one of the applicable laws to GLT Maju Sdn Bhd. In addition, GLT Maju Sdn Bhd adopted GLT Sdn Bhd's Employee Policy that specify the commitment on zero tolerance to sexual harassment as sighted under section 9.0 Diversity (h), *not tolerate any form of intimidation, bullying, victimization, vilification or harassment and to take disciplinary action against those who violate this policy*. Based on the justification provided by PP and evidence verified, the CAR-01 is closed.

CAR ID	CAR-02	Reference to Section:	Indicator 1.1	Date: 29.05.2024
Description of CAR				
Current version of the PDD template is 'VCS-Project-Description-Template-v4.4-FINAL2' and latest VCS standard is 'VCS Standard V4.7' updated on 16.04.2024.				
However, the PDD last updated on 20-Dec-2023 is using 'VCS Project Description Template V4.3' and 'VCS Standard V4.5' which is the outdated version.				
Project Proponent response				Date: 20.06.2024
The revised VCS PDD, version 2.0 is using the latest project description template, version 4.4, and project standard version 4.7.				
Documents submitted by Project Proponent				
VCS PDD, version 2.0				
VVB assessment				Date: 27.06.2024

The PDD has been revised by using the latest template which is version 4.4 and in accordance with the latest VCS standard version 4.7 as verified in the PDD report updated on 20.06.2024.

Observed that, the following are among changes (additional requirement) in the latest version of the PDD (V4.4) compared to the previous version of the PDD (v4.3):

Section 2.2 Risks to Stakeholders and the Environment

- 2.2.1 Management Experience
- 2.2.2 Risk Assessment
 - o Natural and human-induced risks to stakeholders' wellbeing
 - o Risks to stakeholder participation
 - o Working conditions
 - o Safety of women and girls
 - o Safety of minority and marginalized groups, including children
 - o Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)

Section 2.3.1 Labour and Work

- Discrimination
- Sexual harassment
- Equal pay for equal work
- Gender equity in labor and work
- Forced labor
- Child labor
- Human trafficking

As the PDD has been revised to the latest version of the report (v4.4) and in accordance with the latest VCS standard (v4.7) on 20.06.2024, therefore the CAR-02 identified during the validation activities is successfully closed.

CAR ID	CAR-03	Reference to Section:	Indicator 2.1.1 (3)	Date: 29.05.2024
Description of CAR				

Section 3.2, Applicability of Methodology for grouped project, page 44, has *listed the CDM Methodology of AMS III O*.

According to the VCS Standard Version 4.7, under indicator 2.1.1 (3) stated that “*Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.*”

However, the CDM Methodology specified in the PDD, AMS III O is one of the inactive CDM methodologies listed in the Verra Website. In addition, there is no evidence available to be verified during the validation activities to ensure the *PP had send a request and get approval from Verra to include the inactive CDM Methodology of AMS III O for the project.*

Project Proponent response

Date: 20.06.2024

AMS-III.O. is removed from the revised VCS PDD, version 2.0.

Documents submitted by Project Proponent

VCS PDD, version 2.0

VVB assessment

Date: 25.06.2024

GLT has removed the inactive CDM methodology AMS-III.O from the list of Application of Methodology for Grouped Project and PAI1 Project in the new revised VCS PDD, Version 2.0, page 34, 35 & 36.

Based on the above evidence, the corrective action is considered appropriate. Hence, the CAR-03 is successfully closed.

CAR ID	CAR-04	Reference to Section:	Indicator 3.1.2	Date: 29.05.2024				
Description of CAR								
1) Section 4.2 of the PDD, under project emissions for PAI1, page 110, PP has calculated the $PE_{www,treatment,y}$ based on its project scenario according to the 5.4 Project Emissions, paragraph 39 (b) under AMS-III.H where the $PE_{www,treatment,y}$ calculated value as follows:								
$PE_{www,treatment,y}$	5,274	7,000	7,000	7,000	7,000	7,000	7,000	1,726
According to the AMS-III.H Version 19.0, section 5.4, paragraph 39 (b) states that “Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery in the project scenario ($PE_{ww,treatment,y}$);”								

However, in the project scenario observed the entire wastewater treatment system is equipped with Biogas recovery as all wastewater passes through the anaerobic digester. Therefore, the companies interpretation on the methodology applied is not accurate.

- 2) Section 4.2 of the PDD, under project emissions for PAI1, page 112, under methodology AMS-III.H Version 19.0, the formula calculated for Project Emission flaring per year based on the provided emissions calculation sheet titled Tian Siang CER Version 1.0 excel file was $PE_{flare,y} = BE_{ww,treatment,y} / GWP_{CH4}$.

According to the AMS-III.H Version 19.0, TOOL 6 V4.0, paragraph 39 the formula for $PE_{flare,y}$ as follows:

$$PE_{flare,y} = GWP_{CH4} \times \sum_{m=1}^{525600} F_{CH4,RG,m} \times (1 - \eta_{flare,m}) \times 10^{-3} \quad \text{Equation (15)}$$

Therefore, the formula used by the company is not the same as an equation specified under AMS-III.H.

Project Proponent response

Date: 20.06.2024

- 1) $PE_{www,treatment,y}$

The treated wastewater from the anaerobic digester is channelled back to the existing aerobic pond for further treatment. In accordance with the description for $PE_{ww,treatment,y}$ which is to calculate methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery in the project scenario, in our case is aerobic pond, $PE_{ww,treatment,y}$ is calculated to be 0 since $MCF_{ww,treatment,PJ,k} = 0$ (aerobic treatment, well managed). As a result, the project emission for $PE_{www,treatment,y}$ is removed from the revised emission reduction calculation sheet.

- 2) $PE_{flare,y}$

According to AMS-III.H., version 19, for $PE_{flaring}$, for ex-ante estimation, baseline emissions calculation for wastewater and/or sludge treatment (i.e. equation (2) and/or equation (3)) can be used but without the consideration of GWP for CH_4 . As a result, for $PE_{flaring}$, calculation for ex-ante is using equation 2 from AMS-III.H.

Documents submitted by Project Proponent

Emission reduction calculation sheet, version 2.0

VVB assessment

Date: 29.06.2024

- 1) $PE_{www,treatment,y}$

GLT has revised the value of $PE_{www,treatment,y}$ to zero as the wastewater is channelled back to the aerobic pond for further treatment (due to $MCF_{ww,treatment,PJ,k} = 0$ (aerobic treatment, well managed)). This was sighted in the revised emission reduction calculation sheet, version 2.0 and also has been removed from

the Project Emission calculation under AMS-III.H ($PE_{ww,discharge,y}$ - Methane emissions from degradable organic carbon in treated wastewater) in page 108 of the revised VCS PDD, version 2.0.

2) $PE_{flare,y}$

GLT remains the calculation of ex-ante estimation of methane emissions due to incomplete flaring by using baseline emission calculation for wastewater (equation 2) as per stated in the AMS-III.H Version 19 page 22 and 13 of the document.

I. In page 22 quoted as follows:

$PE_{flaring,y}$ Methane emissions due to incomplete flaring in year y (t CO₂e). For ex ante estimation, baseline emission calculation for wastewater and/or sludge treatment (i.e. equation (2) and/or equation (3)) can be used but without the consideration of GWP for CH₄. However, the ex post emission reduction shall be calculated as per methodological tool "Project emissions from flaring"

II. In page 13 quoted as follows:

27. Methane emissions from the baseline wastewater treatment systems affected by the project ($BE_{ww,treatment,y}$) are determined using the COD removal efficiency of the baseline plant:

$$BE_{ww,treatment,y} = \sum_i (Q_{ww,i,y} \times COD_{inflow,i,y} \times \eta_{COD,BL,i} \times MCF_{ww,treatment,BL,i}) \times B_{o,ww} \times UF_{BL} \times GWP_{CH_4}$$

Equation (2)

In GLT scenario, they are not considering the GWP for CH₄ in the calculation.

Based on the above justification and submitted evidence, the corrective action is considered appropriate. Hence, the CAR-04 (1) & (2) is successfully closed.

CAR ID	CAR-05	Reference to Section:	Indicator 3.1.4	Date:29.05.2024
Description of CAR				
According to PDD Section 1.15,				
Grouped Project				

Grouped Project specifies PAIs within the boarder of Malaysia. Malaysia consists of 3 regions, Peninsular Malaysia, Sabah and Sarawak.

- Legislations highlighted needs to be discussed with changes between regions of Sabah and Sarawak who has the rights to enact its own legislative frameworks.

PAI 1

Currently, the following legal requirement are identified in the PDD:

- Environmental Quality Regulations (Designated Premises) (Crude Palm Oil) 1977 – P.U.(A) 342/77 Regulation 6 - Making Changes that change the quality of the effluent.
- Written Declaration on the Design and Construction of Air Pollution Control Systems [Regulation 7(5)] DOE ASPUB
- Construction Site Registration - DOSH JKJ 103
- Environmental Quality (Prescribed Premises) (Crude Palm Oil) Regulations 1977 – DOE
- Labour Act 1995

However, the following legal requirement are not identified though applicable to GLT, for example:

- EQA (Clean Air) Regulation 2014
- EQA (Scheduled Waste) Regulations 2005
- ACT 514, OSHA 1994
- Minimum Wages Order
- Employment Act 1955
- Employees Provident Fund Act 1991 (Act 452)
- ACT 177, Industrial Relations Act 1967

In addition, the following legal requirement is incorrectly cited:

- Environmental Quality Regulations (Designated Premises) (Crude Palm Oil) 1977, whereas it should 'Prescribed Premises'
- Labour Act 1995, whereas it should be 'Employment Act 1955'

Other license/ approval verified during the validation as below specific to PAI 1 is not specified in the PDD:

- Company registration, Companies Commission of Malaysia (SSM) – 900336 X, dated 10.05.2010

- Licensing from Energy Commission (ST) – still in progress as the engine is yet to run. Expected in Jun 2024
- Registration of construction site with CIDB – 0120230418-WP117607, dated 19.04.2023-18.04.2025
- Feed-In Approval (FiAH) – Reference no. G2022050010 dated 11.05.2022, Net export capacity (MW)-1.000 (BG1), installed capacity (MW)-1.200
- Scheduled Feed-in Tariff (FiT) commencement date – 11.05.2025, 21 years at 0.3966 RM/kWh
- Approved plan for POME discharge under Section 18(1), Act 127, Regulation 14, P.U (A) 342/1977 (JP/KKS/2018/2019/003238) dated 12.06.2018 signed by DOE, Pahang

Also, PAI 1 “Grid connection feasibility study and power purchase agreement from TNB” should be identified as a related license/approval (Not obtained yet due to the gas engine as not commenced operation)

Project Proponent response

Date: 05.06.2024

The description of the laws, statutes, and other regulatory frameworks for the grouped project is revised to consider the legislation for both Sabah and Sarawak in the revised VCS PDD, version 2.0 section 1.15.

The missing, incorrect and not specified legal requirement, license/approval for PAI1 is updated in the revised VCS PDD, version 2.0 section 1.15.

Documents submitted by Project Proponent

VVB assessment

Date: 27.06.2024

Referring to the revised PDD dated 20.06.2024 under section 1.15 ‘Compliance with Laws, Statutes and Other Regulatory Frameworks’, GLT has listed the following legal requirement that is applicable to its operation:

- Environmental Quality Regulations (Prescribed Premises) (Crude Palm Oil) 1977 – P.U.(A) 342/77 Regulation 6 - Making Changes that change the quality of the effluent
- Written Declaration on the Design and Construction of Air Pollution Control Systems [Regulation 7(5)] DOE ASPUB
- Construction Site Registration - DOSH JKJ 103
- Environmental Quality (Clean Air) Regulations 2014

- Environmental Quality (Scheduled Wastes) Regulations 2005
- Occupational Safety and Health Act 1994 (Act 514)
- Minimum Wage Order 2022
- Employment Act 1955
- Employees Provident Fund Act 1991 (Act 452)
- ACT 177, Industrial Relations Act 1967

In addition, GLT has listed the licenses and approval related to its project as below:

- Company registration with Companies Commission of Malaysia (SSM)
- Licensing from Energy Commission (ST)
- Registration of construction site with CIDB
- Feed-In Approval (FiAH)
- Feed-in Tariff (FiT)
- Approved plan for POME discharge under Section 18(1), Act 127, Regulation 14, P.U (A) 342/1977 (JP/KKS/2018/2019/003238) dated 12.06.2018 signed by DOE, Pahang
- Grid connection feasibility study and power purchase agreement from TNB

According to the revised PDD Template 'VCS-Project-Description-Template-v4.4-FINAL2' updated on 16.04.2024 under section 1.15 has specified that the PDD have to *identify and demonstrate the compliance of the project with all and any relevant local, regional and national laws, statutes and regulatory framework*.

Therefore, GLT shall identify all laws and regulations that applicable to its operation as the revised PDD still absent with the applicable laws such as Act 840 Anti-Sexual Harassment Act 2022, Act 800 Employment Insurance System Act 2017 and Act 4 Employees' Social Security Act 1969 (Amended 2003). Moreover, on 04.03.2022, Act A1648 OSH Act 2022 is passed with royal assent and come into force on 01.06.2024. The amended act supersedes the OSH Act 1994.

Referring to the above, CAR-05 raised during the validation activities remains open.

Project Proponent response II

Date: 17.07.2024

The Act 840 Anti-Sexual Harassment Act 2022, Act 800 Employment Insurance System Act 2017 and Act 4 Employees' Social Security Act 1969 (Amended 2003) are included under Section 1.15 in the revised PDD, version 3.0

The OSH Act 1994 is updated to Act A1648 OSH Act 2022 under Section 1.15 in the revised PDD, version 3.0

Documents submitted by Project Proponent II

VCS PDD, version 3.0

VVB assessment II

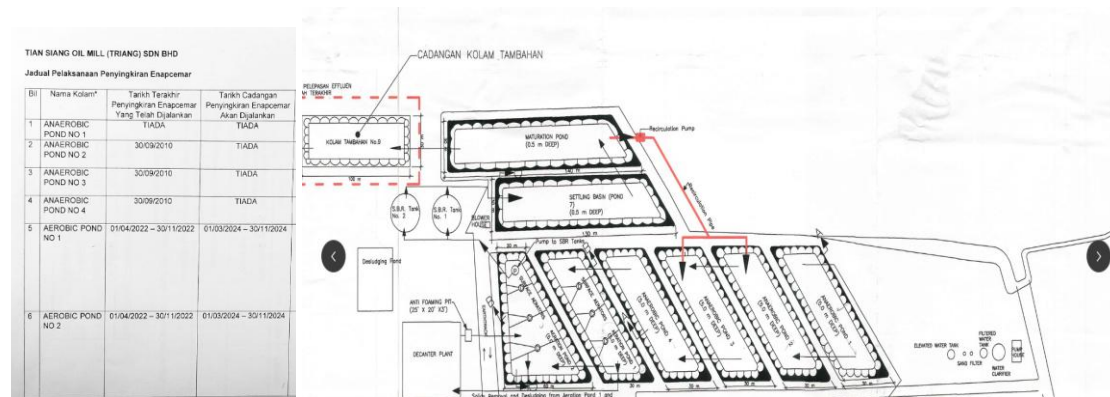
Date: 22.07.2024

Reviewed and verified the revised PDD, version 3.0 found, the Act 840 Anti-Sexual Harassment Act 2022, Act 800 Employment Insurance System Act 2017 and Act 4 Employees' Social Security Act 1969 (Amended 2003) have been identified as applicable laws to GLT Maju Sdn Bhd. Also, the OSH Act 1994 has been updated according to Act A1648 OSH Act 2022. Therefore, CAR-05 is successfully closed.

CAR ID	CAR-06	Reference to Section:	Indicator 3.6.16 (1)	Date: 29.05.2024
Description of CAR				
Section 3.2 of the PDD, Applicability Methodology, page 50, under methodology AMS-III.H, Justification of compliance, point (C) has stated that <i>the sludge removal interval is more than 30 days</i>. According to the AMS-III.H Version 19.0, under section 2.2, Applicability (C) stated that <i>the minimum interval between two consecutive sludge removal events shall be 30 days</i>. Even though the PDD is consistent with the methodology, there is no evidence available to be verified during validation activities to ensure "two consecutive sludge removals are more than 30 days". As per the VCS standard V4.7 requirement, 3.6.16 (1) that stated the <i>PP shall ensure the new PAI meet the applicability conditions set out in the methodology applied to the project</i>.				
Project Proponent response				Date: 20.06.2024
The sludge removal interval is more than 30 days, which is tabulated in 47. Desludging record from Tian Siang POM. The supporting description is updated as a footnote in the VCS PDD, version 2.0 section 3.2.				
Documents submitted by Project Proponent				
47. Desludging record from Tian Siang POM				
VVB assessment				Date: 26.06.2024

GLT had submitted the Desludging Record for Tian Siang POM as evidence to the activity of “two consecutive sludge removals are more than 30 days”.

Based on the document submitted, the records states that the desludging only carried out one time for 5 ponds (3 anaerobic ponds & 2 aerobic ponds) of Tian Siang, and no desludging activity done for Anaerobic Pond 1. For GLT Biogas Plant, the supply of POME is from Anaerobic Pond 1 which is no record of desludging in previous and it is proved that the sludge removal activity is more than 30 days.



With the appropriate justification from the desludging record of Anaerobic Pond 1 as supporting evidence, therefore the CAR-06 is successfully closed.

CAR ID	CAR-07	Reference to Section:	Indicator 3.6.17	Date:29.05.2024
Description of CAR				
Section 3.6.17 of the ‘VCS Standard V4.7’ updated on 16.04.2024 specify that new PAI shall have the following, among others: - Have evidence of the project ownership, in respect of each PAI. - Only be eligible for crediting from the later of start date of the project activity instance or the start of the verification period in which they were added to the grouped project, through to the end of the total project crediting period. - Not be or have been enrolled in another VCS project. - Adhere to the clustering and capacity limit requirements for multiple project activity instances set out in 3.6.8 – 3.6.9. Specific to PAI 1, under section “additionality” under eligibility requirements the reference made to section 3.5 Additionality, which does not discuss Regulatory surplus with requirements for mandatory installation of methane capture by the Malaysian Palm Oli Board (MPOB). Discussion on the mandatory				

installation of methane capture is only discussed under section 1.15 Compliance with Laws, Statutes, and Other Regulatory Framework	
Project Proponent response	Date: 20.06.2024
The description for the regulatory surplus requirement is included in the revised VCS PDD, version 2.0, section 3.5.	
Documents submitted by Project Proponent	
VCS PDD, version 2.0	
VVB assessment	Date: 27.06.2024
Section 3.5.1 of the revised PDD specified that the PAI1 is not mandated by any law, statute or other regulatory framework. Sighted under section 3.5 Additionality, Table 9 - Additionality for PAIs, GLT has explained the necessity to install wastewater treatment systems with methane capture for the new palm-oil mills constructed after 01.01.2014 as set by Malaysian Palm Oil Board (MPOB). Further explained that the requirement is applicable for the palm-oil mill with annual processing capacity equivalent to 270,000 MT and for the mills applied for capacity extensions. Referring to the above, the corrective action established is considered appropriate and accepted. Hence, the CAR-07 identified during the validation assessment is successfully closed.	

CAR ID	CAR-08	Reference to Section:	Indicator 3.7.1	Date:29.05.2024
Description of CAR				
Section 1.8 Ownership (Grouped Project) of PDD has stated that Green Lagoon Technology Sdn Bhd is the sole project owner of the project. While under PAI of the same section stated that GLT Maju Sdn Bhd is the owner of the biogas plant. Green Lagoon Technology Sdn Bhd has signed an agreement with GLT Maju Sdn Bhd, who has the legal rights to operate PAI1. The mill operator, Tian Siang POM has signed an agreement with GLT Maju Sdn Bhd. for ongoing access to the palm oil mill effluent (POME), in addition to permitting the construction and operation of PAI1. The following agreement were verified during the validation assessment: BUILD-OWN-OPERATE-TRANSFER AGREEMENT FOR BIOGAS-TO-POWER PROJECT AT TIAN SIANG (TRIANG) PALM OIL MILL, TRIANG, PAHANG dated 28.05.2021 – The agreement is between Tian Siang Oil Mill (Triang) Sdn Bhd and GLT Maju Sdn Bhd to permitting the construction and operation of PAI1 and for ongoing access to the palm oil mill effluent (POME).				

Lease Agreement for Biogas-To-power Project at Tian Siang (Triang) Palm Oil Mill, Triang, Pahang, dated 28.05.2021 – The agreement is between Tian Siang Oil Mill (Triang) Sdn Bhd PDD specifies “Green Lagoon Technology (GLT) Sdn Bhd has signed an agreement with GLT Maju Sdn Bhd, who has the legal rights to operate PAI1.”

Evidence provided with this regard is GLTM – Addendum BOOT Agreement between Tian Siang and GLT Maju specifies Tian Siang is transferring the Ownership of Environmental Attributes (Carbon Credits) to the developer which is GLT Maju. As GLT Maju irrespective of being a wholly owned subsidiary of GLT it’s its own legal entity with capped liability as a Private Limited Company. And no evidence that GLT Maju has transferred the rights for Environmental Attributes (Carbon Credits) to GLT.

Project Proponent response

Date: 20.06.2024

The evidence that GLT Maju has transferred the rights for Environmental Attributes (Carbon Credits) to GLT is provided as an agreement in 50. GLT2304-MCH_GLTM-Ownership Transfer Agreement_RO.

Documents submitted by Project Proponent

50. GLT2304-MCH_GLTM-Ownership Transfer Agreement_RO

VVB assessment

Date: 27.06.2024

Review the ‘50. GLT2304-MCH_GLTM-Ownership Transfer Agreement_RO’ document found, GLT Maju Sdn Bhd has established the ‘Ownership Transfer Agreement’ between GLT Sdn Bhd on 01.04.2024 and signed by both parties. Observed the agreement under section 1, Asset Description has specified that GLT Maju Sdn Bhd agrees to transfer to the GLT Sdn Bhd all rights, and interest in the following environmental attributes (Asset) – Carbon credits and/ or Renewable Energy Certificates (RECs) generated by GLT Maju Sdn Bhd 1MW Biogas Power Plant located at Triang, Pahang.

Based on the ownership transfer agreement verified, the CAR-08 identified during the verification activities is successfully closed.

CAR ID	CAR-09	Reference to Section:	Indicator 3.8	Date:29.05.2024
Description of CAR				
Start date of Project, various references within the PDD and the associated calculations, crediting period has to be revised as start date of the Project is with commission the digester by pumping in raw POME on the 24th April 2024. PDD sections 1.9, 1.10, 1.11 and other related references needs revision with the Project Start date. Also as evidenced during the validation Gas Engine is yet to start so all gas goes to flaring. So, AMS-1.D scope has not commenced. So, from 24-04-2024 to 29-05-2024 (to date) is only AMS-III.H scopes. Project Scale and Estimated GHG Emission Reductions or Removals has to be re-done accordingly.				

Project Proponent response

Date: 20.06.2024

Start date of Project for various references within the PDD (section 1.9, 1.10, 1.11) and the associated calculations, and crediting period were revised in the VCS PDD, version 2.0.

The initial operation date (IOD) for the gas engine is expected on 19/07/2024, one (1) month after the Interconnection Operation Manual (IOM) meeting (20/06/2024). Hence, the Project Scale and Estimated GHG Emission Reductions or Removals for AMS-I.D. has been redone and revised to 19/07/2024.

Documents submitted by Project Proponent

45. IOM meeting invitation

VVB assessment

Date: 27.06.2024

1. The client has revised the Project Start Date in the VCS PDD Version 2.0 to **24th April 2024**.

1.9 Project Start Date

Project start date	24-Apr-2024
Justification	According to VCS Standard, version 4.7, the project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or carbon dioxide removals. The Grouped Project start date is 24-April-2024, which is the commencement of emissions reductions at PA1 when the biogas plant is in operation.

Start and end date of first or fixed crediting period	24-Apr-2024 to 23-Apr-2031
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Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
24-April-2024 - 31-December-2024	23,335
01-January-2025 - 31-December-2025	35,372
01-January-2026 - 31-December-2026	35,372
01-January-2027 - 31-December-2027	35,372
01-January-2028 - 31-December-2028	35,372
01-January-2029 - 31-December-2029	35,372
01-January-2030 - 31-December-2030	35,372
01-January-2031 - 23- April-2031	10,951
Total estimated ERRs during the first or fixed crediting period	246,518
Total number of years	7
Average annual ERRs	35,217

2. The client has revised the calculation of Baseline Emission for AMS-I.D to start from 19th July 2024 until 31st December 2024 in the VCS PDD Version 2.0

AMS ID

BE_y- Baseline emissions from electricity generation in power plants that are displaced due to the project activity

Parameter	19-July-2024 - 31-Dec-2024	01-Jan-2025 - 31-Dec-2025	01-Jan-2026 - 31-Dec-2026	01-Jan-2027 - 31-Dec-2027	01-Jan-2028 - 31-Dec-2028	01-Jan-2029 - 31-Dec-2029	01-Jan-2030 - 31-Dec-2030	01-Jan-2031 - 23-Apr-2031
EG _{BLy}	3,586	7884	7884	7884	7884	7884	7884	2,441
EF _{CO2}	0.585	0.585	0.585	0.585	0.585	0.585	0.585	0.585
BE _y	2,098	4,612	4,612	4,612	4,612	4,612	4,612	1,428

Based on the submitted evidence and amendment has been made in the VCS PDD Version 2.0, the finding of CAR-09 is successfully closed.

CAR ID	CAR-10	Reference to Section:	Indicator 3.16.1	Date: 29.05.2024
Description of CAR				
Under Section 5.1 of the PDD following missing “Data & Parameters available at validation” are identified, - Applicable for AMS-III.H: 1. Qww.y 2. CODinflow.i.y 3. CODwwtreatment,discharge.ph.y 4. CODremoved.pj.k.y - Applicable for AMS-I.D: 1. ECpj.y In addition, Section 5.2 of the PDD, “Data and Parameters Monitored” for PAI1, Identified there are “Values” (such as 133,257 m3/yr for Qww.i.y) stated in the “Data & Parameters Monitored”: There shall be no values stated under the section but only the parameters to be monitored. - Applicable for AMS III. H 1. Qww.i.y 2. COD ww,untreated,PJ,y or COD inflow, PJ,y 3. CODww.discharge.y Applicable for AMS-I.D: 1. ECpj.y Also under PDD section 5.2 identified the parameter of Sfinal,PJ.y under “Data & parameters monitored”, which is not used for any calculation, performance monitoring and/or reporting purpose for PAI 1.				
Project Proponent response				Date: 05.06.2024
All the missing parameters for AMS-III.H. and AMS-I.D. are included under the revised VCS PDD, version 2.0, section 5.1. AMS-III.H.				

1. $Q_{ww,y}$ ($Q_{ww,i,y}$)
2. $COD_{inflow,i,y}$ ($COD_{ww,untreated,y}$ or $COD_{inflow,y}$)
3. $COD_{ww,treated,y}$ or $COD_{outflow,y}$
4. $COD_{ww,treatment,discharge,ph,y}$ ($COD_{ww,discharged,y}$)
5. $COD_{removed,pj,k,y}$ ($COD_{removed,PJ,k,y}$)

Tool 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 3.0

6. $EC_{pj,y}$ ($EC_{PJ,j,y}$)

All the values included for the parameters stated above for AMS-III.H. and AMS-I.D. are amended to 'value to be monitored ex-post' in the revised VCS PDD, version 2.0, section 5.2.

AMS-III.H.

1. $Q_{ww,i,y}$ ($Q_{ww,i,y}$)
2. $COD_{ww,untreated,PJ,y}$ or $COD_{inflow,PJ,y}$ ($COD_{ww,untreated,y}$ or $COD_{inflow,y}$)
3. $COD_{ww,treated,y}$ or $COD_{outflow,y}$
4. $COD_{ww,discharge,y}$ ($COD_{ww,discharged,y}$)

Tool 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 3.0

1. $EC_{pj,y}$ ($EC_{PJ,j,y}$)

The parameter of $S_{final,PJ,y}$ under "Data & parameters monitored", which is not used for any calculation, performance monitoring and/or reporting purpose for PAI 1 is deleted in the revised VCS PDD, version 2.0, section 5.2.

Documents submitted by Project Proponent

VCS PDD, version 2.0

VVB assessment

Date: 27.06.2024

1. The client had included the monitoring parameters missing for AMS-III.H and Tool 05 under section 5.1 "Data & Parameters available at validation" in the VCS PDD version 2.0 as follows:

AMS-III.H

- i. $Q_{ww,y}$ ($Q_{ww,i,y}$) – page 138 of VCS PDD Version 2.0

- ii. CODinflow.i.y (CODww,untreated,y or CODinflow,y) – page 139 of VCS PDD Version 2.0
- iii. CODww,treated,y or CODoutflow,y – page 139 of VCS PDD Version 2.0
- iv. CODwwtreatment,discharge.ph.y (CODww,discharged,y) – page 140 of VCS PDD Version 2.0
- v. CODremoved.pj.k.y (CODremoved,PJ,k,y) – page 141 of VCS PDD

However, the parameter of $EC_{pj,y}$ ($ECPJ,j,y$) under AMS-III.H (Tool 05) was not listed for PAI 1 Project under section 5.1 “Data & Parameters available at validation” in the VCS PDD Version 2.0.

The client has amended the value of parameters to ‘value to be monitored ex-post’ in the revised VCS PDD, version 2.0, section 5.2.

2. The client has included the monitoring parameters missing for AMS-III.H, AMS-I.D (Tool 05) under section 5.2 to “Data & Parameters Monitored” in the VCS PDD version 2.0 as follows:

AMS-III.H

- i. Qww.i.y ($Q_{ww,i,y}$) – page 202 of VCS PDD Version 2.0
- ii. CODww,untreated,PJ,y or CODinflow,PJ,y (CODww,untreated,y or CODinflow,y) – page 203 of VCS PDD Version 2.0
- iii. CODww,treated,y or CODoutflow,y – page 203 & 204 of VCS PDD Version 2.0
- iv. CODww.discharge.y (CODww,discharged,y) – page 204 of VCS PDD Version 2.0

Tool 05

- i. $EC_{pj,y}$ ($ECPJ,j,y$) – page 212 of VCS PDD Version 2.0

Based on the justification and amendment made by the client, there is a missing parameter of AMS-III.H Tool 05, $EC_{pj,y}$ ($ECPJ,j,y$) to be listed for PAI 1 project under section 5.1 “Data Parameters available at validation”. Hence, the CAR-10 remains “Open”.

Project Proponent response II

Date: 17.07.2024

The parameter of EC_{pj,y} (EC_{PJ,j,y}) under Tool 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 3.0 is updated in the revised PDD version 2.0, page 144.

Documents submitted by Project Proponent I

VCS PDD, version 3.0

VVB assessment II

Date: 22.07.2024

Sighted the parameter of AMS-III.H Tool 05, EC_{pj,y} (EC_{PJ,j,y}) have been listed for PAI 1 project under section 5.1 "Data Parameters available at validation" page 144, VCS PDD version 3.0. Therefore, the CAR-10 is successfully closed.

CAR ID	CAR-11	Reference to Section:	Indicator 3.17.1	Date:29.05.2024
Description of CAR				
Section 1.18 'Sustainable Development Contributions' of the PDD under SDG 7 and SDG 8 stated the following: PDD text states, "An average of 8,000 MWh per year of power is uploaded to the grid." When interviewed it was specified this is based on Biogas engine maintenance recommendations. However related evidence not provided to confirm the fact. "A total of 8 job opportunities provided to the locals" Evidence provided GLTM – List of employees no specification whether the workers are local. Request for the IC numbers to cross refer they are Malaysian was requested but evidence not provided by the closing meeting.				
Project Proponent response				Date: 05.06.2024
An average of 8,000 MWh is updated to 7,884 MWh in the revised VCS PDD, version 2.0. The supporting document is attached as 48. Average power uploaded to the grid. The supporting document for the 8 job opportunities provided to the local is as attached 8. GLTM – List of employees with IC.				
Documents submitted by Project Proponent				
48. Average power uploaded to the grid (SEDA requirement) 8. GLTM – List of employees with IC				

VVB assessment	Date: 27.06.2024
Section 1.18 'Sustainable Development Contributions' on the revised PDD dated 20.06.2024 has stated the following: - SDG 7 - An average of 7,884 MWh per year of power is uploaded to the grid - SDG 8 - A total number of 8 job opportunity will be provided to the local SDG 7 GLT has amended the total of power uploaded to the grid from 8000 MWh to 7884 MWh as sighted in the 'Note of Changes on Procedure on Biogas Power Plant Acceptance Test And Performance Assessment For Feed-In Tariff (FiT) Project In Malaysia' (reference file: 48. <i>Average power uploaded to the grid (Seda Requirement)</i>) sent to Sustainable Energy Development Authority (seda). Section 3.2.2, point number 2 of the Note of Changes explained that the <i>test shall be conducted for 336 hours duration (approximately for 14 days, 24 hours operation) and shall achieve minimum of 90% uptime.</i> SDG 8 Review the 'List of Employees' available with Name of The Employees, IC Number, Position, Employment Type and Date Join. Currently, there are 7 local employees were hired by GLT. The youngest is 20 years old and the oldest is 44 years old. Based on the revised PDD, the project will provide 8 job opportunity to the local people. Based on the above, the corrective action established is considered appropriate and accepted. Therefore, CAR-11 identified during the verification activities is successfully closed.	

CAR ID	CAR-12	Reference to Section:	Indicator 3.18.1	Date: 29.05.2024
Description of CAR				
GLT has identified their stakeholders within 10 KM radius from the project site. Government agencies, schools, nearby estates and communities and palm oil mill are among stakeholders identified. Examples are as follows: - Pejabat Daerah dan Tanah Bera - Communities residing within MRSM Bera, SK LKTP Triang II, SJK (C) Mengkuang - Felda Triang, Felda Triang II, Felda Mengkuang				

- Tiang Siang Oil Mill (Triang)
- Bera town and residents

However, other stakeholders such as Police, Hospital, Labour Department (Kuantan), Department of Occupational Safety and Health (DOSH), Malaysian Palm Oil Board (MPOB), BOMBA and Department of Environment (DOE), Kampong Baharu, Kg. Jawi-Jawi, Kg. Kuala Jerang, Kg. Tanjung Kundang, Triang Hilir, Kg. Hayat, Kg. Ayer Bening, etc. are not identified though they are within 10 KM from the project site.

Project Proponent response

Date: 05.06.2024

Department of Environment (DOE)

Invitation to Department of Environment (DOE) has been extended on 27th October 2023 via email for the stakeholder consultation purposes. Follow-up via phone calls have been made. However, no participants from DOE has attended the stakeholder consultation.

Kampong Baharu, Kg. Jawi-Jawi, Kg. Kuala Jerang, Kg. Tanjung Kundang, Triang Hilir, Kg. Hayat, Kg. Ayer Bening, etc.

Based on the information obtained from the Penghulu Mukim Triang and Mukim Bera, the settlements mentioned above is not located within 10KM from the project site.

The list of stakeholders mentioned in the VCS PDD were identified based on the Google Map search and cross-checked with Profil Daerah Pahang 2018 but some of the settlements were not shown in the Google Map. Therefore, the list of settlements mentioned above were not identified as part of the stakeholders.

On-going stakeholder consultation with other stakeholders

On-going stakeholder consultation will be continued in the future during the monitoring and hence, if there are any issues raised by the stakeholders, they will be reported in the respective Monitoring Report in the future.

Documents submitted by Project Proponent

41. Stakeholder Invitation Letter to DOE

49. Profil Daerah Pahang 2018

VVB assessment

Date: 27.06.2024

Based on the revised PDD dated 20.06.2024, section 2.1.1 Stakeholders Identification, GLT has identified their stakeholders to be included in the invitation list as below. In addition, GLT has specified that the stakeholders identified are within 10km radius from the project site.

- Department of Environment (DOE) Pahang (invitation for stakeholders' consultation is sent on 27.10.2023 and 13.05.2024)
- Pejabat Daerah Dan Tanah Bera
- Communities residing within MRSB Bera
- Communities residing within SK LKTP Triang II
- Communities residing within SJK (C) Mengkuang
- Nearby local residents – Felda Triang II
- Felda Triang 01 and Felda Mengkuang millworkers and community residing within the mills
- Tian Siang Oil Mill (Triang) Sdn Bhd workers and community residing within the mill
- Bera town and housing estates residents

Further explained that the above identified stakeholders were considered adequate to fulfil the criteria to represent the affected communities within 10km radius from the project site.

Based on the 'Profil Daerah Pahang 2018' and justification provided, the concern related to the villages (Kampung Baharu, Kg. Jawi-Jawi, Kg. Kuala Jerang, Kg. Tanjung Kundang, Triang Hilir, Kg. Hayat, Kg. Ayer Bening, etc.) is successfully closed.

However, Labour Department (Kuantan) and Department of Occupational Safety and Health (DOSH) are yet to be identified as a GLT's stakeholder though they are related in term of the applicable law and regulation to GLT. Hence, the concern related to the government agencies as a GLT's stakeholders remains open.

Project Proponent response II

Date: 17.07.2024

An email was sent to Labour Department (Kuantan) and Department of Occupational Safety and Health (DOSH) on 17 July 2024 to inform them regarding the project and mentioned that they can submit any comment to GLT if they have any comment. Labour Department (Kuantan) and Department of Occupational Safety and Health (DOSH) has been listed as the stakeholder and was included in the revised PDD, version 3.0 under section 2.1.1.

Documents submitted by Project Proponent II

PDD version 3.0

VVB assessment II

Date: 22.07.2024

Review the revised PDD, version 3.0 under section 2.1.1 found, Labour Department (Kuantan) and Department of Occupational Safety and Health have been identified as a GLT's stakeholders. Therefore, CAR-12 is successfully closed.

CAR ID	CAR-13	Reference to Section:	Indicator 3.18.2	Date: 29.05.2024
Description of CAR				
Stakeholders such as Police, Hospital, Labour Department (Kuantan), Department of Occupational Safety and Health (DOSH), Malaysian Palm Oil Board (MPOB), BOMBA and Department of Environment (DOE), Kampong Baharu, Kg. Jawi-Jawi, Kg. Kuala Jerang, Kg. Tanjung Kundang, Triang Hilir, Kg. Hayat, Kg. Ayer Baning, etc. were not identified though they are within 10 KM from the project site, hence not consulted prior to the implementation of project activity.				
Project Proponent response				Date: 05.06.2024
Department of Environment (DOE) Invitation to Department of Environment (DOE) has been extended on 27th October 2023 via email for the stakeholder consultation purposes. Follow-up via phone calls have been made. However, no participants from DOE has attended the stakeholder consultation. Kampong Baharu, Kg. Jawi-Jawi, Kg. Kuala Jerang, Kg. Tanjung Kundang, Triang Hilir, Kg. Hayat, Kg. Ayer Baning, etc. Based on the information obtained from the Penghulu Mukim Triang and Mukim Bera, the settlements mentioned above is not located within 10KM from the project site. The list of stakeholders mentioned in the PDD were identified based on the Google Map search and cross-checked with Profil Daerah Pahang 2018 and some of the settlements were not shown in the Google Map. Therefore, the list of settlements mentioned above were not identified as part of the stakeholders. On-going stakeholder consultation On-going stakeholder consultation will be continued in the future during the monitoring and hence, if there are any issues raised by the stakeholders, they will be reported in the respective Monitoring Report in the future.				
Documents submitted by Project Proponent				
41. Stakeholder Invitation Letter to DOE 49. Profil Daerah Pahang 2018				
VVB assessment				Date: 27.06.2024

Review the 'Stakeholder Consultation Report' dated Oct 2023 found, GLT has announced their stakeholders meeting will be conducted on 31.10.2023 at Bera Rest House, Pahang. The announcement is made via GLT's webpage, LinkedIn Profile and Facebook. According to the report, the identified stakeholders is located within 8-9km from Tian Siang Oil Mill Sdn Bhd (TSOMSB). An invitation letter is provided to the identified stakeholders prior to the meeting. The following is an example of the stakeholders identified and invited by GLT:

- Department of Environment (DOE) Pahang (invitation for stakeholders' consultation is sent on 27.10.2023 and 13.05.2024)
- Pejabat Daerah Dan Tanah Bera
- Communities residing within MRSB Bera
- Communities residing within SK LKTP Triang II

Refer to the CAR-12

As Labour Department (Kuantan) and Department of Occupational Safety and Health (DOSH) are yet to be identified as a GLT's stakeholder, thus there is no consultation made to both government agencies. Hence, CAR-13 remains open.

Project Proponent response II

Date: 17.07.2024

An email was sent to Labour Department (Kuantan) and Department of Occupational Safety and Health (DOSH) on 17th July 2024 to inform them regarding the project and mentioned that they can submit any comment to GLT if they have any comment

Documents submitted by Project Proponent II

Email attachment

VVB assessment II

Date: 22.07.2024

Review the revised PDD, version 3.0 under section 2.1.1 found, Labour Department (Kuantan) and Department of Occupational Safety and Health have been identified as a GLT's stakeholders. On 17.07.2024, an email communication has been sent to both Labour Department and DOSH. Both government agencies were informed of the project description and location of the project. Any inquiry, they may contact GLT directly. Based on the evidence provided, CAR-12 is successfully closed.

CAR ID	CAR-14	Reference to Section:	Indicator 3.18.3	Date: 29.05.2024
Description of CAR				

The Global Public Stakeholder Consultation resulted in one comment on 15-02-2024 with the request for baseline scenario and what is the share of the practice mentioned in baseline scenario in Malaysia or in the mentioned region? However, the comment is not available in the public comment summary.	
Project Proponent response	Date: 05.06.2024
The description for the public comment is included in revised VCS PDD, version 2.0, section 2.1.5.	
Documents submitted by Project Proponent	
VCS PDD, version 2.0	
VVB assessment	Date: 27.06.2024
Review the revised PDD dated 20.06.2024 under section 2.1.5 Public Comments found, the comment received on 15.02.2024 is included in the PDD under section 1.1 Summary Description of the Project. However, according to the 'VCS Project Description Template V4.4' under section 2.1.5, all public comments received should be summarized under section 2.1.5 Public Comments. Hence, CAR-14 remains open.	
Project Proponent response II	Date: 17.07.2024
The public comment received, and action taken are tabulated according to the VCS PDD project description template, version 4.4 under section 2.1.5 in the revised PDD, version 3.0	
Documents submitted by Project Proponent II	
VCS PDD, version 3.0	
VVB assessment II	Date: 22.07.2024
Reviewed and verified the revised PDD, version 3.0 under section 2.1.5 found, the public comment is included and tabulated accordingly. Therefore, CAR-14 is successfully closed.	

CAR ID	CAR-15	Reference to Section:	Indicator 3.18.4	Date: 29.05.2024
Description of CAR				
Customer Complaint Handling (GLTSB-CM-04) dated 01,05,2021 is established with the objective to <i>provide mechanism for handling complaints received from customers, being the key interested parties</i> . Section 7.0 Procedure describes the process flow from receiving until resolving of the complaint. A sample of <i>Complaint Record</i> and <i>Complaint Log</i> are included in the complaint procedure as well. However, the PDD stated that <i>comment/feedback/ complaints from stakeholders will be channeled to POM by GLT and the result will be incorporated in POM's complaint procedure</i> .				

Project proponent shall handle the complaints directly and not entrust other party who is not liable for the Project.	
Project Proponent response	Date: 20.06.2024
The Grievance Redress Procedure in Section 2.1.4 has been amended accordingly based on the 18. GLTSB Customer complaint procedure	
Documents submitted by Project Proponent	
VCS PDD, version 2.0	
VVB assessment	Date: 27.06.2024
Review the revised PDD dated 20.06.2024 under section 2.1.4 Grievance Redress Procedure found, all comment or feedback or complaints received from stakeholders will be channeled directly to GLT for further action. The comments or complaints can be made verbally or written to any of GLT's personnel, and the complaints will be channeled to the Management Representatives (MR) for further and necessary action. In addition, GLT has established a 'Customer Complaint Handling' (GLTSB-CM-04) dated 01.05.2021 that explained the procedure of handling complaints. Section 7 of the procedure illustrated the process flow from receiving until closure of the complaint which include the person in-charge. Based on the above, corrective action established is considered appropriate and accepted. Therefore, CAR-15 identified during the verification assessment is successfully closed.	

CAR ID	CAR-16	Reference to Section:	Indicator 3.26.2	Date:29.05.2024
Description of CAR				
5.3 of the PDD, Quality Control and Quality Assurance, (Training), paragraph 237, PP has stated a training plan will be set out for the operator. This will ensure that monitoring staff involved in the project are properly trained so that they are able to undertake the tasks required by this monitoring plan. Appropriate staff training must be provided before the project starts operating and generating emission reductions.				
Project Proponent response				Date: 20.06.2024
The biogas plant is yet to be in full operation. The training plan will be developed once the biogas plant is fully in operation and hence, staff training will be provided before the project starts operating and generating emission reductions. In addition, there is monitoring manual for the carbon credit project which is attached in 19. GLT_VCS Monitoring Manual 1.0_050424				
Documents submitted by Project Proponent				

19. GLT_VCS Monitoring Manual 1.0_050424	
VVB assessment	Date: 27.06.2024
Referring to the 'Wastewater to Biogas Grouped Project – Monitoring Manual' dated April 2024, under section 4. Training Plan stated the following: <i>The first training will be led by the appointed Carbon Consultant who will prepare all the training documents as well as data recording and documentation templates. Subsequent training will be performed by the engineer.</i> Among training to be provided to the GLT's personnel are as below: - General training on monitoring – Plant Manager, Engineer and Supervisor, General workers - Training on data recording and troubleshooting - Engineer and Supervisor, General workers - Refresher training program (to maintain staff competency in monitoring) - Engineer and Supervisor, General workers In addition, the monitoring manual also specified that perioding training on the operations of the biogas system, POME treatment and power generators to respective employees shall be conducted. Referring to the revised PDD dated 20.06.2024, under section 5.3 (c) Training of Monitoring Personnel has specified the following: <i>All plant staff will be trained by Green Lagoon Technology Sdn Bhd, prior to the commissioning of the plants. The involved personnel shall receive adequate and regular training on the project performances and the monitoring plan involving methane capture up to utilization of the biogas</i> Moreover, under section 5.3 (f) - Quality Control, and Quality Assurance, point number 5 Training has explained that <i>a training plan will be set out for the operator. Appropriate staff training must be provided before the project starts operating and generating emission reductions.</i> Based on the above, corrective action established is considered appropriate and accepted. Therefore, CAR-16 identified during the verification assessment is successfully closed.	

Table 3: Forward Action Requests (FAR)

FAR ID	FAR-01	Reference to Section:	Indicator 3.1.4	Date: 29.05.2024
Description of FAR				
The biogas plant is built on the Tian Siang POM's land Lot PT 7952 and PT 7953, which is for the purpose of planting oil palms.				

Referring to the Tian Siang POM's memo to Bera Land Department dated 05.12.2023, the mill has appointed BroUkur as a consultant to change the term and condition of the land. The memo is signed by the mill manager and acknowledged by Bera Land Department on 05.12.2023.

The PP needs to ensure the land is converted to the industrial land to ensure comply with the National Land Code, Act 828.

Raised as a FAR as the POM (The owner of the land which is leased to the PP) is in process of converting the current land title of Agricultural to Industrial. And also considering the government process takes considerable time to be completed.

Project Proponent response

Date: 05.06.2024

The owner of the land which is leased to the project proponent is in the process to convert the land title of agriculture to industrial. The status will be updated once the owner receive the approval.

Documents submitted by Project Proponent

VVB assessment

Date: 27.06.2024

As the process of changing the term and condition of the land is still in progress, FAR-01 raised during the validation assessment remains open. To be verified in the next verification assessment.

Table 4: Observations (OBS)

OBS ID	OBS-01	Reference to Section:	PDD 1.18	Date:29.05.2024
Description of OBS				
Section 1.18 'Sustainable Development Contributions' identified 3 contributions to the Sustainable Development Goals (SDG) which is SDG 7, 8 and 13. SDG 7 - Ensure access to affordable, reliable, sustainable and modern energy for all SDG 8 - Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all SDG 13 - Take urgent action to combat climate change and its impacts The PP could consider includes the following SDG as it is related to the project as well: SDG 3 – Ensuring healthy lives and promoting well-being at all ages. Capture of harmful emissions, improving air quality of communities				

SDG 6 – Clean water and sanitation – improvement treatment of wastewater
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